

IDENTIFICATION OF PARENT STRAINS

Strain	Parentage or Source
A72-507	Amsoy x Wayne
A72-512	Amsoy x Wayne
A73-19084	IVR Ex 5003 x Wells
A74-102011	M62-263 x IVR Ex4426
A74-104030	[Provar x (AX56P64-1 x PI9110-1)] x Wells
A74-204034	[Grant x (Lincoln x Hawkeye)] x [Amsoy ⁸ x (Blackhawk x Harosoy)]
Agripro 25	Beeson x Calland
AP68-1016	Clark ⁵ x PI84946-2
AP68-1022	Clark ⁵ x PI84946-2
Asgrow A3300	Beeson x Calland
AX56P64-1	Amsoy (Adams x Harosoy)
AX900-4-3	CX407BC7-255 x AP68-1022
AX901-40-2	Beeson x AP68-1022
C1070	Lincoln x Ogden
C1317-71	C1223 ⁸ x Mukden
C1223	C1070 x Adams
C1253	Blackhawk x Harosoy
C1421	Adelphia ⁸ x Mukden
C1426	(Blackhawk x Harosoy) x Kent
C1477	Amsoy 71 (Amsoy ⁸ x C1253)
C1520	Bonus x Cutler
D49-2573	Roanoke x N45-745
C1523	Beeson x L63-1397
C1528	Calland x L63-1397
CX407BC7-255	Amsoy ⁸ x C1253
CX456-90	Amsoy x PI219782
Dortchsoy 110	Ogden x Wabash
D49-2510	S100 x CNS
D60-9647	FC31745 x D49-2510
D61-5141	Dorman ⁵ x PI181537
D68-18	Dyer x Bragg
IVR Ex4426	Amsoy x Wayne

IDENTIFICATION OF PARENT STRAINS

Strain	Parentage or Source
IVR Ex5003	Provar x (AX56P64-1 x PI19110-1)
J74-5	Forrest x (D68-18 x PI88788)
K1001	Wayne x C1317-17
K1028	Williams x Calland
L12	[(Clark ⁸ x CNS) x (Clark ⁸ x Blackhawk)] x [(Clark ⁶ x T201) x (Clark ⁶ x T145)] <u>I</u> <u>r</u> <u>Rps1</u> <u>rxp1</u>
L15	Wayne x Clark 63 (<u>Rps1</u>)
L57-0034	Clark x Adams
L60-347-1-60-2B	Harosoy x Higan
L61-344	Harosoy ⁶ x T117 (<u>Dt₂</u>)
L62-535	Harosoy ⁶ x T145 (<u>dt₁</u>)
L63-3297	Clark ⁶ x T141 (<u>dt₁</u>)
L63-1397	Harosoy ⁶ x T207 (<u>Dt₂</u>)
L63-3534	(Clark ⁶ x T201) x (Clark ⁶ x T145) (<u>I₁</u> <u>P₁</u> <u>r</u>)
L66-531	(Clark ⁶ x T245) <u>dt₁</u> <u>e₂</u> x (Clark ⁶ x T175) <u>E₁</u> <u>t</u>
L66-1359	Wayne x L57-0034
L66L-137	Wayne x L57-0034
L66-154	Wayne x L57-0034
L66L-177	Wayne x L57-9819
L68-4096	(L15 ⁵ x L63-3534) x (Wayne ¹⁰ x Kanrich) <u>r</u> <u>Rpm</u> <u>Rps</u>
L69-5343	L12 ⁶ x Hawkeye (<u>I</u> <u>r</u> <u>Im</u>)
L69U40-19-1	Calland x Amsoy
L70-2283	Chippewa ⁶⁴ x Custer
L72-844c-1	Williams ⁵ x L68-4096
L72D-549	L62-3929 x Rampage
L72U-547	L62-535 x SRF300
L72U-640	L66-531 x C1426
L72U-2567	Williams x Ransom
L73-6626	R62-659 x L66-531
L75-8020	Williams x L70-2283
LL4102	[Wayne ² x (Clark x Adams)] x Cutler
M10	Lincoln x Richland
M53-117	M10 x PI 180501
M54-139	Renville x Capital

IDENTIFICATION OF PARENT STRAINS

Strain	Parentage or Source
M54-240	(Lincoln x Richland) x Korean
M61-224	Merit x Harosoy
M62-263	Grant x M319W
M62-275	Norchief x Harosoy
M61-345	M319W x Harosoy
M63-194	Corsoy x PI 132207
M63-2174	Corsoy x M53-117
M64-157	Merit x Amsoy
M64-185	Chippewa 64 x Amsoy
M65-69	M384 x Corsoy
M65-115	Anoka x Amsoy
M66-18	Clay x Altona
M67-141	Corsoy x Wayne
M68-48	Evans x M54-240 x M54-139)
M319W	Lincoln X Hawkeye
M384	Capital x Renville
Mitchell	Amsoy x Wayne
ML7293-4	Merit x Lee
NKS 1492	Corsoy x Wayne
N45-745	Ogden x CNS
N45-1497	Ral soy x Ogden
Pride B216	Corsoy x Wayne
R62-659	(R54-168 x Hill) x Lee x Dortchsoy 110)
R54-168	D49-2573 x N45-1497
SL12	Wayne ¹⁰ x Kanrich (<u>I</u> <u>r</u> <u>Rpm</u> <u>Rps</u>)
SRF300	Wayne ⁷ x D61-5141
SRF350	Wayne ³ x D61-5141
Y68-1034	York x PI71506
Y68-1033	York x PI71506

UNIFORM 00, 1980

Strain	Parentage	Previous Testing*	Generation Compositd
Clay (0)	Capital x Renville	3	F ₅
Maple Arrow	Harosoy 63 x 840-7-3	3	F ₅
Maple Presto	(Amsoy x Portage) x 840-7-3	2	F ₅
McCall	(Acme x Chippewa) x Hark	7	F ₅
Portage (00)	Acme x Comet	20	F ₅
M71-148	Clay x Evans	-	F ₅
OT80-1	840-7-3 x (Harosoy 63 x Altona)	-	F ₆
OT80-2	Maple Presto x Evans	-	F ₅
OT80-3	(Amsoy x Portage) x 840-7-3	-	F ₆

*Number of years in test or name of 1979 test

The regional data for 1980 and the three-year mean show that McCall is the highest yielding Group 00 variety in the test. M71-148 was the highest yielding entry in the test, had excellent shattering resistance, and is resistant to races 1 and 2 of phytophthora root rot. This strain is probably too late for the 00 test and should be considered a Group 0 strain.

UNIFORM TEST 00, 1980

Descriptive and Other Data

Strain	Descriptive Code		Chlorosis Score		Hypocotyl Score	Shattering
			Ames	Lamberton	Ames	Manhattan 2 Weeks
Clay (0)	PGBr	SY	3.3	2.5	1	3
Maple Arrow	PTBr	SYBr	2.3	3.0	2	3
Maple Presto	PTBr	DY	3.3	3.5	1	2
McCall	PGBr	DY	3.0	2.0	1	2
Portage (00)	PGBr	SY	2.3	4.0	1	4
M71-148	WGBr	SY	2.7	4.0	1	1
OT80-1	PTBr	DYBr	3.5	4.0	1	2
OT80-2	PGBr	DY	3.3	4.0	1	2
OT80-3	PTBr	DYBr	2.2	4.0	1	-

Disease Data

Strain	BSR	GERM SMV PSB PS				PR
	Laf. Ind. n	* Lafayette, Indiana				Laf. Ind. a
	%	%	score	%	score	-Reaction - - - - -
Clay (0)	100	67	3M	10	5S	S
Maple Arrow	40	70	3M	3	3E	R
Maple Presto	0	37	3E	34	3E	R
McCall	0	69	3M	20	5S	S
Portage (00)	0	80	4M	7	5S	S
M71-148	40	78	1	9	5S	R
OT80-1	60	63	3E	18	5S	R
OT80-2	60	81	4E	7	3E	S
OT80-3	20	62	4E	23	3E	S

* Petri dish germination on potato dextrose agar

UNIFORM TEST 00, 1980

Regional Summary

Strain	Yield	Rank	Matu- rity	Lodg- ing	Height	Seed Quality	Seed Size	Composition	
No. of Tests	7	7	7	7	7	6	7	5	5
	bu/a	No.	Date	Score	In.	Score	g/100	%	%
Clay (0)	38.6	3	+9	2.5	29	2.5	16.9	41.8	19.9
Maple Arrow	37.1	4	+9	2.1	31	2.2	18.5	40.3	20.1
Maple Presto	25.0	9	-7	1.3	23	3.4	15.7	39.2	20.7
McCall	39.0	2	+6	2.1	30	2.3	15.6	40.8	19.0
Portage (00)	32.4	6	9-5*	1.9	27	2.6	18.2	39.7	19.3
M71-148	39.1	1	+9	2.3	30	2.3	14.7	41.2	20.1
OT80-1	34.1	5	-1	2.1	28	2.2	16.5	41.5	20.6
OT80-2	31.4	7	-6	1.5	24	1.9	14.7	40.1	20.6
OT80-3	29.9	8	-7	2.5	27	2.7	18.0	42.0	19.6

* 112 days after planting

1978-1980, 3-year mean

No. of Tests	24	24	24	24	24	22	24	15	15
Clay (0)	38.3	1	+11.5	1.9	28	2.3	16.5	40.7	20.0
Maple Arrow	34.9	3	+8.0	1.8	30	1.9	17.9	40.4	20.0
Maple Presto	25.9	5	-8.3	1.2	24	3.0	15.4	39.1	20.2
McCall	38.2	2	+4.5*	1.7	29	2.1	15.2	39.9	19.2
Portage (00)	31.8	4	9/6*	1.6	28	2.3	17.0	39.6	19.3

* 108 days after planting

UNIFORM TEST 00, 1980

Strain	Mean 7 Tests	Ont.		Wisc.	N.D.
		Ottawa	Elora	Ashland	Fargo
YIELD (bu/a)					
Clay (0)	38.6	53.9	44.5	28.9	32.1
Maple Arrow	37.1	53.5	47.3	30.0	25.3
Maple Presto	25.0	38.6	25.2	24.8	19.4
McCall	39.0	56.4	44.8	27.5	31.4
Portage (00)	32.4	46.0	38.0	27.2	24.8
M71-148	39.1	55.4	38.0	29.4	32.0
OT80-1	34.1	44.4	40.6	28.9	25.7
OT80-2	31.4	49.4	23.6	26.5	28.1
OT80-3	29.9	39.3	32.9	24.7	23.6
C.V. (%)		10.2	12.7	7.12	7.8
L.S.D. (5%)		7.2	6.9	3.4	3.6
Row sp (in.)		10	7	24	18
Rows/plot		4	8	4	4
Reps		4	4	3	3

Strain	7 Tests	<u>YIELD RANK</u>			
		Ottawa	Elora	Ashland	Fargo
Clay (0)	3	3	3	4	1
Maple Arrow	4	4	1	1	6
Maple Presto	9	9	8	8	9
McCall	2	1	2	5	3
Portage (00)	6	6	5	6	7
M71-148	1	2	6	2	2
OT80-1	5	7	4	3	5
OT80-2	7	5	9	7	4
OT80-3	8	8	7	9	8

UNIFORM TEST 00, 1980

<u>Man.</u>		<u>Minn.</u>	
Morden	Brandon*	Rosemount	Morris
<u>YIELD (bu/a)</u>			
38.7	1.8	31.5	40.6
39.0	3.7	26.6	37.9
36.8	8.9	13.7	16.7
49.6	3.8	29.0	34.2
41.5	3.4	19.4	30.2
43.0	1.8	32.3	43.5
47.1	6.1	23.4	28.8
46.6	6.9	18.1	27.2
43.2	11.1	18.7	27.2
12.4	13.3	8.1	9.3
7.8	1.3	3.3	5.1
12	9	30	30
4	4	4	4
4	3	3	3

<u>YIELD RANK</u>			
8	9*	2	2
7	6	4	3
9	2	9	9
1	5	3	4
5	7	6	5
4	8	1	1
2	4	5	6
3	3	8	7
6	1	7	7

*Not included in mean

UNIFORM TEST 00, 1980

Strain	Mean 7 Tests	Ont.		Wisc.	N.D.
		Ottawa	Elora	Ashland	Fargo
MATURITY (date)					
Clay (0)	+9	+5	+5	+6	+12
Maple Arrow	+9	+4	+3	+8	+12
Maple Presto	-7	-15	-5	+11	-6
McCall	+6	+4	+4	+4	+9
Portage (00)*	9/5	9/20	9/9	9/14	8/28
M71-148	+9	+5	+5	+5	+13
OT80-1	-1	-2	-5	-5	0
OT80-2	-6	-14	-3	-3	-3
OT80-3	-7	-13	-5	-7	-4
Date planted	5/16	5/21	5/21	5/21	5/20
*Days to maturity	112	122	111	116	100

Strain	7 Tests	LODGING (score)			
		Ottawa	Elora	Ashland	Fargo
Clay (0)	2.5	3.8	1.6	2.7	1.0
Maple Arrow	2.1	4.0	1.4	1.7	1.0
Maple Presto	1.3	2.0	1.0	1.3	1.0
McCall	2.1	4.2	1.4	2.3	1.0
Portage (00)	1.9	3.0	1.4	2.3	1.0
M71-148	2.3	3.0	1.6	2.3	2.0
OT80-1	2.1	4.2	1.3	2.0	1.0
OT80-2	1.5	3.2	1.0	1.0	1.0
OT80-3	2.5	5.0	1.9	3.0	1.0

UNIFORM TEST 00, 1980

Man.		Minn.	
Morden	Brandon*	Rosemount	Morris
YIELD (bu/a)			
+17		+11	+9
+15		+10	+14
-10		-10	-11
+12		+6	+1
9/11		8/24	8/22
+17		+8	+8
-2		+3	+3
-4		-7	-7
-6		-5	-9
5/8	6/5	5/15	5/8
126	-	101	106

LODGING (score)			
4.0	1.0*	3.0	1.7
3.5	1.0	2.3	1.0
1.0	1.0	2.0	1.0
2.8	1.0	2.3	1.0
2.2	1.0	2.3	1.0
4.0	1.0	2.3	1.0
2.8	1.0	2.3	1.0
1.2	1.0	2.0	1.0
2.5	1.0	3.3	1.0

*Not included in mean

UNIFORM TEST 00, 1980

Strain	Mean 7 Tests	Ont.		Wisc.
		Ottawa	Elora	Ashland
		<u>HEIGHT (inches)</u>		
Clay (0)	29	36	29	30
Maple Arrow	31	40	30	34
Maple Presto	23	29	24	28
McCall	30	40	33	32
Portage (00)	27	35	28	34
M71-148	30	39	32	31
OT80-1	28	34	29	34
OT80-2	24	33	26	24
OT80-3	27	34	28	31

Strain	Mean 6 Tests	QUALITY (score)		
		Ottawa	Elora	Ashland
Clay (0)	2.5	2.5	3.0	2.0
Maple Arrow	2.2	2.0	1.5	3.0
Maple Presto	3.4	4.0	3.5	2.7
McCall	2.3	3.0	3.0	2.0
Portage (00)	2.6	3.0	3.0	2.3
M71-148	2.3	2.0	3.0	2.0
OT80-1	2.2	1.5	3.0	1.0
OT80-2	1.9	2.0	2.5	1.3
OT80-3	2.7	3.0	4.0	1.7

UNIFORM TEST 00, 1980

Man.		N.D.	Minn.	
Morden	Brandon*	Fargo	Rosemount	Morris
HEIGHT (inches)				
26	20	26	29	25
28	22	28	29	26
22	21	23	20	17
27	23	27	30	21
24	21	27	23	20
26	24	28	31	25
24	23	26	25	23
23	22	23	18	18
24	22	25	24	22

QUALITY (score)				
	5.0*	1.5	3.0	3.0
	5.0	1.5	2.3	3.0
	2.0	4.0	3.0	3.0
	5.0	1.0	2.7	2.3
	3.0	1.0	3.0	3.0
	5.0	1.0	3.0	2.7
	2.0	2.0	3.0	2.7
	3.0	1.0	2.3	2.3
	2.0	1.5	2.7	3.0

*Not included in mean

UNIFORM TEST 00, 1980

Strain	Mean 7 Tests	Ont.		Wisc.
		Ottawa	Elora	Ashland
		SEED SIZE (g/100)		
Clay (0)	16.9	20.8	17.0	19.0
Maple Arrow	18.5	21.8	18.7	19.7
Maple Presto	15.7	18.8	17.8	16.2
McCall	15.6	20.0	15.7	16.2
Portage (00)	18.2	23.0	19.5	19.1
M71-148	14.7	18.5	14.5	15.6
OT80-1	16.5	19.8	16.8	18.3
OT80-2	14.7	17.0	15.8	15.2
OT80-3	18.0	22.2	19.6	18.5

Strain	Mean 5 Tests	Ont.		Wisc.	Man.	Minn.
		Ottawa	Elora	Ashland	Morden	Rosemount
% PROTEIN						
Clay (0)	41.8	41.4	41.6	42.7	41.4	42.1
Maple Arrow	40.3	41.8	39.9	40.1	38.8	41.0
Maple Presto	39.2	37.6	38.9	40.1	38.4	41.0
McCall	40.8	42.5	41.6	41.7	38.5	39.9
Portage (00)	39.7	39.9	39.8	42.0	37.8	39.2
M71-148	41.2	41.3	40.6	41.3	38.7	41.6
OT80-1	41.5	41.3	41.3	41.5	41.2	42.1
OT80-2	40.1	41.4	38.7	39.5	39.0	42.1
OT80-3	42.0	42.9	41.1	42.1	39.8	44.0

UNIFORM TEST 00, 1980

Man.		N.D.	Minn.	
Morden	Brandon*	Fargo	Rosemount	Morris
SEED SIZE (g/100)				
17.0	13.1	14.0	15.1	14.6
18.6	10.4	16.7	14.8	19.2
16.1	10.0	14.2	13.3	13.6
17.0	8.5	12.5	14.1	13.5
18.1	8.9	15.2	16.6	15.9
14.0	9.5	12.5	14.0	13.7
17.2	11.0	14.5	13.8	14.8
14.9	10.1	12.5	14.3	13.0
19.1	11.8	14.5	17.2	15.2

*Not included in mean

		<u>Ont.</u>		<u>Wisc.</u>	<u>Man.</u>	<u>Minn.</u>
Strain	Mean 5 Tests	Ottawa	Elora	Ashland	Morden	Rosemount
		<u>% OIL</u>				
Clay (0)	19.9	20.3	20.0	19.8	19.9	19.6
Maple Arrow	20.1	20.5	20.8	20.5	19.1	19.6
Maple Presto	20.7	21.0	21.1	20.4	21.2	19.6
McCall	19.0	19.1	18.9	17.8	20.0	19.4
Portage (00)	19.3	20.1	19.4	17.7	20.1	19.0
M71-148	20.1	18.8	20.6	20.8	20.1	20.3
OT80-1	20.6	20.3	20.8	20.1	21.2	20.4
OT80-2	20.6	21.6	21.5	20.9	19.3	19.8
OT80-3	19.6	20.8	19.5	18.4	20.5	19.0

UNIFORM TEST 0, 1980

Strain	Parentage	Previous Testing*	Generation Compositd
Clay	Capital x Renville	13	F ₅
Evans (0)	Merit x Harosoy	10	F ₅
Hodgson 78 (I)	Hodgson x Merit	3	F ₅
McCall (00)	(Acme x Chippewa) x Hark	-	F ₅
M70-153	Steele x Hodgson	2	F ₅
M71-25	Clay x Evans	UT 00	F ₅
M71-38	Wilkin x M62-263	UT 00	F ₅
M71-39	Wilkin x M62-275	UT 00	F ₅
M71-43	Wilkin x M63-217Y	1	F ₅
M71-52	Evans x M62-345	1	F ₅
M71-65	Steele x M63-194	1	F ₅
M72-24	Evans x Wells	-	F ₅
M72-37	Wilkin x M65-115	-	F ₅
M72-51	Evans x M64-185	-	F ₅
M72-52	Evans x M65-115	-	F ₅
M72-107	Wilkin x M63-194	-	F ₅
M73-37	Evans x XK505	-	F ₅
M73-62	M61-224 x Nag. Feher	-	F ₅
M73-93	M65-69 x M66-18	-	F ₅

* Number of years in test or name of 1979 test

Several strains in this test were higher in yield than the check variety Evans. M70-153 has been equal to Hodgson 78 in yield, but matures 5 days earlier than Hodgson 78. M71-25, evaluated in UT00 in 1978, was 2 bushels higher in yield, 1 day earlier in maturity, and had better lodging resistance than Evans. M71-43, also evaluated in UT00 in 1979, was 2 bushels higher in yield, matured 3 days earlier, and had much better lodging resistance than Evans. All three of the above strains are resistant to races 1 and 2 of phytophthora root rot and have excellent shattering resistance in the Kansas evaluations.

UNIFORM TEST 0, 1980

Descriptive and Other Data

Strain	Descriptive Code	Chlorosis Score		Hypocotyl Score	Shattering
		Ames	Lamberton	Ames	Manhattan 2 Weeks
Clay	PGBr SYy	3.3	2.5	1	2
Evans (0)	WGBr DYY	3.2	3.0	1	1
Hodgson 78 (I)	PGBr DYBf	2.5	3.0	5	1
McCall (00)	PGBr DYY	3.0	2.0	1	1
M70-153	PGBr DYBf	2.2	4.5	2	1
M71-25	WGBr DYY	2.7	4.0	1	1
M71-38	WGBr SYBf	2.3	2.5	1	2
M71-39	P+WGBr DYY	2.2	3.5	1	1
M71-43	PGBr DYY	2.7	2.5	5	1
M71-52	WGBr SYy	2.8	4.5	3	1
M71-65	PGBr DYY	2.7	4.5	3	1
M72-24	WGBr DYBf	2.5	2.0	1	1
M72-37	PGTn DYIb	2.0	1.5	1	1
M72-51	P+WGBr DYY	3.3	5.0	1	1
M72-52	PGBr DYGr	3.0	3.0	1	2
M72-107	PGBr DYY	2.5	4.0	1	1
M73-37	PGBr DYGr	2.3	3.5	1	1
M73-62	WGTn DYY	2.0	2.0	2	1
M73-93	PGBr DYBf	2.8	3.0	2	-

UNIFORM TEST 0, 1980

Disease Data

Strain	BSR Laf. Ind. n	GERM * Lafayette, d	SMV a	PSB d	PS a	PR Laf. Ind. a
	%	%	score	%	score	Reaction
Clay	100	64	3E	18	5S	S
Evans (0)	0	86	1	2	5S	R
Hodgson 78 (1)	20	87	2E	18	3M	R
McCall (00)	0	65	2M	20	5S	S
M70-153	20	81	1	8	3E	R
M71-25	20	82	1	5	4S	R
M71-38	0	78	1	8	5S	R
M71-39	40	61	1	25	5S	R
M71-43	0	88	1	8	5S	R
M71-52	20	85	1	6	4E	R
M71-65	0	68	2M	16	5S	R
M72-24	60	82	1	5	5E	R
M72-37	0	72	1	15	5S	S
M72-51	0	85	1	5	5E	R
M72-52	0	76	2E	8	5E	R
M72-107	0	90	1	2	5S	R
M73-37	0	88	3M	2	5S	R
M73-62	20	63	2E	8	4S	S
M73-93	0	99	3M	1	3E	R

* Petri dish germination on potato dextrose agar

UNIFORM TEST 0, 1980

Regional Summary

Strain	Yield	Rank	Matu- rity	Lodg- ing	Height	Seed Quality	Seed Size	Composition	
No. of Tests	8 bu/a	8 No.	7 Date	8 Score	8 In.	7 Score	7 g/100	5 %	5 %
Clay	37.4	16	-9	1.7	30	2.3	16.9	42.6	19.9
Evans (0)	41.7	11	9-19*	2.5	38	2.0	16.4	41.8	19.8
Hodgson 78 (I)	43.0	5	+8	2.9	40	2.3	17.6	40.7	19.1
McCall (00)	36.0	18	-11	2.0	31	2.0	14.9	40.0	19.4
M70-153	43.4	3	+4	2.3	35	2.0	16.8	42.7	19.3
M71-25	43.9	1	-1	2.1	33	2.3	18.0	41.1	20.3
M71-38	36.9	17	-6	1.8	32	2.1	16.2	40.8	20.0
M71-39	35.5	19	-7	1.8	31	2.1	16.6	41.0	19.1
M71-43	43.6	2	-3	1.5	33	2.1	16.7	42.2	19.3
M71-52	43.1	4	+6	2.8	34	2.2	20.6	41.9	19.1
M71-65	41.9	9	+1	2.5	36	2.4	19.8	41.6	19.7
M71-24	40.5	13	+3	1.6	35	2.2	18.5	42.4	19.2
M72-37	39.5	14	-4	1.9	30	2.2	17.1	41.7	19.9
M72-51	42.2	8	+3	2.7	36	2.3	18.0	42.2	18.9
M72-52	40.9	12	+6	3.0	37	2.6	17.6	41.5	20.5
M72-107	42.8	6	+2	1.9	36	2.3	18.1	41.1	19.6
M73-37	41.9	9	+3	2.4	36	2.3	17.0	42.0	19.1
M73-62	42.5	7	+2	2.0	30	2.0	17.1	42.4	19.3
M73-93	39.3	15	+9	2.2	32	2.6	16.7	42.9	18.2

* 124 days after planting

1979-1980 2-year mean

No. of Tests	15	15	14	15	14	13	14	9	9
Clay	37.4	7	-7.0	1.8	30	2.2	16.8	41.5	19.5
Evans (0)	41.0	6	9-21*	2.1	37	2.1	16.4	41.0	19.4
Hodgson 78 (I)	43.0	1	+8.2	2.5	39	2.2	17.3	40.5	19.0
M70-153	43.0	1	+3.4	2.5	34	2.0	16.4	41.6	19.3
M71-43	41.9	5	-2.2	1.5	33	2.0	17.2	41.7	19.1
M71-52	42.4	3	+5.8	2.5	34	2.2	20.6	41.4	18.9
M71-65	42.0	4	+0.6	2.4	35	2.3	19.9	41.1	19.4

* 122 days after planting

UNIFORM TEST 0, 1980

Strain	Mean 8 Tests	Ont.			Wisc.
		Ridge- town	Elora	Ottawa	Spooner
YIELD (bu/a)					
Clay	37.4	45.7	42.7	42.4	31.3
Evans (0)	41.7	43.9	44.9	47.0	33.6
Hodgson 78 (I)	43.0	56.5	41.4	37.7	33.4
McCall (00)	36.0	35.4	43.9	45.4	36.7
M70-153	43.4	46.0	46.7	42.2	34.0
M71-25	43.9	51.4	43.0	58.8	35.7
M71-38	36.9	31.9	41.2	48.9	31.7
M71-39	35.5	39.4	36.7	48.2	30.6
M71-43	43.6	45.7	48.3	57.4	37.8
M71-52	43.1	53.9	49.2	42.0	29.6
M71-65	41.9	49.5	47.7	43.0	34.9
M72-24	40.5	39.8	45.5	43.4	35.5
M72-37	39.5	49.7	40.4	49.2	28.7
M72-51	42.2	47.1	46.7	45.4	35.0
M72-52	40.9	45.0	45.7	41.4	34.5
M72-107	42.8	47.5	50.4	47.4	35.0
M73-37	41.9	46.1	49.4	42.4	32.3
M73-62	42.5	45.9	49.6	51.3	35.2
M73-93	39.3	46.8	37.6	32.3	27.7
C.V. (%)		9.0	8.5	12.5	10.3
L.S.D. (5%)		5.8	5.3	8.1	5.5
Row sp (in.)		24	7	10	36
Rows/plot		4	8	4	4
Reps.		4	4	4	3

UNIFORM TEST 0, 1980

<u>Mich.</u>	<u>Minn.</u>		<u>N.D.</u>
<u>Ithaca</u>	<u>Rosemount</u>	<u>Morris</u>	<u>Fargo</u>
<u>YIELD (bu/a)</u>			
42.7	32.6	37.8	24.2
52.3	38.3	48.9	24.4
58.5	37.4	49.0	29.9
42.4	28.3	36.0	20.2
52.8	38.6	58.3	28.8
50.1	35.7	49.5	26.7
41.5	31.3	46.3	22.4
42.0	29.9	39.6	17.3
47.2	35.1	51.1	26.1
49.3	42.9	54.3	23.3
49.8	37.6	50.3	22.4
50.1	38.8	49.0	22.2
45.8	33.4	45.0	23.9
53.9	36.6	51.0	21.6
51.0	36.7	48.6	23.9
50.3	35.9	50.2	25.4
51.6	37.9	53.5	21.6
47.7	37.9	47.7	24.3
55.2	34.7	52.7	27.1
8.3	7.3	8.2	22.3
6.8	4.4	6.5	8.9
28	30	30	18
4	4	4	4
3	3	3	3

UNIFORM TEST 0, 1980

Strain	Mean 7 Tests	Ridge- town	Ont.	
			Elora	Ottawa
MATURITY (date)				
Clay	-9	-12	-7	-10
Evans (0)*	9/19	9/21	9/19	10/1
Hodgson 78 (I)	+8	+3	+10	+8
McCall (00)	-11	-14	-9	-11
M70-153	+4	0	+4	+4
M71-25	-1	0	0	-1
M71-38	-6	-4	-4	-5
M71-39	-7	-5	-4	-10
M71-43	-3	-6	-3	-2
M71-52	+6	+10	+8	+5
M71-65	+1	0	+2	+2
M72-24	+3	-3	+3	+2
M72-37	-4	-4	-5	-4
M72-51	+3	+2	+6	+2
M72-52	+6	0	+10	+5
M72-107	+2	0	+3	+4
M73-37	+3	-1	+6	+4
M73-62	+2	+3	+4	0
M73-93	+9	+4	+8	+11
Date planted	5/18	5/22	5/21	5/21
*Days to maturity	124	122	121	133

UNIFORM TEST 0, 1980

<u>Wisc.</u>	<u>Mich.</u>	<u>Minn.</u>		<u>N.D.</u>
<u>Spooner</u>	<u>Ithaca</u>	<u>Rosemount</u>	<u>Morris</u>	<u>Fargo*</u>
<u>MATURITY (date)</u>				
-10	-7	-6	-9	-9
9/29	9/15	9/8	9/9	9/22
+7	+5	+6	+15	frost
-11	-8	-9	-13	-16
+6	+7	+1	+7	frost
-3	-2	-3	-1	0
-11	-5	-6	-4	-13
-10	-6	-7	-10	-13
-4	-2	-1	0	-4
+7	+6	+2	+3	frost
+3	+1	0	-1	-6
+2	+5	+1	+9	0
-2	-5	-5	-4	-3
+3	+6	+3	+1	frost
+6	+8	+2	+9	0
+2	+4	+2	0	-1
+3	+3	+1	+7	frost
+2	+5	0	-1	frost
+3	+8	+15	+13	frost
5/19	5/22	5/15	5/8	5/20
133	116	116	124	125

* Not included in the mean

UNIFORM TEST 0, 1980

Strain	Mean 8 Tests	<u>Ont.</u>			<u>Wisc.</u>	<u>Mich.</u>	<u>Minn.</u>		<u>N.D.</u>
		Ridge- town	Elora	Ottawa	Spooner	Ithaca	Rose- mount	Morris	Fargo
<u>LODGING (score)</u>									
Clay	1.7	1.2	2.4	3.2	1.0	1.0	2.7	1.3	1.0
Evans (0)	2.5	1.0	3.2	5.0	2.7	1.8	3.7	1.0	1.5
Hodgson 78 (I)	2.9	1.8	3.7	5.0	2.7	3.0	3.3	2.3	1.5
McCall (00)	2.0	1.0	2.5	4.8	2.0	1.2	2.3	1.0	1.0
M70-153	2.3	1.0	3.0	4.8	2.3	2.0	2.7	1.7	1.0
M71-25	2.1	1.0	3.0	4.8	2.3	1.0	2.0	1.3	1.0
M71-38	1.8	1.0	1.5	4.2	1.7	1.0	2.3	1.0	1.5
M71-39	1.8	1.0	1.8	4.5	1.7	1.0	2.3	1.0	1.0
M71-43	1.5	1.0	1.3	3.0	1.0	1.0	2.3	1.3	1.0
M71-52	2.8	1.0	3.5	5.0	3.7	1.8	3.7	2.0	2.0
M71-65	2.5	1.0	3.3	4.8	2.7	1.5	3.3	2.0	1.5
M72-24	1.6	1.0	1.8	3.0	1.8	1.2	2.0	1.0	1.0
M72-37	1.9	1.0	3.7	3.5	2.0	1.0	2.3	1.0	1.0
M72-51	2.7	1.2	3.8	5.0	2.7	2.2	4.0	1.7	1.0
M72-52	3.0	1.0	4.0	5.0	3.3	2.8	3.3	2.3	2.0
M72-107	1.9	1.0	2.0	4.5	1.7	1.3	2.7	1.0	1.0
M73-37	2.4	1.0	2.8	4.5	2.3	1.0	3.7	2.0	2.0
M73-62	2.0	1.0	3.5	3.8	2.0	1.0	2.7	1.0	1.0
M73-93	2.2	1.2	3.3	4.2	1.5	1.5	3.3	1.3	1.0

UNIFORM TEST 0, 1980

Strain	Mean 8 Tests	Ridge- town	Ont.		Wisc.	Mich.	Minn.	N.D.	
			Elora	Ottawa	Spooner	Ithaca	Rose- mount	Morris	Fargo
HEIGHT (inches)									
Clay	30	26	32	37	34	29	29	25	25
Evans (0)	38	32	40	55	47	38	36	36	28
Hodgson 78 (I)	40	36	43	56	44	40	37	35	29
McCall (00)	31	24	33	45	38	31	30	23	24
M70-153	35	30	35	49	39	38	34	29	26
M71-25	33	28	35	47	36	32	30	28	29
M71-38	32	25	33	47	36	29	30	27	28
M71-39	31	28	30	43	37	29	31	25	23
M71-43	33	29	35	46	39	33	33	28	24
M71-52	34	31	40	42	39	34	32	33	24
M71-65	36	30	38	50	43	34	35	31	30
M72-24	35	28	35	45	39	37	35	31	27
M72-37	30	30	31	38	33	31	30	25	21
M72-51	36	35	38	43	42	39	35	30	24
M72-52	37	30	42	47	44	38	36	34	28
M72-107	36	32	38	50	40	35	35	30	30
M73-37	36	33	38	46	42	36	35	32	27
M73-62	30	26	31	38	36	32	29	24	21
M73-93	32	28	34	42	36	34	32	29	23

UNIFORM TEST 0, 1980

Strain	Mean 7 Tests	Ridge- town	Ont.		Wisc.	Minn.		N.D.
			Elora	Ottawa	Spooner	Rose- mount	Morris	Fargo
QUALITY (score)								
Clay	2.3	3.0	3.0	2.0	2.3	2.3	2.3	1.5
Evans (0)	2.0	2.0	2.5	2.0	2.0	2.3	2.0	1.0
Hodgson 78 (I)	2.3	3.0	3.0	2.0	2.7	2.0	2.5	1.0
McCall (00)	2.0	2.0	2.5	2.0	2.3	2.0	2.5	1.0
M70-153	2.0	2.0	3.5	2.0	2.0	1.3	2.5	1.0
M71-25	2.3	3.0	3.0	2.0	2.3	2.3	2.3	1.0
M71-38	2.1	3.0	2.5	1.5	1.3	2.3	2.7	1.5
M71-39	2.1	2.0	2.0	2.0	2.3	1.7	3.0	1.5
M71-43	2.1	3.0	1.5	2.0	2.3	2.0	2.7	1.0
M71-52	2.2	2.0	3.0	2.5	2.3	2.0	2.0	1.5
M71-65	2.4	2.0	3.5	3.0	2.3	2.0	3.3	1.0
M72-24	2.2	3.0	3.0	2.0	2.0	1.7	2.3	1.5
M72-37	2.2	3.0	2.5	2.0	2.0	2.3	2.3	1.5
M72-51	2.3	2.0	3.5	2.0	3.0	2.7	2.0	1.0
M72-52	2.6	3.0	3.5	2.5	3.7	2.3	2.0	1.0
M72-107	2.3	3.0	2.5	2.0	2.7	2.3	2.7	1.0
M73-37	2.3	2.0	3.0	2.5	3.0	2.0	2.5	1.0
M73-62	2.0	2.0	2.0	2.0	2.0	2.3	2.5	1.0
M73-93	2.6	2.0	3.5	3.0	3.7	3.0	2.0	1.0

UNIFORM TEST 0, 1980

Mean 7 Tests	Ont.		Wisc.		Minn.		N.D.
	Ridge- town	Elora	Ottawa	Spooner	Rose- mount	Morris	Fargo
SIZE (g/100)							
16.9	17.7	16.8	19.2	19.6	14.8	15.0	15.5
16.4	16.5	15.8	19.5	19.7	13.9	17.5	11.7
17.6	18.3	17.2	19.5	19.9	14.6	16.8	16.6
14.9	15.2	13.7	17.0	18.6	13.3	14.0	12.2
16.8	16.9	15.7	20.2	18.8	14.3	17.8	13.8
18.0	18.3	16.3	20.8	21.7	14.3	18.0	16.3
16.2	16.9	16.0	19.2	18.9	13.7	15.9	12.5
16.6	17.8	16.0	19.0	19.8	14.5	16.1	12.8
16.7	17.8	16.4	19.5	19.6	14.4	17.4	12.0
20.6	21.3	23.0	24.2	23.3	18.3	20.7	13.4
19.8	21.0	19.7	21.8	23.0	16.5	19.4	17.2
18.5	18.7	16.7	21.8	20.9	16.3	17.9	17.4
17.1	18.1	14.8	20.5	19.9	14.6	17.4	14.1
18.0	17.3	19.1	21.2	21.1	15.2	17.4	14.4
17.6	18.3	19.0	21.2	19.6	15.6	18.1	11.7
18.1	18.3	18.5	21.2	21.3	15.3	17.1	14.9
17.0	16.1	15.7	20.8	19.6	15.5	18.2	13.2
17.1	16.9	17.2	20.8	19.3	13.5	17.8	13.9
16.7	16.9	15.4	20.8	16.7	14.8	17.0	15.3

UNIFORM TEST 0, 1980

Strain	Mean 5 Tests	Ont.		Wisc.	Minn.	
		Elora	Ottawa	Spooner	Rose- mount	Morris
% PROTEIN						
Clay	42.6	42.1	42.3	42.7	43.4	42.6
Evans (0)	41.8	41.2	42.0	43.6	40.7	41.6
Hodgson 78 (1)	40.7	42.0	39.6	42.3	39.0	40.6
McCall (00)	40.0	40.5	40.5	39.9	39.2	40.1
M70-153	42.7	42.2	41.6	45.1	42.7	42.0
M71-25	41.1	40.2	40.2	43.3	42.1	39.9
M71-38	40.8	41.8	40.0	42.0	40.2	40.1
M71-39	41.0	41.0	41.3	41.6	40.5	40.5
M71-43	42.2	41.5	42.0	43.7	41.5	42.2
M71-52	41.9	42.7	42.1	41.6	42.6	40.6
M71-65	41.6	42.0	41.8	42.5	40.8	40.7
M72-24	42.4	42.1	43.2	42.2	42.2	42.2
M72-37	41.7	41.8	42.8	42.8	42.7	38.5
M72-51	42.2	42.0	43.6	42.5	42.9	39.8
M72-52	41.5	41.4	40.8	42.0	42.4	40.7
M72- 107	41.1	41.3	41.7	43.4	40.0	39.1
M73-37	42.0	41.2	42.9	42.2	41.6	42.2
M73-62	42.4	41.8	42.6	43.6	42.5	41.6
M73-93	42.9	43.2	44.7	42.2	43.2	41.0

UNIFORM TEST 0, 1980

Mean 5 Tests	Ont.		Wisc.	Minn.	
	Elora	Ottawa	Spooner	Rose- mount	Morris
% OIL					
19.9	20.5	20.0	19.8	19.5	19.7
19.8	20.1	20.1	18.8	19.7	20.1
19.1	19.5	18.4	18.6	19.7	19.5
19.4	19.9	19.8	19.0	19.5	18.6
19.3	19.5	19.2	19.8	18.7	19.4
20.3	20.2	21.0	20.8	19.3	20.4
20.0	19.8	21.0	20.1	19.6	19.5
19.1	19.6	19.4	17.8	19.4	19.3
19.3	20.3	18.8	19.7	18.6	19.0
19.1	19.5	18.9	19.1	19.3	18.6
19.7	20.4	20.3	19.6	18.5	19.6
19.2	18.2	19.6	19.4	19.5	19.5
19.9	20.3	19.9	19.7	19.7	19.8
18.9	19.4	19.3	18.0	18.8	19.1
20.5	21.1	20.6	19.9	20.0	20.7
19.6	20.0	19.4	19.5	20.1	19.1
19.1	20.0	19.2	18.7	18.6	19.2
19.3	18.8	19.8	19.4	18.9	19.7
18.2	18.0	18.6	17.7	18.2	18.7

UNIFORM TEST I, 1980

Strain	Parentage	Previous Generation Testing*	Composited
Corsoy 79 (II)	Corsoy ⁶ x Lee 68	UTII	F ₃
Evans (0)	Merit x Harosoy	3	F ₅
Hardin			
(A76-102009)	Corsoy ³ x Cutler 71	2	F ₃
Hodgson 78 (I)	Hodgson ⁷ x Merit	6	F ₅
A77-112023	AP6M(SI)C1	1	F ₄
A78-121014	Pride B-216 x Hodgson	PI	F ₄
A78-123018	Pride B-216 x Hodgson	PI	F ₄
M71-80	Merit x M62-263	PI	F ₅
M72-3	Evans x Hodgson	UTO	F ₅
M75-2	Hodgson ⁴ x [M67-141 x (Chippewa x Higan)]	PI	F ₅

* Number of years in this test or name of 1979 test

Descriptive and Other Data

Strain	Descriptive Code	Chlorosis Score		Hypocotyl Score	Shattering
		Ames	Lamberton	Ames	Manhattan 2 Weeks
Corsoy 79 (II)	PGBr DYY	4.0	4.0	1	3
Evans (0)	WGBr DYY	3.2	3.0	1	1
Hardin	PGBr DYY	3.3	4.5	1	1
Hodgson 78 (I)	PGBr DYBf	2.5	3.0	5	3
A77-112023	PTTn DYB1	2.0	1.2	2	1
A78-121014	PGBr DYBf	4.0	5.0	2	1
A78-123018	PGBr DYBf	3.7	4.5	1	1
M71-80	WG+TBr DYB1	2.2	1.2	1	1
M72-3	WGBr DYBf	2.5	1.2	2	1
M75-2	PGBr DYBf	2.0	1.5	5	4

UNIFORM TEST I, 1980

Disease Data

Strain	Laf.	Ames	GERM	SMV	PSB	PS	PR	Race 1
	Ind.	Ia.	* Lafayette, IN	a	d	a	Laf. Ind.	Ames Ia.
	n	n	d	a	d	a	a	a
	%	Reac.	%	Score	%	Score	---Reaction---	
Corsoy 79 (II)	0	S	89	5E	5	4E	R	R
Evans (0)	0	S	81	1	7	3S	R	R
Hardin	0	I	88	4E	8	3E	R	R
Hodgson 78 (I)	20	S	71	2M	25	3E	R	R
A77-112023	20	S	88	4M	10	3E	R	R
A78-121014	60	S	82	1	17	2E	S	S
A78-123018	40	S	95	1	4	2M	S	S
M71-80	0	S	92	1	5	2M	R	R
M72-3	0	S	94	1	2	2E	H	H
M75-2	0	S	96	1	3	2E	R	R

* Petri dish germination on potato dextrose agar

UNIFORM TEST 1, 1980

Regional Summary

Strain	Yield	Rank	Matu- rity	Lodg- ing	Height	Seed Quality	Size	Protein	Oil
No. of Tests	13	13	11	12	13	10	10	5	5
	bu/a	No.	Date	Score	In.	Score	g/100	%	%
Corsoy 79 (11)	43.2	6	+8	2.9	41	2.2	15.2	42.6	19.7
Evans (0)	37.8	10	-8	1.9	31	2.2	15.6	41.2	21.4
Hardin	44.8	4	+5	2.5	38	2.1	15.1	41.0	20.0
Hodgson 78 (I)	42.0	8	9-17*	2.1	34	2.0	16.3	41.7	20.8
A77-112023 <i>Lakota</i>	47.7	3	+2	2.9	42	2.3	16.8	44.1	19.7
A78-121014	48.7	1	+2	2.2	34	2.1	18.3	43.4	20.1
A78-123018	48.5	2	+4	2.3	33	2.0	15.6	42.0	20.0
M71-80	42.1	7	0	1.9	32	1.8	16.7	42.1	20.9
M72-3	43.6	5	-2	2.1	32	1.9	16.3	40.6	20.8
M75-2	41.4	9	+1	2.2	33	2.1	16.9	42.2	20.8

* 123 days after planting

1979-1980, 2-year mean

No. of Tests	27	27	23	25	26	21	22	10	10
Evans (0)	38.9	4	-6.4	1.9	32	2.0	16.0	40.7	20.8
Hardin	46.3	2	+4.1	2.5	38	1.9	15.7	40.5	19.8
Hodgson 78 (I)	42.9	3	9-21*	2.2	36	2.0	16.9	41.1	20.5
A77-112023 <i>Lakota</i>	47.1	1	+1.4	3.0	42	2.4	16.4	43.3	19.1

* 125 days after planting

Yields of the three Iowa strains exceeded the yield of Hardin by 3 to 4 bushels per acre in 1980. A77-112023 averaged 1 bushel per acre higher in yield than Hardin in the two year summary, averaged 3 days earlier in maturity, but was more prone to lodging than Hardin. The strains A78-121014 and A78-123018 have slightly better lodging resistance than Hardin but both strains are susceptible to phytophthora root rot.

UNIFORM TEST I, 1980

		Ont.	S.D.	Iowa		Wisc.
	Mean	Ridge-				
Strain	13 Tests	town	Brookings	Corwith	Knierim	Arlington
YIELD (bu/a)						
Corsoy 79 (II)	43.2	53.8	25.1	50.7	55.2	22.4
Evans (0)	37.8	44.0	18.3	47.9	39.3	31.8
Hardin	44.8	52.6	19.9	49.3	53.6	31.6
Hodgson 78 (I)	42.0	50.4	18.4	51.1	46.6	31.6
A77-112023	47.7	47.8	36.9	54.1	58.2	33.9
A78-121014	48.7	51.2	35.0	54.1	55.2	36.7
A78-123018	48.5	53.0	39.3	54.6	56.8	38.2
M71-80	42.1	43.1	26.6	45.8	46.7	36.2
M72-3	43.6	47.3	31.6	51.2	46.4	34.7
M75-2	41.4	50.4	16.4	51.0	45.6	32.6
C.V. (%)		5.6	12.4	4.3	5.1	7.9
L.S.D. (5%)		3.9	5.5	3.2	3.7	4.4
Row sp. (in.)		24	30	27	27	30
Rows/plot		4	4	4	4	4
Reps.		4	3	4	4	3

	<u>YIELD RANK</u>					
	Mean 13 Tests					
Corsoy 79 (II)	6	1	6	7	3	10
Evans (0)	10	9	9	9	10	7
Hardin	4	3	7	8	5	8
Hodgson 78 (I)	8	5	8	5	7	9
A77-112023	3	7	2	2	1	5
A78-121014	1	4	3	2	3	2
A78-123018	2	2	1	1	2	1
M71-80	7	10	5	10	6	3
M72-3	5	8	4	4	8	4
M75-2	9	5	10	6	9	6

UNIFORM TEST I, 1980

Mich.		Minn.		Ill.	N.D.	Ind.	Neb.
Dundee	Ithaca	Waseca	Lamber- ton	Dekalb	Oakes	Lafayette	Mead
YIELD (bu/a)							
42.3	48.7	47.1	45.2	37.2	34.1	46.8	52.4
43.8	44.9	37.3	38.6	33.3	39.9	27.2	45.7
48.4	48.5	50.5	44.9	42.6	41.6	41.1	57.4
48.9	53.5	46.2	41.6	39.0	25.9	40.7	52.6
53.9	54.8	46.8	43.1	45.7	40.3	50.1	54.6
49.3	52.1	51.2	49.6	40.0	43.2	54.9	60.5
50.6	52.4	51.1	47.2	43.4	31.8	49.5	63.0
44.1	45.2	43.0	44.8	36.8	43.9	39.7	51.6
49.4	51.5	46.2	44.4	36.4	34.2	39.2	54.4
45.5	41.4	48.9	42.1	35.9	36.7	42.5	49.7*
8.1	13.4	7.4	8.7	9.4	25.7	7.2	5.8
6.6	N.S.	5.9	6.6	6.3	16.4	5.1	5.4
30	23	30	30	30	12	24	30
4	4	4	4	4	4	4	4
3	3	3	3	3	3	3	3

YIELD RANK

10	6	5	3	6	8	4	7
9	9	10	10	10	5	10	10
6	7	3	4	3	3	6	3
5	2	7	9	5	10	7	6
1	1	6	7	1	4	2	4
4	4	1	1	4	2	1	2
2	3	2	2	2	9	3	1
8	8	9	5	7	1	8	8
3	5	7	6	8	7	9	5
7	10	4	8	9	6	5	9*

*Herbicide injury

UNIFORM TEST I, 1980

Strain	Mean 11 Tests	Ont.	S.D.	Iowa		Wisc.
		Ridge- town	Brookings	Corwith	Knierim	Arling- ton
MATURITY (date)						
Corsoy 79 (II)	+8	+6	+10		+9	+16
Evans (0)	-8	-6	-13		-8	-15
Hardin (I)	+5	+2	+5		+6	+10
Hodgson 78 (II)*	9/17	9/25	9/25		9/8	9/20
A77-112023	+2	-3	+7		+4	+5
A78-121014	+2	0	+6		+2	+5
A78-123018	+4	+2	+9		+4	+6
M71-80	0	0	+1		+1	0
M72-3	-2	-2	-1		-2	0
M75-2	+1	-1	0		-1	+3
Date planted	5/16	5/22	5/20	5/15	5/14	5/20
*Days to matur- ity	123	126	128	-	117	123

<u>LODGING (score)</u>						
	<u>12 Tests</u>					
Corsoy 79 (II)	2.9	2.0		4.5	3.2	2.7
Evans (0)	1.9	1.0		4.2	1.6	1.2
Hardin	2.5	1.2		4.7	2.9	2.3
Hodgson 78 (I)	2.1	1.0		4.3	1.9	1.5
A77-112023	2.9	1.8		5.0	3.2	2.7
A78-121014	2.2	1.0		4.2	2.3	1.7
A78-123018	2.3	1.8		3.9	2.4	1.8
M71-80	1.9	1.0		4.0	1.6	1.5
M72-3	2.1	1.2		4.6	1.4	1.8
M75-2	2.2	1.2		4.4	1.7	1.7

UNIFORM TEST I, 1980

Mich.		Minn.		Ill.	N.D.	Ind.	Neb.
Dundee	Ithaca	Waseca	Lamberton	Dekalb	Oakes	Lafayette	Mead
MATURITY (date)							
+11	+6	+6	+2	+3		+10	+7
-5	-5	-10	-8	-7		-6	-8
+8	+3	+4	+2	0		+6	+8
9/14	9/23	9/22	9/18	9/16		8/31	9/14
+5	+2	0	-2	0		+4	+5
+1	+1	0	0	0		+7	+4
+5	+2	+2	+2	0		+5	+4
+2	-2	-1	-2	0		+1	+4
-1	+1	0	-1	-2		-3	-6
+1	+3	-1	0	0		+1	+5*
5/14	5/22	5/20	5/14	5/2	5/6	5/7	5/27
123	124	125	127	137	-	116	110
LODGING (score)							
3.7	2.7	2.7	2.3	3.5	2.0	2.0	3.0
1.8	1.6	1.3	1.0	4.5	2.0	1.0	1.8
3.0	2.6	2.3	2.0	3.5	1.5	1.5	2.7
1.9	2.3	2.0	1.3	3.3	1.5	1.5	2.7
4.0	3.0	2.0	2.0	3.8	3.0	1.8	3.0
2.2	2.1	2.0	1.7	3.7	2.5	1.3	2.0
2.5	2.2	2.0	1.3	3.7	2.5	1.5	2.2
2.3	1.7	1.7	1.0	3.3	2.0	1.0	1.5
2.6	2.1	1.7	1.0	3.8	1.5	1.5	2.2
2.2	2.3	2.0	1.3	3.5	2.0	1.5	2.5*

* Herbicide injury

UNIFORM TEST I, 1980

Strain	Mean 13 Tests	Ont.	S.D.	Iowa		Wisc.
		Ridge- town	Brook- ings	Corwith	Knierim	Arling- ton
<u>HEIGHT (inches)</u>						
Corsoy 79 (II)	41	42	27	48	44	35
Evans (0)	31	32	29	36	24	28
Hardin	38	37	21	46	41	34
Hodgson 78 (I)	34	36	26	40	30	30
A77-112023	42	41	32	56	43	34
A78-121014	34	32	29	38	33	32
A78-123018	33	33	31	38	34	30
M71-80	32	32	28	38	32	27
M72-3	32	36	29	40	27	29
M75-2	33	34	23	40	30	29

<u>QUALITY (score)</u>						
	<u>10 Tests</u>					
Corsoy 79 (II)	2.2	3.0	2.0	1.4		4.0
Evans (0)	2.2	2.0	3.0	2.2		3.7
Hardin	2.1	2.0	2.0	1.3		3.7
Hodgson 78 (I)	2.0	2.0	2.0	1.3		4.3
A77-112023	2.3	3.0	4.0	1.4		3.7
A78-121014	2.1	2.0	3.0	1.5		3.7
A78-123018	2.0	2.0	2.0	1.4		3.0
M71-80	1.8	2.0	2.0	1.8		2.0
M72-3	1.9	2.0	2.0	1.8		2.7
M75-2	2.1	3.0	2.0	1.6		4.3

UNIFORM TEST I, 1980

Mich.		Minn.		Ill.	N.D.	Ind.	Neb.
Dundee	Ithaca	Waseca	Lam- berton	Dekalb	Oakes	Lafayette	Mead
HEIGHT (inches)							
44	40	36	36	43	53	37	42
31	36	29	28	31	43	24	28
41	42	33	36	40	52	35	37
35	38	30	34	35	48	28	30
47	41	35	39	44	48	39	42
35	37	29	33	34	49	29	29
32	37	30	32	30	47	29	30
36	36	27	30	35	39	27	31
35	37	27	31	32	37	26	30
36	38	30	33	34	42	28	29*

QUALITY (score)							
		2.0	2.3	1.3	1.0	1.5	3.3
		2.7	2.0	1.3	1.0	2.5	2.0
		2.3	2.7	1.3	1.0	2.0	3.0
		1.7	2.3	1.3	1.0	1.5	2.2
		2.0	1.7	1.2	2.0	1.5	2.3
		1.7	1.7	2.0	1.0	1.5	2.5
		2.3	2.3	2.0	1.0	1.5	2.8
		1.7	2.0	1.2	1.5	1.5	2.5
		2.3	2.3	1.4	1.0	1.5	2.3
		2.0	2.0	1.2	1.0	1.5	2.5*

* Herbicide injury

UNIFORM TEST I, 1980

Strain	Mean 10 Tests	Ont.	S.D.	Iowa		Wisc.
		Ridge- town	Brook- ings	Corwith	Knierim	Arling- ton
SIZE (g/100)						
Corsoy 79 (II)	15.2	16.0	15.4	14.2		15.7
Evans (0)	15.6	16.7	15.7	14.6		15.5
Hardin	15.1	15.3	14.9	12.6		15.3
Hodgson 78 (I)	16.3	17.3	16.9	15.2		16.7
A77-112023	16.8	15.9	15.8	15.6		17.6
A78-121014	18.3	17.5	17.7	16.8		19.6
A78-123018	15.6	15.6	16.2	14.8		16.3
M71-80	16.7	16.4	17.8	15.9		18.6
M72-3	16.3	15.7	16.2	15.4		17.0
M75-2	16.9	18.4	16.6	15.8		17.6

		<u>Ont.</u>	<u>S.D.</u>	<u>Iowa</u>	<u>Wisc.</u>	<u>Minn.</u>
<u>Strain</u>	<u>Mean</u>	<u>Ridge-</u>	<u>Brook-</u>			
	<u>5 Tests</u>	<u>town</u>	<u>ings</u>	<u>Corwith</u>	<u>Arlington</u>	<u>Waseca</u>

UNIFORM TEST I, 1980

<u>Minn.</u>		<u>Ill.</u>	<u>N.D.</u>	<u>Ind.</u>	<u>Neb.</u>
Waseca	Lamberton	Dekalb	Oakes	Lafayette	Mead
<u>SIZE (g/100)</u>					
16.2	16.7	13.3	14.9	14.5	14.8
17.8	16.6	13.5	14.9	15.2	15.4
17.1	17.3	13.6	16.1	14.0	15.2
17.7	17.8	13.9	16.8	15.3	15.2
18.4	18.0	16.0	16.8	16.4	17.4
19.6	19.4	16.9	19.6	17.6	18.2
15.6	16.9	14.4	15.6	15.0	15.3
18.8	18.2	15.2	16.9	13.6	16.0
19.1	17.8	13.8	16.5	15.2	16.7
18.3	18.8	14.1	17.6	15.7	15.8*

* Herbicide injury

	<u>Ont.</u>	<u>S.D.</u>	<u>Iowa</u>	<u>Wisc.</u>	<u>Minn.</u>
Mean				Arling-	
5 Tests	Ridgetown	Brookings	Corwith	ton	Waseca
<u>% OIL</u>					
19.7	20.0	20.1	20.7	19.3	18.6
21.4	20.1	22.2	22.5	21.9	20.4
20.0	20.2	21.7	20.2	19.6	18.2
20.8	20.1	21.8	22.1	21.0	19.1
19.7	18.9	20.9	20.6	19.5	18.7
20.1	19.7	21.0	21.0	20.1	18.6
20.0	19.9	20.2	21.6	19.5	18.8
20.9	20.0	21.5	21.9	21.2	19.7
20.8	20.8	22.4	20.6	21.3	18.8
20.8	20.4	21.5	21.8	21.2	18.9

PRELIMINARY TEST I, 1980

Strain	Parentage	Generation Composited
Corsoy 79 (II)	Corsoy ⁶ x Lee 68	F ₃
Evans (0)	Merit x Harosoy	F ₅
Hardin	Corsoy ³ x Cutler 71	F ₃
Hodgson 78 (I)	Hodgson ⁷ x Merit	F ₅
A79-131010	L69U40-19-1 x [AP68-1016 x (C1426 x AP68-1016)]	F ₄
A79-133019	AP6 (ZYT)(F ₄)C1	F ₄
A79-134008	AP6 (1YT)(F ₄)C2	F ₄
A79-134018	Pellax A73-19084	F ₄
A79-134026	Pride B216 x LL4102	F ₄
A79-134034	C1520 x L69U40-19-1	F ₄
A79-135010	Pride B216 x Cumberland	F ₄
A79-135012	Pride B216 x LL4102	F ₄
A79-136010	Oakland x L69U40-19-1	F ₄
A79-136012	Pride B216 x LL4102	F ₄
A79-136030	AP6 (1YT)(F ₄)C2	F ₄
A79-138014	NKS1492 x Asgrow A3300	F ₄
A79-138015	NKS1492 x Asgrow A3300	F ₄
A79-138024	A74-102011 x C1523	F ₄
A79-287041	AP6 (1YT)(F ₄)C2	F ₄
M70-620	Hill x Steele	F ₅
M72-38	Wilkin x M65-115	F ₅
M72-79	M62-263 x Wells	F ₅
M72-95	Evans x Wells	F ₅
M72-124	ML7293-4 x Wells	F ₅
M73-32	Evans x XK505	F ₅
M73-80	M64-157 x M63-217Y	F ₅
M73-92	M65-69 x M66-18	F ₅
W417	[Merit x (Hawkeye x Wis. Manchu 3)] x Wells	F ₄
W442	Evans x Steele	F ₄

PRELIMINARY TEST I, 1980

Descriptive and Other Data

Strain	Descriptive Code	Chlorosis Score Ames	Shattering Manhattan 2 Weeks
Corsoy 79 (II)	PGBr DYY	4.0	2
Evans (0)	WGBr DYY	3.2	1
Hardin	PGBr DYY	3.3	2
Hodgson 78 (I)	PGBr DYBf	2.5	3
A79-131010	PGBr DYGr	3.2	1
A79-133019	PTBr DYB1	3.7	1
A79-134008	WG+TBr DYBf+B1	1.8	2
A79-134018	PGBr DYIb	3.2	2
A79-134026	PGBr DYIb	3.3	3
A79-134034	WGBr DYY	2.5	4
A79-135010	WTT DYB1	4.0	3
A79-135012	WTBr SYGr	3.7	2
A79-136010	PTTn DYGr	3.7	1
A79-136012	WTBr DYBr	4.5	2
A79-136030	PGBr DYIb	3.2	1
A79-138014	PTBr SYBr	4.7	2
A79-138015	PTBr DYB1	4.2	1
A79-138024	PGBr DYGr	3.0	1
A79-287041	WTBr SYB1	2.7	4
M70-620	WGTn DYBf	2.7	4
M72-38	PGTn DYGr+Y	1.7	4
M72-79	WGBr SYBf	2.7	1
M72-95	WGBr DYY	2.8	3
M72-124	PGBr SYIb	2.8	2
M73-32	P+WGBr DYGr+Y	3.0	2
M73-80	PGBr DYGr+Ib+Y	2.5	3
M73-92	PGBr DYY	2.7	2
W417	PGBr DYIb	2.0	2
W442	WGBr DYY	2.5	1

PRELIMINARY TEST I, 1980

Disease Data

Strain	BSR		GERM	SMV	PSB	PS	PR	Race 1
	Laf.	Ames	*				Laf.	Ames
	Ind. n	Ia. n	Lafayette, IN d	a	d	a	Ind. a	Ia. a
	%	Reac.	%	Score	%	Score	--Reaction--	
Corsoy 79 (II)	0	S	89	5E	6	3E	R	R
Evans (0)	0	S	75	1	9	5S	R	R
Hardin	0	S	81	5M	7	3E	R	R
Hodgson 78 (I)	20	S	85	2E	11	3E	R	R
A79-131010	20	I	93	4M	6	2E	H	H
A79-133019	0	S	93	5E	7	4S	S	S
A79-134008	20	S	97	2M	8	4E	S	S
A79-134018	0	S	83	3M	16	3E	S	S
A79-134026	40	S	85	4E	8	4S	S	S
A79-134034	80	S	83	2M	12	4E	S	S
A79-135010	40	S	85	5E	10	3E	S	S
A79-135012	80	S	85	5E	10	3E	S	S
A79-136010	0	S	87	5E	2	3E	H	S
A79-136012	100	S	77	5E	15	4E	H	S
A79-136030	80	S	92	3M	7	4E	R	R
A79-138014	20	S	84	5E	9	3E	S	S
A79-138015	100	S	87	4E	9	4E	S	S
A79-138024	20	S	92	4E	6	4S	H	S
A79-287014	100	S	85	3M	12	3S	S	S
M70-620	60	S	73	1	24	4E	R	R
M72-38	20	S	68	2M	31	3S	R	R
M72-79	20	S	72	3M	20	3E	R	R
M72-95	20	S	88	3M	10	4S	R	R
M72-124	0	S	89	2M	8	4E	H	S
M73-32	0	S	79	3M	15	2E	R	R
M73-80	0	S	70	2M	25	2E	H	S
M73-92	0	S	87	4M	12	5E	R	R
W417	60	S	77	1	18	4S	R	R
W442	0	S	82	1	5	3E	R	R

* Petri dish germination on potato dextrose agar

PRELIMINARY TEST I, 1980

Regional Summary

Strain	No. of Tests	Yield	Rank	Matu- rity	Lodg- ing	Height	Seed Quality	Seed Size	Composition	
		9	9	8	8	9	6	6	5	
		bu/a	No.	Date	Score	In.	Score	g/100	%	%
Corsoy 79 (II)		45.3	20	+8	3.0	40	2.1	15.7	42.5	20.1
Evans (O)		39.1	29	-8	1.8	31	2.4	16.3	41.8	20.8
Hardin		46.1	17	+3	2.7	36	2.1	16.1	41.1	20.3
Hodgson 78 (I)		42.0	28	9-20*	2.2	33	2.3	17.3	42.1	20.9
A79-131010		46.3	15	+5	2.0	37	2.5	17.5	41.1	20.2
A79-133019		50.0	2	+7	2.7	34	2.3	18.2	41.7	20.0
A79-134008		46.8	11	0	2.2	38	2.7	17.3	41.7	20.6
A79-134018		45.4	19	+6	2.5	39	2.1	16.0	43.0	20.1
A79-134026		47.4	8	+3	2.4	33	2.4	17.6	42.8	19.4
A79-134034		46.9	10	+4	2.7	38	2.0	19.3	42.2	20.5
A79-135010		49.5	5	+4	2.3	36	2.2	18.2	42.5	20.1
A79-135012		46.7	12	+4	1.9	34	2.8	18.6	44.6	19.7
A79-136010		48.6	6	+5	2.1	33	2.0	21.2	42.5	20.2
A79-136012		50.3	1	+3	2.2	36	2.2	18.1	43.7	19.6
A79-136030		45.7	18	+5	2.0	37	2.1	17.2	41.8	20.0
A79-138014		49.8	4	+6	2.2	34	1.7	20.0	43.8	19.8
A79-138015		48.5	7	+7	2.2	36	1.6	20.5	41.7	19.9
A79-138024		49.9	3	+8	2.3	36	2.4	20.1	42.0	19.6
A79-287041		46.7	12	+1	2.0	28	2.5	18.2	41.1	19.9
M70-620		44.6	23	+2	2.4	34	2.5	17.2	41.1	21.1
M72-38		42.3	27	-1	1.9	31	2.5	18.4	42.3	20.3
M72-79		47.1	9	+2	2.0	34	3.1	18.3	41.8	20.8
M72-95		46.4	14	+4	1.9	37	2.4	16.9	40.8	21.0
M72-124		44.0	24	+1	1.6	35	2.1	16.3	44.6	19.8
M73-32		45.1	22	+2	2.0	34	2.0	18.5	42.3	20.9
M73-80		46.3	15	-3	1.7	31	2.6	18.8	41.7	20.7
M73-92		44.0	24	+5	2.3	35	2.6	18.8	42.6	19.8
W417		45.2	21	+1	1.6	33	2.0	17.7	41.1	20.4
W442		43.3	26	+2	2.2	36	2.1	18.4	41.4	20.2

* 125 days after planting

Several strains in this test averaged higher than the check varieties Hardin and Hodgson 78. The strains A79-135010, A79-136010, and A79-136012 all averaged at least 2 bushels higher in yield than Hardin but all were susceptible or heterogeneous in reaction to phytophthora root rot. The strains A79-133019, A79-138014, A79-138015, and A79-138024 all yielded well but averaged 1 to 3 days late in maturity for continued testing in the Group I tests.

PRELIMINARY TEST I, 1980

Strain	Mean 9 Tests	Ont.	S.D.	Iowa	
		Ridge- town	Brook- ings	Corwith	Knierim
YIELD (bu/a)					
Corsoy 79 (II)	45.3	59.3	37.8	51.3	55.1
Evans (O)	39.1	47.9	20.5	49.7	36.0
Hardin	46.1	54.5	20.0	49.8	53.6
Hodgson 78 (I)	42.0	53.4	22.8	48.4	44.0
A79-131010	46.3	50.8	29.4	49.3	49.8
A79-133019	50.0	57.1	32.0	55.6	54.4
A79-134008	46.8	49.6	27.6	53.7	51.8
A79-134018	45.4	47.8	31.2	52.7	51.9
A79-134026	47.4	54.4	26.3	51.6	55.2
A79-134034	46.9	52.6	34.5	48.9	51.5
A79-135010	49.5	47.3	37.0	53.2	58.4
A79-135012	46.7	53.0	33.4	50.7	52.7
A79-136010	48.6	51.2	39.1	47.5	57.1
A79-136012	50.3	53.4	29.7	56.4	55.2
A79-136030	45.7	48.4	37.8	42.8	54.1
A79-138014	49.8	51.1	46.0	49.7	53.1
A79-138015	48.5	49.3	43.7	50.5	56.3
A79-138024	49.9	57.3	34.2	59.3	56.0
A79-287041	46.7	53.8	33.7	54.4	50.7
M70-620	44.6	53.3	19.6	49.5	48.8
M72-38	42.3	49.9	23.2	53.1	46.5
M72-79	47.1	51.7	30.3	48.0	53.6
M72-95	46.4	47.4	27.7	49.6	51.5
M72-124	44.0	50.8	26.0	47.5	44.7
M73-32	45.1	53.3	26.1	45.2	52.2
M73-80	46.3	53.7	22.2	54.4	45.9
M73-92	44.0	48.0	29.5	51.3	49.0
W417	45.2	53.4	33.6	50.1	46.3
W442	43.3	51.3	14.1	45.5	51.9
C.V. (%)		4.0	11.6	5.4	3.9
L.S.D. (5%)		4.3	7.0	5.6	4.1
Row sp. (in.)		24	30	27	27
Rows/plot		4	4	4	4
Reps		2	2	2	2

PRELIMINARY TEST I, 1980

Wisc.	Mich.		Minn.	
Arlington	Dundee	Ithaca	Waseca	Lamberton
<u>YIELD (bu/a)</u>				
23.8	43.5	43.0	48.9	44.6
27.5	40.8	50.2	44.8	34.8
32.1	47.1	59.7	49.8	48.4
33.0	44.4	44.5	47.4	40.2
35.7	59.4	48.5	45.9	48.2
39.4	55.5	52.3	54.8	48.6
34.7	57.3	48.9	47.0	50.7
33.3	50.3	46.1	48.4	46.7
36.1	52.6	52.7	49.0	49.0
37.7	52.3	54.3	46.8	43.4
37.7	54.0	57.7	53.3	46.8
37.5	49.1	51.7	49.8	42.2
39.5	49.8	59.9	50.8	42.1
38.9	54.0	64.4	52.4	48.4
39.5	43.5	56.3	44.3	44.2
37.1	51.0	64.2	51.4	44.8
37.7	46.1	50.1	49.8	53.0
37.3	50.8	51.5	51.6	51.4
35.9	46.9	54.5	47.6	42.4
33.3	48.7	54.9	50.0	43.2
30.0	42.3	52.9	43.0	39.4
35.8	51.8	53.3	48.6	50.7
30.7	54.9	58.2	47.9	49.7
38.0	48.9	51.4	45.2	43.6
37.8	48.0	53.3	47.8	42.3
34.5	49.7	58.8	50.1	47.4
34.8	45.4	44.1	49.5	44.4
34.0	47.8	54.6	43.4	43.3
37.7	45.0	57.4	43.8	43.1
7.5	9.0	12.1	6.1	9.0
5.4	9.1	13.1	6.0	8.4
30	30	28	30	30
4	4	4	2	2
2	2	2	2	2

PRELIMINARY TEST I, 1980

Strain	Mean 8 Tests	Ont.	S.D.	Iowa	
		Ridge- town	Brook- ings	Corwith	Knierim
MATURITY DATE					
Corsoy 79 (II)	+8	+6	+7		+8
Evans (0)	-8	-6	-13		-8
Hardin	+3	+2	+1		+6
Hodgson 78 (I)*	9/20	9/25	9/26		9/8
A79-131010	+5	0	+11		+4
A79-133019	+7	+7	+8		+8
A79-134008	0	-2	+4		+2
A79-134018	+6	+4	+8		+8
A79-134026	+3	0	+8		+6
A79-134034	+4	+4	+8		+2
A79-135010	+4	+2	+8		+5
A79-135012	+4	+2	+7		+6
A79-136010	+5	+1	+8		+6
A79-136012	+3	0	+8		+5
A79-136030	+5	+2	+8		+6
A79-138014	+6	+4	+10		+7
A79-138015	+7	+4	+13		+8
A79-138024	+8	+5	+9		+8
A79-287041	+1	-2	+5		+3
M70-620	+2	+2	+2		+2
M72-38	-1	-2	+1		0
M72-79	+2	+6	+4		+2
M72-95	+4	+2	+4		+2
M72-124	+1	-2	+2		+2
M73-32	+2	-2	+9		+2
M73-80	-3	-4	-9		-2
M73-92	+5	+7	+7		+6
W417	+1	0	+6		0
W442	+2	0	+7		+2
Date planted	5/18	5/22	5/20	5/15	5/14
*Days to matur- ity	125	126	129	-	117

PRELIMINARY TEST I, 1980

Wisc.	Mich.		Minn.	
Arlington	Dundee	Ithaca	Waseca	Lamberton
<u>MATURITY DATE</u>				
+8	+8	+11	+7	+6
-8	-5	-6	-10	-6
-1	+5	+6	+4	+4
9/27	9/16	9/21	9/22	9/18
0	+4	+7	+7	+4
+2	+8	+9	+7	+5
-6	+1	+3	0	0
+1	+7	+8	+8	+3
-4	+2	+5	+4	+2
-1	+6	+9	+2	+2
0	+2	+7	+6	+2
+6	+2	+7	+3	+2
+5	+4	+8	+7	+4
+2	+3	+4	+6	-1
+1	+4	+8	+8	+1
+2	+8	+8	+7	+2
+2	+6	+7	+8	+4
+8	+6	+11	+8	+6
-7	-2	+5	+4	0
0	+4	+6	+2	-1
-1	-2	+2	-6	-2
0	+1	+4	-1	-1
+4	+5	+10	0	+1
0	+4	+6	-2	+1
0	0	+6	+1	0
-2	-3	0	0	-2
+2	+6	+8	+2	+3
-6	+4	+5	-2	-1
+3	+3	+4	-1	-2
5/20	5/14	5/22	5/20	5/14
130	125	122	125	127

PRELIMINARY TEST I, 1980

Strain	Mean 8 Tests	Ont.	S.D.	Iowa	
		Ridge- town	Brook- ings	Corwith	Knierim
LODGING (score)					
Corsoy 79 (II)	3.0	1.5		4.4	3.3
Evans (0)	1.8	1.0		4.2	1.7
Hardin	2.7	2.0		4.6	2.8
Hodgson 78 (I)	2.2	1.5		4.0	2.2
A79-131010	2.0	1.5		3.6	2.0
A79-133019	2.7	3.5		3.7	2.4
A79-134008	2.2	1.5		4.0	1.8
A79-134018	2.5	2.0		4.0	3.0
A79-134026	2.4	2.0		4.0	2.2
A79-134034	2.7	2.5		4.4	2.6
A79-135010	2.3	1.5		3.4	2.1
A79-135012	1.9	1.0		2.8	2.0
A79-136010	2.1	1.5		3.5	2.1
A79-136012	2.2	1.5		4.0	2.0
A79-136030	2.0	1.0		3.9	1.8
A79-138014	2.2	1.5		3.2	2.2
A79-138015	2.2	2.0		3.7	1.9
A79-138024	2.3	2.0		3.6	2.2
A79-287041	2.0	1.0		3.4	2.4
M70-620	2.4	2.0		4.2	1.8
M72-38	1.9	1.0		4.2	2.2
M72-79	2.0	1.0		3.8	2.4
M72-95	1.9	1.0		3.7	2.2
M72-124	1.6	1.0		3.6	1.3
M73-32	2.0	2.0		4.6	1.6
M73-80	1.7	1.0		4.2	1.8
M73-92	2.3	1.5		4.1	2.5
W417	1.6	1.0		3.2	1.7
W442	2.2	1.5		4.6	2.0

PRELIMINARY TEST I, 1980

Wisc.	Mich.		Minn.	
Arlington	Dundee	Ithaca	Waseca	Lamberton
LODGING (score)				
2.8	3.5	3.3	2.5	2.5
1.5	2.3	2.0	1.0	1.0
2.2	3.0	2.8	2.0	2.0
1.5	2.3	2.4	1.5	2.0
1.5	2.3	2.4	1.0	2.0
1.8	3.5	2.4	2.0	2.0
1.2	2.8	2.5	1.5	2.0
1.5	3.3	2.8	2.0	1.0
1.5	2.5	2.3	1.5	3.0
2.0	4.3	2.8	1.0	2.0
1.5	2.5	2.5	2.0	2.5
1.5	2.3	2.5	1.5	1.5
1.5	2.5	2.8	1.5	1.0
2.0	2.5	2.4	2.0	1.0
1.5	2.0	2.4	2.0	1.0
1.5	3.0	2.3	2.0	1.5
1.2	2.5	2.8	2.0	1.5
2.0	2.8	2.8	2.0	1.0
1.5	2.0	2.3	2.0	1.5
2.0	3.0	2.9	1.5	2.0
1.2	2.0	1.8	1.5	1.0
1.5	2.8	1.9	1.5	1.0
1.2	2.3	2.4	1.0	1.5
1.0	2.0	2.0	1.0	1.0
1.2	2.0	2.4	1.0	1.5
1.2	2.0	1.5	1.0	1.0
1.8	2.8	2.2	2.0	1.5
1.0	2.5	1.6	1.0	1.0
1.5	2.8	2.3	1.5	1.5

PRELIMINARY TEST I, 1980

Strain	Mean 9 Tests	Ont.	S.D.	Iowa	
		Ridge- town	Brook- ings	Corwith	Knierim
HEIGHT (inches)					
Corsoy 79 (II)	40	44	31	45	39
Evans (0)	31	36	26	40	20
Hardin	36	42	24	46	35
Hodgson 78 (I)	33	36	28	41	27
A79-131010	37	38	31	42	36
A79-133019	34	38	31	38	32
A79-134008	38	44	30	45	36
A79-134018	39	45	33	40	38
A79-134026	33	36	27	38	30
A79-134034	38	39	29	44	36
A79-135010	36	34	33	42	35
A79-135012	34	36	28	40	33
A79-136010	33	32	32	40	32
A79-136012	36	40	28	44	34
A79-136030	37	38	36	41	32
A79-138014	34	38	35	40	32
A79-138015	36	36	35	42	34
A79-138024	36	39	29	41	34
A79-287041	28	32	25	34	26
M70-620	34	40	27	38	28
M72-38	31	32	26	35	28
M72-79	34	41	28	40	32
M72-95	37	38	35	44	34
M72-124	35	38	32	36	32
M73-32	35	35	31	40	30
M73-80	31	34	28	34	24
M73-92	35	40	31	38	31
W417	33	36	30	38	32
W442	36	40	30	42	34

PRELIMINARY TEST I, 1980

Wisc.	Mich.	Minn.		
Arlington	Dundee	Ithaca	Waseca	Lamberton
<u>HEIGHT (inches)</u>				
36	44	43	40	35
30	32	37	28	28
30	39	38	34	34
31	34	38	32	32
34	41	42	36	34
28	35	36	36	34
36	37	40	40	38
33	43	44	40	37
28	34	34	34	38
32	42	43	36	38
28	36	40	36	38
30	34	37	32	34
30	35	35	34	30
34	39	37	40	28
31	33	39	42	38
28	37	36	32	32
32	37	39	36	34
33	37	38	40	35
24	28	33	28	26
31	37	36	32	36
29	32	35	30	28
28	32	34	34	33
34	38	40	38	34
32	35	41	34	36
30	35	41	36	33
30	29	35	34	30
32	36	39	36	30
27	34	39	31	30
33	38	39	35	35

PRELIMINARY TEST I, 1980

Strain	Mean 6 Tests	Ont.	S.D.	Iowa	Wisc.	Minn.	
		Ridge- town	Brook- ings	Corwith	Arling- ton	Waseca	Lamber- ton
QUALITY (score)							
Corsoy 79 (II)	2.1	2.0	2.0	1.3	3.5	1.5	2.0
Evans (O)	2.4	2.0	3.0	1.3	3.0	2.5	2.5
Hardin	2.1	2.0	2.0	1.3	2.5	2.5	2.5
Hodgson 78 (I)	2.3	2.0	3.0	1.5	3.0	2.0	2.0
A79-131010	2.5	2.0	4.0	1.7	3.0	2.0	2.0
A79-133019	2.3	2.0	5.0	1.4	2.0	2.0	1.5
A79-134008	2.7	2.0	3.0	1.8	2.5	3.0	4.0
A79-134018	2.1	2.0	2.0	1.9	3.0	1.5	2.0
A79-134026	2.4	2.0	4.0	1.9	3.0	1.5	2.0
A79-134034	2.0	2.0	2.5	1.4	2.5	1.5	2.0
A79-135010	2.2	2.0	5.0	1.3	2.0	1.0	2.0
A79-135012	2.8	3.0	5.0	1.8	3.5	1.5	2.0
A79-136010	2.0	2.0	4.0	1.3	2.0	1.0	1.5
A79-136012	2.2	3.0	3.0	1.4	2.0	1.5	2.0
A79-136030	2.1	2.0	2.0	1.5	3.0	2.0	2.0
A79-138014	1.7	2.0	1.0	1.3	3.0	1.5	1.5
A79-138015	1.6	1.0	1.0	1.6	2.0	1.5	2.5
A79-138024	2.4	2.0	2.0	1.6	3.5	2.5	3.0
A79-287041	2.5	2.0	3.0	1.7	3.5	2.5	2.0
M70-620	2.5	2.0	3.0	1.7	3.5	2.5	2.0
M72-38	2.5	2.0	2.0	1.7	4.5	2.0	2.5
M72-79	3.1	2.0	3.0	2.4	3.5	3.5	4.0
M72-95	2.4	2.0	2.0	1.8	4.0	2.5	2.0
M72-124	2.1	2.0	3.0	1.3	2.0	2.0	2.0
M73-32	2.0	3.0	2.0	1.7	1.5	2.0	2.0
M73-80	2.6	2.0	2.0	1.7	3.0	4.0	3.0
M73-92	2.6	2.0	3.0	1.5	2.5	3.5	3.0
W417	2.0	2.0	3.0	1.7	2.0	1.5	2.0
W442	2.1	2.0	2.0	1.8	3.0	1.5	2.0

PRELIMINARY TEST I, 1980

Strain	Mean 6 Tests	Ont.	S.D.	Iowa	Wisc.	Minn.	
		Ridge- town	Brook- ings	Corwith	Arling- ton	Waseca	Lamber- ton
SIZE (g/100)							
Corsoy 79 (II)	15.7	15.5	16.5	13.4	15.1	17.3	16.6
Evans (0)	16.3	17.0	15.3	14.4	15.6	18.2	17.3
Hardin	16.1	16.0	15.7	13.0	15.2	17.1	19.3
Hodgson 78 (I)	17.3	17.9	16.7	14.2	17.3	18.8	19.0
A79-131010	17.5	16.0	18.3	15.2	17.9	17.8	19.7
A79-133019	18.2	19.5	16.9	17.0	18.8	18.8	18.2
A79-134008	17.3	16.5	17.4	14.9	17.9	17.9	18.9
A79-134018	16.0	15.2	16.1	14.0	16.8	16.5	17.6
A79-134026	17.6	18.4	17.1	16.4	18.1	17.5	18.2
A790134034	19.3	19.8	17.0	17.6	20.8	20.3	20.4
A79-135010	18.2	16.1	17.4	18.2	19.2	19.8	18.7
A79-135012	18.6	18.8	16.3	18.1	21.2	19.1	18.3
A79-136010	21.2	20.2	19.9	18.6	22.7	23.3	22.4
A79-136012	18.1	16.8	16.1	16.7	19.4	18.9	20.7
A79-136030	17.2	17.9	16.9	14.8	18.6	17.4	17.4
A79-138014	20.0	20.2	20.3	18.5	20.6	20.1	20.5
A79-138015	20.5	18.9	20.3	17.6	23.2	21.2	21.7
A79-138024	20.1	20.6	19.3	17.2	21.2	21.4	20.9
A79-287041	18.2	18.5	19.1	16.8	17.8	18.1	18.9
M70-620	17.2	18.1	16.0	15.0	17.9	18.0	18.2
M72-38	18.4	17.3	18.7	15.4	20.8	18.8	19.3
M72-79	18.3	17.3	16.9	17.0	20.4	18.0	20.4
M72-95	16.9	17.4	16.4	14.8	18.6	17.0	17.2
M72-124	16.3	17.0	16.2	14.6	17.2	16.4	16.1
M73-32	18.5	18.2	18.2	15.8	19.7	20.0	18.8
M73-80	18.8	19.1	20.2	16.4	20.4	20.1	16.6
M73-92	18.8	17.5	18.6	15.6	19.2	20.5	21.4
W417	17.7	17.0	18.3	14.4	19.4	18.2	18.8
W442	18.4	17.4	18.1	15.4	20.2	19.0	20.1

PRELIMINARY TEST I, 1980

Strain	Mean 5 Tests	Ont. Ridge- town	S.D. Brook- ings	Iowa Corwith	Wisc. Arling- ton	Minn. Waseca
% PROTEIN						
Corsoy 79 (II)	42.5	42.5	42.7	39.5	44.8	43.1
Evans (0)	41.8	41.8	41.6	37.2	47.0	41.2
Hardin	41.1	42.1	41.2	39.6	40.3	42.3
Hodgson 78 (I)	42.1	42.4	42.8	38.8	42.8	43.5
A79-131010	41.1	43.2	41.8	39.2	42.0	39.4
A79-133019	41.7	43.5	41.5	39.7	41.6	42.4
A79-134008	41.7	41.6	42.3	39.6	42.8	42.0
A79-134018	43.0	43.2	42.5	42.3	44.2	42.7
A79-134026	42.8	42.8	43.2	41.3	42.7	44.1
A79-134034	42.2	44.7	42.7	38.8	42.8	41.8
A79-135010	42.5	42.2	42.3	41.7	43.8	42.4
A79-135012	44.6	43.3	44.9	41.6	48.1	45.0
A79-136010	42.5	44.0	41.9	41.6	42.7	42.1
A79-136012	43.7	45.7	44.1	39.9	45.3	43.5
A79-136030	41.8	43.3	41.6	39.1	43.3	41.5
A79-138014	43.8	42.8	44.2	41.9	45.9	44.3
A79-138015	41.7	42.9	42.9	38.5	43.5	40.9
A79-138024	42.0	43.2	42.6	39.9	42.4	42.0
A79-287041	41.1	41.8	42.0	38.6	41.4	41.6
M70-620	41.1	40.8	40.7	39.4	43.5	40.9
M72-38	42.3	44.9	42.7	40.0	43.0	40.8
M72-79	41.8	41.4	41.1	41.9	42.7	41.8
M72-95	40.8	43.3	40.2	38.2	42.0	40.5
M72-124	44.6	42.0	45.3	43.6	46.1	46.1
M73-32	42.3	43.3	42.6	40.5	42.9	42.4
M73-80	41.7	41.0	43.7	39.3	42.4	42.2
M73-92	42.6	43.7	42.2	41.7	43.5	42.1
W417	41.1	42.3	41.9	37.4	42.0	41.8
W442	41.4	-	41.5	38.7	43.8	41.5

PRELIMINARY TEST I, 1980

Strain	Mean 5 Tests	Ont. Ridge- town	S.D. Brook- ings	Iowa Corwith	Wisc. Arling- ton	Minn. Waseca
% OIL						
Corsoy 79 (II)	20.1	19.8	21.2	21.8	19.0	18.9
Evans (O)	20.8	20.6	22.7	21.3	19.6	20.0
Hardin	20.3	20.2	21.3	22.0	19.2	19.0
Hodgson 78 (I)	20.9	20.1	22.1	21.8	21.1	19.2
A79-131010	20.2	19.7	21.5	21.8	19.3	18.5
A79-133019	20.0	18.7	20.9	21.4	20.7	18.4
A79-134008	20.6	19.9	21.4	21.5	20.4	19.6
A79-134018	20.1	19.4	21.4	20.5	19.5	19.6
A79-134026	19.4	19.9	19.7	20.6	17.9	18.8
A79-134034	20.5	19.1	21.0	22.3	20.5	19.5
A79-135010	20.1	19.8	21.2	21.8	19.5	18.1
A79-135012	19.7	19.1	20.3	21.2	19.4	18.3
A79-136010	20.2	19.3	21.3	21.3	19.9	19.0
A79-136012	19.6	18.7	20.7	20.0	19.8	18.6
A79-136030	20.0	20.0	21.1	21.8	18.4	18.8
A79-138014	19.8	19.0	21.2	21.1	19.3	18.3
A79-138015	19.9	19.5	20.7	21.4	19.5	18.3
A79-138024	19.6	19.6	20.4	21.1	18.9	18.1
A79-287041	19.9	20.2	22.0	19.9	19.0	18.5
M70-620	21.1	20.6	23.1	21.3	20.8	19.8
M72-38	20.3	19.5	21.8	21.7	19.2	19.3
M72-79	20.8	20.8	21.5	21.6	20.8	19.4
M72-95	21.0	19.8	22.6	23.3	20.4	18.8
M72-124	19.8	19.9	21.0	20.7	19.7	17.5
M73-32	20.9	20.1	22.2	21.6	21.2	19.4
M73-80	20.7	20.3	21.6	21.3	20.9	19.4
M73-92	19.8	20.6	20.5	21.3	17.6	18.8
W417	20.4	21.1	21.3	21.5	19.8	18.4
W442	20.2	-	21.7	21.0	19.6	18.4

UNIFORM TEST II, 1980

Strain	Parentage	Previous Testing*	Generation Compositd
Amcor	C1477 x Corsoy	4	F ₆
Century	Calland x Bonus	3	F ₆
Corsoy 79 (II)	Corsoy ⁶ x Lee 68	2	F ₃
Hardin (I)	Corsoy ³ x Cutler 71	UTI	F ₃
Pella (III)	L66L-137 x Calland	1	F ₄
A77-211021	Beeson x A72-507	1	F ₄
A78-122028	Hodgson x Sloan	PII	F ₄
A78-122031	SRF 350 x Pride B-216	PI	F ₄
A78-123002	C1520 x Coles	PI	F ₄
A78-125029	Pride B-216 x AX900-4-3	PI	F ₄
A78-227013	Pride B-216 x AX901-40-2	PII	F ₄
A78-227015	Pride B-216 x AX901-40-2	PII	F ₄
A78-227016	Pride B-216 x AX901-40-2	PII	F ₄
L27	Corsoy ⁸ x Kingwa	-	F ₃
L73-4673	Corsoy x L66L-154	1	F ₅
U20325	C1317-71 x C1253	1	F ₅
U56355	C1477 x C1421	PII	F ₅

*Number of years in this test or name of 1979 test

Descriptive and Other Data

Strain	Descriptive Code	Chlorosis Score		Hypocotyl Score	Shattering Manhattan
		Ames	Lamberton	Ames	2 Weeks
Amcor	PGBr SYy	3.7	4.0	1	2
Century	PTBr DYB1	3.0	3.5	5	3
Corsoy 79 (II)	PGBr DYY	4.0	4.0	1	2
Hardin (I)	PGBr DYY	3.3	4.5	1	2
Pella (III)	PTTn DYB1	3.2	4.0	1	2
A77-211021	WGBr DYBf	3.7	4.5	3	2
A78-122028	WGBr SYBf	2.5	3.0	5	2
A78-122031	WGBr DYBf	3.8	4.5	2	2
A78-123002	PTBr DYBr	2.5	3.0	2	2
A78-125029	WGBr DYBf	4.0	2.5	1	2
A78-227013	WGBr DYBf	4.2	4.0	4	2
A78-227015	WGTn DYBf	3.3	2.5	1	2
A78-227016	WGTn DYBf	4.3	3.5	2	2
L27	PGBr DYY	4.0	4.0	2	2
L73-4673	PGTn DYY	4.0	4.0	1	2
U20325	PGBr SYIb	2.7	3.0	2	2
U56355	PGBr SYy	2.7	5.0	5	2

UNIFORM TEST II, 1980

Strain	BSR		GERM	SMV	PSB	PS	PR	PR	Race 1
	Laf.	Ames	*				Vickery	Laf.	Ames
	Ind.	Ia.	Lafayette, IN				Ohio	Ind.	Ia.
	n	n	d	a	d	a	a	a	a
	%	Reac.	%	Score	%	Score	--Reaction--		
Ancor	60	S	93	3M	3	5S	2.0	R	R
Century	60	S	92	4E	8	3S	2.3	R	R
Corsoy 79 (II)	0	S	82	5E	11	5E	2.1	R	R
Hardin (I)	0	S	80	4S	16	4E	2.9	R	R
Pella (III)	0	S	95	3E	1	3S	2.2	R	R
A77-211021	0	S	85	1	4	5E	3.1	R	R
A78-122028	20	S	88	1	2	3S	2.4	S	S
A78-122031	20	S	92	3E	4	3S	2.5	S	S
A78-123002	20	S	66	4E	30	2E	2.7	H	H
A78-125029	60	R	95	3M	3	4E	2.4	S	S
A78-227013	0	R	99	1	1	5E	2.2	R	R
A78-227015	60	R	95	1	5	5E	2.5	R	R
A78-227016	60	R	91	1	7	4E	3.2	S	S
L27	0	S	93	5E	5	3S	2.0	R	R
L73-4673	40	S	93	2M	5	4E	2.7	S	S
U20325	60	S	91	1	2	3E	2.8	R	R
U56355	20	S	97	3M	3	5E	2.4	R	R

* Petri dish germination on potato dextrose agar

Two strains, A78-227015 and A78-227016 averaged 1 bushel higher in yield than Century in the 1980 tests. Both strains have good lodging resistance, A78-227015 is resistant to races 1 and 2 of phytophthora, A78-227016 is susceptible to these races of the pathogen. The strain L27, which is resistant to all but races 12 and 16 of phytophthora, averaged about two bushels below Corsoy 79 in yield but was similar to Corsoy 79 in other characteristics.

The two-year mean yields of A77-211021, Ancor, and Century were essentially the same with A77-211021 maturing 3 to 4 days earlier than Ancor or Century.

UNIFORM TEST II, 1980

Regional Summary

Strain No. of Tests	Yield	Rank	Matu- rity	Lodg- ing	Height	Seed Quality	Seed Size	Composition	
	22	22	21	21	22	19	19	5	5
	bu/a	No.	Date	Score	In.	Score	g/100	%	%
Amcor	46.9	6	+4	2.8	42	2.3	16.6	40.5	22.0
Century	47.6	3	+3	1.7	36	2.1	18.5	42.9	21.0
Corsoy 79 (II)	46.5	8	9-22*	2.6	39	2.1	15.3	41.2	22.3
Hardin (I)	44.4	15	-4	2.5	35	2.4	14.8	40.8	23.2
Pella (III)	45.9	12	+6	1.8	38	2.0	19.4	39.3	22.1
A77-211021	47.5	4	-1	2.5	37	2.7	18.9	41.9	21.6
A78-122028	44.0	17	-2	3.0	37	2.1	14.8	39.5	23.5
A78-122031	46.1	11	-4	2.1	31	2.1	16.2	41.0	22.7
A78-123002	46.5	8	-4	2.3	36	2.2	17.1	41.4	21.8
A78-125029	46.5	8	-2	1.8	36	2.0	14.4	41.1	23.4
A78-227013	47.3	5	0	2.2	35	2.0	16.0	41.8	21.4
A78-227015	48.6	2	+2	2.0	32	2.0	17.8	43.2	21.1
A78-227016	48.7	1	+2	1.9	32	2.1	16.2	42.7	21.7
L27	44.6	14	-1	2.8	38	2.0	14.9	41.9	22.5
L73-4673	46.7	7	+5	2.0	38	1.9	16.8	42.7	20.8
U20325	44.4	15	+2	1.6	36	2.2	17.8	42.3	21.3
U56355	44.7	13	+2	1.5	35	2.2	15.9	42.2	21.7

* 126 days after planting

1979-1980 2-year mean

No. of Tests	44	44	42	43	44	39	40	11	11
Amcor	48.0	2	+4.6	2.9	43	2.3	17.1	39.8	20.9
Century	48.0	2	+3.5	1.9	38	2.0	19.0	42.1	20.2
Corsoy 79 (II)	47.0	6	9-23*	2.5	40	2.0	15.8	40.5	21.3
Pella	47.6	5	+6.0	1.9	39	1.9	19.8	39.2	21.1
A77-211021	48.2	1	+0.5	2.6	39	2.6	19.3	40.8	20.7
L73-4673	47.7	4	+4.8	2.2	38	1.8	17.0	41.6	20.0
U20325	44.1	7	+2.9	1.9	38	2.1	18.1	41.3	20.4

* 126 days after planting

Strain	Mean 22 Tests	N.J.	Penn.	Ont.		Ohio	
		Adel- phia	Landis- ville	Ridge- town	Harrow	Hoyt- ville	Wooster
YIELD (bu/a)							
Amcor	46.9	44.9	39.7	52.9	55.3	55.3	46.8
Century	47.6	42.9	29.7	54.3	53.9	58.4	47.1
Corsoy 79 (II)	46.5	43.4	34.2	56.7	53.4	57.4	45.3
Hardin (I)	44.4	44.7	29.2	55.1	57.2	59.3	49.5
Pella (III)	45.9	41.7	32.2	51.4	55.9	51.6	48.5
A77-211021	47.5	43.2	31.8	55.8	60.4	55.4	48.4
A78-122028	44.0	41.4	31.2	52.8	50.4	44.7	42.8
A78-122031	46.1	41.0	16.8	52.7	54.5	60.4	50.1
A78-123002	46.5	45.2	28.4	55.7	61.6	55.5	52.2
A78-125029	46.5	44.7	22.1	50.3	51.1	56.5	47.1
A78-227013	47.3	44.4	27.5	56.9	57.1	52.9	46.1
A78-227015	48.6	43.2	31.2	52.6	57.2	58.2	46.1
A78-227016	48.7	45.7	29.8	54.9	58.0	62.9	50.3
L27	44.6	41.2	26.6	52.0	54.4	55.9	45.9
L73-4673	46.7	48.7	30.2	50.7	53.4	55.9	47.8
U20325	44.4	41.6	22.0	49.3	51.9	52.7	46.6
U56355	44.7	42.9	21.5	49.4	50.0	53.7	45.0
C.V. (%)		5.1	17.5	5.4	7.6	6.1	5.9
L.S.D. (5%)		4.5	8.3	4.0	5.9	5.5	4.6
Row sp. (in.)		30	30	24	24	30	30
Rows/plot		3	4	4	4	4	4
Reps		4	3	4	4	3	3

	22 Tests		YIELD RANK				
Amcor	6	4	1	8	8	12	10
Century	3	11	9	7	11	4	9
Corsoy 79 (II)	8	8	2	2	12	6	15
Hardin (I)	15	5	10	5	4	3	4
Pella (III)	12	13	3	13	7	16	5
A77-211021	4	9	4	3	2	11	6
A78-122028	17	15	5	9	16	17	17
A78-122031	11	17	17	10	9	2	3
A78-123002	8	3	11	4	1	10	1
A78-125029	8	5	14	15	15	7	8
A78-227013	5	7	12	1	6	14	12
A78-227015	2	9	5	11	4	5	12
A78-227016	1	2	8	6	3	1	2
L27	14	16	13	12	10	8	14
L73-4673	7	1	7	14	12	8	7
U20325	15	14	15	17	14	15	11
U56355	13	11	16	16	17	13	16

UNIFORM TEST II, 1980

Strain	Ind.			Mich.		Wisc.	S.D.
	Lafayette	Bluffton	Greenfield	Dundee	Ithaca	Arlington	Brookings
YIELD (bu/a)							
Amcor	59.9	53.8	47.3	50.6	54.0	27.1	33.0
Century	56.7	54.3	49.0	55.5	51.4	31.9	33.5
Corsoy 79 (II)	48.1	45.5	45.4	52.8	56.7	28.2	34.8
Hardin (I)	40.2	44.5	35.1	51.6	58.1	34.2	19.5
Pella (III)	49.8	-	47.1	50.8	53.5	34.7	20.1
A77-211021	52.0	51.6	36.7	57.2	62.2	39.6	38.5
A78-122028	55.5	42.9	37.3	44.6	51.5	35.3	38.2
A78-122031	52.9	45.6	42.1	54.2	52.0	37.4	41.3
A78-123002	49.5	52.8	37.2	57.1	57.1	34.4	34.8
A78-125029	56.9	46.1	44.4	51.9	51.6	37.1	40.9
A78-227013	60.2	49.5	46.9	56.1	54.3	31.6	36.0
A78-227015	61.8	53.4	51.0	56.1	52.4	38.6	35.5
A78-227016	58.9	53.0	41.6	56.5	50.8	38.8	36.1
L27	49.0	50.6	40.2	45.7	54.3	27.1	28.0
L73-4673	59.1	47.4	35.3	51.2	51.2	29.3	33.0
U2020325	46.7	49.2	37.9	51.9	52.0	35.4	34.4
U56355	52.2	50.1	36.9	51.4	54.5	35.8	33.9
C.V. (%)	7.9	10.8	21.5	7.8	8.5	6.6	12.3
L.S.D. (5%)	6.8	8.7	14.7	6.8	7.5	3.6	6.8
Row sp. (in.)	24	30	30	30	28	30	30
Rows/plot	4	3	3	4	4	4	4
Reps	3	3	3	3	3	3	3

YIELD RANK							
Amcor	3	2	3	15	8	17	13
Century	7	1	2	6	15	12	12
Corsoy 79 (II)	15	14	6	8	4	15	8
Hardin (I)	17	16	17	11	2	11	17
Pella (III)	12	-	4	14	9	9	15
A77-211021	11	6	15	1	1	1	4
A78-122028	8	17	12	17	14	8	3
A78-122031	9	13	8	7	11	4	1
A78-123002	13	5	13	2	3	10	9
A78-125029	6	12	7	9	13	5	2
A78-227013	2	10	5	4	6	13	6
A78-227015	1	3	1	4	10	3	7
A78-227016	5	4	9	3	17	2	5
L27	14	7	10	16	6	16	16
L73-4673	4	11	16	13	16	14	14
U20325	16	10	11	9	11	7	10
U56355	10	8	14	12	5	6	11

UNIFORM TEST II, 1980

Ill.		Ill.		Minn.		Iowa		Neb.
Dekalb	Pontiac	Urbana	Girard	Waseca	Lamber- ton	Marshall- town	Ames	Mead
YIELD (bu/a)								
45.7	38.8	45.7	37.4	41.6	42.8	47.2	56.5	54.9
43.8	37.7	48.3	38.2	51.5	50.3	48.7	58.5	51.4
44.0	37.0	40.8	35.4	48.6	50.0	50.8	58.6	56.9
42.6	33.5	40.8	26.6	52.5	46.5	48.4	56.8	51.2
41.8	37.8	52.8	37.8	49.0	50.1	50.9	55.5	50.7
43.8	34.6	44.1	25.8	51.7	48.8	48.5	60.1	55.3
41.9	32.7	42.4	29.1	50.9	44.7	47.0	56.8	53.4
48.1	28.9	45.3	24.0	54.8	46.5	51.5	61.1	53.9
44.4	29.6	42.7	28.4	49.4	49.2	52.2	55.3	49.4
45.4	28.1	47.6	28.7	53.4	53.8	51.5	57.2	56.4
46.4	31.6	47.5	30.2	47.0	49.1	52.5	59.6	56.3
49.5	31.1	48.8	30.9	54.0	51.1	52.3	62.5	51.4
48.7	31.7	47.4	29.1	53.0	51.5	52.7	65.7	55.0
44.3	30.8	43.7	32.5	52.2	49.0	48.2	60.5	48.7
50.6	35.6	46.9	38.5	49.0	48.1	50.0	59.6	55.7
44.3	32.8	45.6	32.2	44.4	47.9	47.5	55.7	54.3
42.6	28.3	48.9	28.6	49.5	48.6	47.4	54.4	58.5
5.7	10.3	5.7	10.3	5.9	6.9	3.5	5.2	9.7
4.2	5.6	4.3	5.4	4.9	5.6	2.5	4.3	8.4
30	30	30	30	30	30	27	27	30
4	4	4	4	4	4	4	4	4
3	3	3	3	3	3	4	4	3

YIELD RANK								
6	1	9	4	17	17	16	13	7
12	3	4	2	8	4	10	9	11
11	4	16	5	14	6	8	8	1
14	7	16	15	5	14	12	11	13
17	2	1	3	12	5	7	15	14
12	6	12	16	7	10	11	5	5
16	9	15	10	9	16	17	11	10
4	15	11	17	1	14	5	3	9
8	14	14	14	11	7	4	16	15
7	17	5	12	3	1	5	10	2
5	11	6	9	15	8	2	6	3
2	12	3	8	2	3	3	2	11
3	10	7	10	4	2	1	1	6
9	13	13	6	6	9	13	4	16
1	5	8	1	12	12	9	6	4
9	8	10	7	16	13	14	14	8
14	16	2	13	10	11	15	17	17

UNIFORM TEST II, 1980

Strain	Mean 21 Tests	N.J.	Penn.	Ont.		Ohio	
		Adel- phia	Landis- ville	Ridge- town	Harrow	Hoyt- ville	Wooster
MATURITY (date)							
Amcor	+4	+3	+3	+7	+3	+7	+4
Century	+3	+2	+3	+2	+2	+4	+1
Corsoy 79 (II)	9/22	9/15	9/19	10/1	9/30	9/19	9/20
Hardin (I)	-4	-3	-4	-5	-5	-4	-2
Pella (III)	+6	+7	+5	+7	+2	+8	+7
A77-211021	-1	-2	-7	0	-1	0	-1
A78-122028	-2	-1	-7	-1	-4	-4	-2
A78-122031	-4	-3	-9	-5	-5	-4	-4
A78-123002	-4	-4	-9	-4	-6	-4	-5
A78-125029	-2	-1	-7	-2	-3	-3	-1
A78-227013	0	+3	-4	-2	-2	+1	+1
A78-227015	+2	+3	-4	-1	+1	+4	+1
A78-227016	+2	+3	-7	+5	+3	+5	+2
L27	-1	-3	-7	0	-2	-1	-1
L73-4673	+5	+7	0	+8	+3	+4	+3
U20325	+2	0	-4	0	+1	+4	+3
U56355	+2	0	-4	-1	0	+4	+3
Date planted	5/19	5/28	6/5	5/22	5/27	5/6	5/16
*Days to maturity	126	110	106	132	126	136	127

	21 Tests	LODGING (score)					
Amcor	2.8	3.0	1.0	2.5	3.6	2.8	3.5
Century	1.7	1.0	1.0	1.2	2.1	2.1	2.4
Corsoy 79 (II)	2.6	3.0	1.0	1.8	3.1	3.4	3.6
Hardin (I)	2.5	3.0	1.0	2.0	2.6	2.8	3.4
Pella (III)	1.8	1.0	1.0	1.8	2.4	2.0	1.8
A77-211021	2.5	3.2	1.0	2.0	2.8	3.3	3.1
A78-122028	3.0	4.0	1.0	2.8	3.0	3.8	4.1
A78-122031	2.1	2.0	1.0	1.5	2.9	2.5	2.0
A78-123002	2.3	2.2	1.0	2.2	2.4	2.8	3.1
A78-125029	1.8	1.5	1.0	2.0	2.3	2.5	2.2
A78-227013	2.2	2.2	1.0	2.0	2.3	3.1	3.0
A78-227015	2.0	1.2	1.0	2.8	2.6	2.7	2.4
A78-227016	1.9	1.8	1.0	2.2	2.4	2.4	2.5
L27	2.8	3.0	1.0	2.5	2.9	3.8	3.5
L73-4673	2.0	2.2	1.0	1.5	2.0	2.2	2.8
U20325	1.6	1.5	1.0	1.0	1.8	1.5	1.8
U56355	1.5	1.2	1.0	1.5	1.6	1.7	2.4

UNIFORM TEST II, 1980

Ill.		Minn.		Iowa		Neb.		S.D.	Mich.	
Urbana	Girard	Waseca	Lamber- ton	Marshall- town	Ames	Mead	Brook- ings		Dundee	Ithaca
MATURITY (date)										
+6	+5	+3	+6		+6	+4	+6		+6	+6
+7	+6	+3	+3		+3	-1	+6		+5	+6
9/12	8/30	9/30	9/23		9/18	9/23	10/6		9/24	10/5
0	-4	-1	-1		-6	-2	-6		-2	-6
+11	+11	+6	+6		+7	+6	+9		+5	+8
+1	-1	+1	0		+2	0	+2		0	+2
0	+2	-1	+1		-1	-1	+2		-2	-5
-3	-6	-2	-1		-7	-1	0		-4	-4
-2	-2	-2	-3		-5	-3	+1		-2	-5
0	-3	-1	0		-4	-1	+2		-2	-3
+4	+3	+2	+1		0	-1	+5		+1	+1
+7	+5	+2	+3		+3	+2	+4		+3	+4
+2	+3	+2	+4		+2	+2	+6		+4	+6
0	-1	0	+2		0	-1	+1		0	-1
+9	+7	+7	+6		+7	+4	+6		+6	+4
+8	+2	+5	+7		+4	+1	+6		+1	+8
+6	+3	+5	+6		+2	0	+7		+1	+4
5/7	5/6	5/20	5/14	5/16	5/13	5/27	5/20		5/14	5/22
128	116	133	132	--	128	119	139		133	136

LODGING (score)										
2.6	2.8	2.0	3.3	3.4	3.2	1.8			3.4	4.0
1.3	1.5	1.7	1.3	2.1	1.9	1.0			2.5	3.3
2.4	3.2	2.0	2.7	3.3	3.0	1.2			3.2	3.3
2.8	2.9	2.0	2.3	3.5	3.1	1.3			3.2	2.5
1.4	1.4	1.7	1.7	2.0	2.0	1.2			2.2	3.8
2.2	2.3	2.0	2.3	3.6	3.3	1.5			3.5	2.8
2.8	3.4	2.3	3.3	3.8	3.2	1.8			3.7	3.9
2.0	2.5	1.7	2.0	2.9	2.4	1.3			3.0	2.7
1.9	1.9	2.0	2.0	3.3	2.7	1.3			2.9	3.3
1.2	1.8	1.7	1.3	2.6	2.0	1.2			2.5	2.3
1.6	2.0	2.0	1.7	2.8	2.2	1.5			2.8	3.8
1.3	1.5	1.0	1.7	2.2	2.1	1.0			2.8	3.8
1.2	1.6	1.0	2.0	2.2	2.8	1.2			2.8	3.3
2.8	2.9	2.0	2.7	3.6	3.1	1.5			3.2	3.5
2.1	1.9	2.0	2.0	3.0	2.8	1.2			2.7	2.8
1.4	1.6	1.7	2.0	1.9	1.8	1.2			2.1	3.3
1.2	1.3	1.7	2.0	1.9	1.9	1.2			1.9	2.3

UNIFORM TEST II, 1980

Strain	Ind.			Wisc.	Ill.	
	Lafayette	Bluffton	Greenfield	Arlington	Dekalb	Pontiac
<u>MATURITY (date)</u>						
Amcor	+5	+3	+2	+1	-6	+2
Century	+2	+3	+1	+3	+1	+2
Corsoy 79 (II)	9/10	9/11	9/19	10/3	9/26	9/21
Hardin (I)	-3	-1	-6	-8	-4	-4
Pella (III)	+9	-	+3	+4	+4	+2
A77-211021	0	-1	-2	-4	-3	-2
A78-122028	0	-1	-3	-7	-4	-2
A78-122031	-2	-3	-3	-11	-3	-3
A78-123002	-2	-3	-3	-6	-4	-2
A78-125029	-1	-1	-3	-6	-3	-5
A78-227013	+2	+1	0	-5	-1	0
A78-227015	+4	+2	+1	0	+1	0
A78-227016	+2	+1	0	-1	+1	+1
L27	0	0	-2	0	-4	-2
L73-4673	+7	+3	+2	+3	+5	+1
U20325	+1	+1	-1	+1	-1	+2
U56355	+1	+2	-1	-2	0	0
Date planted	5/7	5/15	5/29	5/20	5/21	5/27
Days to maturity	126	119	113	136	128	117
<u>LODGING (score)</u>						
Amcor	2.2	1.8	2.2	3.5	4.2	1.8
Century	1.0	1.5	1.0	1.5	2.8	1.0
Corsoy 79 (II)	2.0	2.2	2.3	2.5	4.0	1.8
Hardin (I)	2.2	1.8	2.3	2.2	4.0	1.7
Pella (III)	1.5	---	1.0	1.8	2.8	1.0
A77-211021	1.8	2.0	1.5	2.2	4.3	1.5
A78-122028	2.8	2.3	1.7	2.5	4.9	1.7
A78-122031	1.3	1.7	2.0	1.7	3.8	1.0
A78-123002	1.3	1.7	2.2	2.7	4.0	1.2
A78-125029	1.0	1.3	1.0	1.3	3.8	1.0
A78-227013	1.5	1.7	2.3	2.0	3.8	1.2
A78-227015	1.2	1.7	2.0	1.5	3.7	1.0
A78-227016	1.0	1.2	1.3	1.2	3.2	1.0
L27	2.2	2.2	3.3	3.0	4.3	1.7
L73-4673	1.2	1.5	1.0	2.0	4.0	1.0
U20325	1.0	1.0	1.0	1.5	2.2	1.0
U56355	1.0	1.0	1.0	1.2	2.3	1.0

UNIFORM TEST II, 1980

Strain	Mean 22 Tests	N.J.	Penn.	Ont.		Ohio	
		Adel- phia	Landis- ville	Ridge- town	Harrow	Hoyt- ville	Wooster
HEIGHT (inches)							
Amcor	42	40	34	47	45	45	46
Century	36	32	24	42	42	41	36
Corsoy 79 (II)	39	36	28	46	41	43	40
Hardin (I)	35	33	26	40	38	35	41
Pella (III)	38	36	25	41	43	41	41
A77-211021	37	35	26	45	42	44	39
A78-122028	37	35	26	42	41	39	46
A78-122031	31	28	19	35	35	33	30
A78-123002	36	33	23	42	39	41	40
A78-125029	36	33	23	43	42	41	35
A78-227013	35	33	23	40	37	39	36
A78-227015	32	30	20	37	39	36	32
A78-227016	32	30	20	36	36	40	35
L27	38	36	22	45	42	43	44
L73-4673	38	34	22	44	40	37	40
U20325	36	36	20	38	42	39	40
U56355	35	34	19	40	40	40	35
QUALITY (score)							
	19 Tests						
Amcor	2.3	1.2	2.0	2.0	2.0	2.2	2.5
Century	2.1	1.8	2.3	2.0	2.0	1.7	2.6
Corsoy 79 (II)	2.1	1.0	2.0	2.0	1.0	1.3	1.7
Hardin (I)	2.4	1.2	2.0	2.0	2.0	2.5	1.8
Pella (III)	2.0	1.0	2.0	2.0	1.0	2.3	2.1
A77-211021	2.7	2.5	2.2	2.0	4.0	1.7	2.7
A78-122028	2.1	1.2	2.0	2.0	2.0	1.8	1.7
A78-122031	2.1	1.5	2.2	1.0	3.0	2.5	2.0
A78-123002	2.2	1.8	2.0	2.0	3.0	2.1	2.3
A78-125029	2.0	1.0	2.0	2.0	2.0	2.4	2.4
A78-227013	2.0	1.0	2.0	2.0	2.0	1.7	2.3
A78-227015	2.0	1.5	2.0	2.0	2.0	2.0	2.6
A78-227016	2.1	1.8	2.0	2.0	2.0	2.7	2.3
L27	2.0	1.2	2.2	2.0	1.0	1.5	1.9
L73-4673	1.9	1.0	2.0	2.0	1.0	1.4	2.0
U20325	2.2	1.8	2.0	2.0	2.0	1.5	2.1
U56355	2.2	1.2	2.0	2.0	2.0	2.1	1.9

UNIFORM TEST II, 1980

Strain	Ind.			Mich.		Ill.	
	Lafayette	Bluffton	Greenfield	Dundee	Ithaca	Dekalb	Pontiac
HEIGHT (inches)							
Amcor	41	40	35	47	48	46	38
Century	35	35	31	45	43	40	29
Corsoy 79 (II)	39	32	34	43	45	46	34
Hardin (I)	34	31	34	41	43	42	31
Pella (III)	36	--	31	44	46	41	33
A77-211021	36	32	28	43	44	41	30
A78-122028	37	33	29	43	45	42	29
A78-122031	31	28	27	37	37	34	25
A78-123002	34	33	33	40	43	44	29
A78-125029	34	33	29	40	44	42	26
A78-227013	36	31	32	39	43	39	27
A78-227015	31	29	27	38	42	35	24
A78-227016	31	28	27	36	39	33	24
L27	36	33	33	46	43	44	30
L73-4673	40	35	33	43	42	44	34
U20325	34	32	28	43	45	43	30
U56355	34	33	27	40	44	38	30

QUALITY (score)							
Amcor	2.0	1.0	2.0			1.5	2.1
Century	1.5	1.5	2.0			1.3	2.3
Corsoy 79 (II)	1.5	1.0	2.0			1.3	1.9
Hardin (I)	2.0	1.5	2.0			1.3	1.7
Pella (III)	1.5	---	1.5			1.5	1.9
A77-211021	2.5	2.0	1.5			1.5	2.1
A78-122028	1.5	1.5	1.5			1.7	1.8
A78-122031	1.5	1.5	1.5			2.0	1.7
A78-123002	1.5	1.5	1.5			1.5	1.9
A78-125029	1.5	1.5	1.5			1.9	1.7
A78-227013	1.5	1.5	1.5			1.5	1.9
A78-227015	1.0	1.0	2.0			1.3	1.9
A78-227016	1.0	1.0	1.5			1.6	1.6
L27	1.5	1.5	2.0			1.3	1.7
L73-4673	2.0	1.0	1.5			1.3	1.9
U20325	2.0	1.0	1.5			1.4	2.5
U56355	1.5	1.0	1.5			1.5	2.1

UNIFORM TEST II, 1980

Wisc. Arling- ton	Ill.		Minn.		Iowa		Neb.	S.D.
	Urbana	Girard	Waseca	Lamber- ton	Marshall- town	Ames	Mead	Brook- ings
<u>HEIGHT (inches)</u>								
41	36	42	39	44	42	52	41	35
34	31	39	39	36	37	44	26	35
36	33	40	39	40	40	51	33	32
33	30	35	35	36	37	45	28	32
40	36	39	40	40	39	47	31	38
35	31	37	38	40	38	48	33	35
37	33	38	38	40	38	46	31	35
30	28	32	34	33	32	38	23	31
33	30	35	36	37	36	48	28	35
35	30	37	35	35	36	43	30	35
31	30	36	36	36	36	41	30	36
30	26	33	34	35	32	38	27	31
27	28	33	34	34	35	42	29	32
36	33	39	40	39	43	49	32	30
34	34	39	40	39	42	44	33	36
34	30	38	39	38	38	45	31	35
34	31	37	37	42	38	43	29	33

<u>QUALITY (score)</u>								
3.0	4.0	3.2	2.0	2.7		2.7	2.2	3.0
2.0	4.2	3.0	1.3	2.0		2.1	1.7	2.0
3.7	3.0	3.3	2.3	2.7		1.6	2.7	3.0
4.0	3.8	4.0	3.0	3.0		1.8	2.2	3.0
2.0	3.3	2.8	2.0	2.7		1.5	2.0	2.0
3.3	3.7	4.7	2.7	3.7		3.0	3.2	3.0
2.7	2.7	3.7	2.3	2.7		2.0	2.3	2.0
4.3	2.7	4.3	2.0	1.7		1.4	2.0	2.0
3.3	3.2	3.3	2.3	2.0		1.7	2.2	2.0
2.7	2.7	3.2	1.7	1.7		1.3	2.3	3.0
3.7	2.5	3.2	1.3	1.7		1.6	2.3	2.0
3.0	2.7	3.3	2.0	2.7		1.5	2.3	2.0
4.0	2.3	3.3	2.0	2.3		1.6	2.7	2.0
3.0	3.0	3.2	2.7	2.3		2.0	2.0	2.0
3.7	2.8	2.2	1.0	2.3		1.9	2.2	2.0
2.0	3.0	3.2	1.3	3.0		3.6	2.7	3.0
2.3	3.7	3.5	2.3	2.7		2.3	2.7	3.0

UNIFORM TEST II, 1980

Strain	Mean 19 Tests	N.J.	Penn.	Ont.		Ohio	
		Adel- phia	Landis- ville	Ridge- town	Harrow	Hoyt- ville	Wooster
SIZE (g/100)							
Amcor	16.6	12.0	14.9	17.2	17.7	17.2	15.3
Century	18.5	14.0	17.5	19.3	19.8	18.6	17.0
Corsoy 79 (II)	15.3	12.0	13.4	17.2	16.3	15.1	13.9
Hardin (I)	14.8	10.0	12.6	15.8	15.9	15.4	13.4
Pella (III)	19.4	16.0	16.6	20.8	21.1	19.3	18.2
A77-211021	18.9	15.0	16.1	19.4	20.6	18.5	16.2
A78-122028	14.8	12.0	14.3	15.0	15.7	14.6	14.0
A78-122031	16.2	12.0	14.3	16.1	17.4	16.4	14.4
A78-123002	17.1	14.0	15.9	18.7	18.5	16.8	17.2
A78-125029	14.4	13.0	13.1	13.4	14.9	14.3	12.3
A78-227013	16.0	14.0	13.6	15.9	17.7	16.3	15.4
A78-227015	17.8	16.0	17.0	18.2	20.4	18.6	15.9
A78-227016	16.2	15.0	14.3	17.2	18.4	17.7	15.4
L27	14.9	12.0	13.1	16.5	15.9	15.8	13.6
L73-4673	16.8	14.0	15.6	16.9	17.5	16.1	14.7
U20325	17.8	15.0	18.7	17.7	18.9	17.6	16.9
U56355	15.9	13.0	14.6	16.0	17.3	16.7	14.8

UNIFORM TEST II, 1980

Ind.		Wisc.		Ill.	
Lafayette	Bluffton	Greenfield	Arlington	Dekalb	Pontiac
SIZE (g/100)					
15.9	17.1	18.2	14.3	16.1	17.4
19.7	19.4	19.8	18.5	18.2	18.8
15.8	14.7	19.5	14.1	14.3	15.2
14.1	14.5	14.9	14.4	14.5	14.1
19.9	--	20.6	20.1	19.5	19.2
19.6	18.9	19.5	19.5	18.9	17.6
15.3	13.5	15.0	14.3	13.7	14.4
16.6	16.3	16.6	16.3	16.1	15.7
17.2	17.0	16.9	17.8	17.3	16.1
15.1	13.9	14.9	14.7	14.6	13.7
17.0	15.9	16.6	15.4	16.6	15.3
17.7	17.3	19.2	18.5	19.0	17.6
16.6	15.5	16.5	16.5	17.4	16.1
14.4	14.9	15.2	13.7	13.8	15.1
17.8	16.3	18.0	16.6	17.1	17.7
18.9	16.1	17.8	18.1	16.9	19.4
16.7	16.4	16.5	16.5	16.1	15.5

UNIFORM TEST II, 1980

Strain	Ill.		Minn.		Iowa	Neb.	S.D.
	Urbana	Girard	Waseca	Lamber- ton	Ames	Mead	Brookings
	SIZE (g/100)						
Amcor	17.7	12.9	17.8	19.2	18.8	17.9	16.9
Century	19.7	15.4	20.1	18.9	20.6	19.8	16.9
Corsoy 79 (II)	14.2	11.3	17.0	17.1	16.2	17.1	16.3
Hardin (I)	15.1	11.8	17.0	17.2	16.2	18.6	16.6
Pella (III)	20.3	17.0	20.6	20.7	20.4	18.2	20.1
A77-211021	19.1	13.6	21.1	23.6	22.4	21.7	18.3
A78-122028	14.6	11.4	17.4	16.5	17.0	15.7	16.2
A78-122031	16.1	11.3	17.9	18.6	17.6	19.6	17.9
A78-123002	17.2	12.1	18.5	18.5	18.6	18.1	17.6
A78-125029	14.9	9.9	16.2	16.9	16.1	17.7	13.7
A78-227013	16.1	12.2	16.7	17.6	17.4	17.9	16.1
A78-227015	14.9	13.7	20.1	19.5	20.0	18.7	16.5
A78-227016	16.8	12.0	16.2	17.4	17.8	17.0	14.4
L27	15.0	10.4	17.2	16.9	17.0	16.6	16.0
L73-4673	18.1	13.2	18.0	17.5	19.1	19.3	15.0
U20325	19.3	13.3	18.2	18.6	20.0	19.9	17.1
U56355	17.0	10.9	18.1	17.3	17.2	16.4	15.9

UNIFORM TEST II, 1980

		<u>Ohio</u>	<u>Ind.</u>	<u>Ill.</u>	<u>Iowa</u>	<u>Neb.</u>
Strain	Mean 5 Tests	Hoytville	Lafayette	Urbana	Ames	Mead
		<u>PROTEIN (%)</u>				
Amcor	40.5	40.5	41.1	39.5	40.9	40.4
Century	42.9	44.6	42.9	41.8	42.6	42.6
Corsoy 79 (II)	41.2	41.1	42.5	40.9	39.7	41.9
Hardin (I)	40.8	41.2	41.4	42.0	38.3	41.3
Pella (III)	39.3	40.1	--	39.2	38.4	39.3
A77-211021	41.9	40.6	42.3	42.9	41.9	41.9
A78-122028	39.5	41.8	41.7	38.4	37.5	38.0
A78-122031	41.0	42.9	42.3	40.0	40.3	39.5
A78-123002	41.4	41.5	43.2	39.2	40.7	42.2
A78-125029	41.1	43.3	41.6	40.0	40.0	40.5
A78-227013	41.8	43.3	43.0	40.1	40.4	42.1
A78-227015	43.2	44.3	43.3	42.5	42.6	43.1
A78-227016	42.7	43.6	42.9	42.2	42.9	42.1
L27	41.9	43.1	42.5	41.7	41.0	41.4
L73-4673	42.7	42.7	44.2	41.2	42.0	43.2
U20325	42.3	42.2	42.5	41.6	42.0	43.4
U56355	42.2	41.9	42.4	42.7	42.1	41.8

Strain	5 Tests	OIL (%)				
		Ohio	Ind.	Ill.	Iowa	Neb.
Amcor	22.0	23.6	22.1	21.5	21.6	21.3
Century	21.0	22.5	21.5	20.2	19.3	21.6
Corsoy 79 (II)	22.3	23.3	22.1	22.2	22.3	21.5
Hardin (I)	23.2	23.9	23.0	22.8	22.6	23.8
Pella (III)	22.1	23.2	--	22.2	20.6	22.3
A77-211021	21.6	22.9	20.8	21.0	21.0	22.1
A78-122028	23.5	24.7	22.7	23.1	23.0	24.2
A78-122031	22.7	23.2	22.5	22.7	22.5	22.6
A78-123002	21.8	22.4	21.5	21.4	22.0	21.6
A78-125029	23.4	23.6	23.2	24.4	22.8	23.1
A78-227013	21.4	22.7	21.4	21.3	20.2	21.3
A78-227015	21.1	22.4	20.8	21.9	19.9	20.7
A78-227016	21.7	23.7	21.2	21.1	21.2	21.2
L27	22.5	23.7	22.4	23.1	21.6	21.8
L73-4673	20.8	21.7	20.6	20.5	20.7	20.5
U20325	21.3	21.9	22.0	20.9	20.6	20.9
U56355	21.7	22.7	21.7	21.3	20.8	21.9

PRELIMINARY TEST II, 1980

Strain	Parentage	Generation Composited
Century	Calland x Bonus	F ₆
Corsoy 79 (II)	Harosoy x Capital	F ₉
Gnome	Williams x Ransom	F ₅
Hardin (I)	Corsoy ³ x Cutler 71	F ₃
Pella (III)	L66L-137 x Calland	F ₄
A75D15	Amsoy x Harosoy Dt ₂ (L61-344)	F ₆
A75D29	Hark x Harosoy Dt ₂ (L61-344)	F ₆
A75D44	Corsoy x Harosoy Dt ₂ (L61-344)	F ₅
A79-138035	Pride B216 x Cumberland	F ₄
A79-232005	AP6TW 2YTC (F ₄) C1	F ₄
A79-232026	AP6TW 2YTC (F ₄) C1	F ₄
A79-232027	AP6YW 2YTC (F ₄) C1	F ₄
A79-235002	Pride B216 x Cumberland	F ₄
A79-236002	Pride B216 x Cumberland	F ₄
A79-236003	Pride B216 x Cumberland	F ₄
A79-237014	C1523 x Pride B216	F ₄
A79-237034	C1523 x Pride B216	F ₄
A79-238034	M68-48 x Pride B216	F ₄
HC77-869	Woodworth x V68-1034	F ₅
HC77-870	Woodworth x V68-1034	F ₅
HC77-874	Wayne x Dare	F ₅
HC77-876	Wayne x Dare	F ₅
HC77-878	Woodworth x V68-1034	F ₅
HC77-951	Woodworth x V68-1038	F ₅
HC77-955	Calland x York	F ₅
HC77-1489	Hodgson x L72U-547	F ₅
HW79015	A72-512 x Oakland (AX-1733E-18)	F ₅
HW79022	Woodworth x L60-347-1-60-2B (OX750-26)	F ₄
HW79054	Cumberland x Pella (AX-176M-86)	F ₅
U59218	Williams x Amsoy 71	F ₄

PRELIMINARY TEST II, 1980

Descriptive and Other Data

Strain	Descriptive Code		Chlorosis	Shattering
			Score Ames	Manhattan 2 Weeks
Century	PTBr	DYB1	3.0	1
Corsoy 79 (II)	PGBr	DYY	4.0	2
Gnome	PTTn	SYB1	3.2	1
Hardin (I)	PGBr	DYY	3.3	1
Pella (III)	PTTn	DYB1	3.2	2
A75D15	PGBr	SYB1	2.2	1
A75D29	PGBr	DYY	3.3	1
A75D44	PGBr	DYY	2.8	1
A79-138035	WTTn	DYB1	4.0	1
A79-232005	PGBr	DYGr	3.0	2
A79-232026	WTBr	DYY	3.0	1
A79-232027	WTBr	DYBr	4.5	2
A79-235002	WTBr	DYGr	3.3	2
A79-236002	P+WTBr	DYGr	3.7	1
A79-236003	WTBr	DYBr	4.2	1
A79-237014	PGBr	DYGr	3.3	1
A79-237034	WGBr	DYY	4.0	1
A79-238034	WGBr	DYBf	4.2	1
HC77-869	WTTn	DYBr	3.8	1
HC77-870	WTTn	DYB1+Br	3.0	1
HC77-874	WTBr	DYB1	3.5	1
HC77-876	WTBr	DYB1	4.2	1
HC77-878	WTTn	DYB1	3.2	1
HC77-951	WTTn	DYB1	2.8	1
HC77-955	PTTn	SYB1	3.3	4
HC77-1489	PGBr	DYBr	3.3	1
HW79015	PGBr	DYIb	3.2	2
HW79022	PTTn	DYBr	2.5	1
HW79054	PTTn	DYB1	3.8	2
U59218	PGTn	SYB1	3.3	-

PRELIMINARY TEST II, 1980

Strain	Disease Data								Race 1 Ames Ia. a
	BSR		GERM	SMV	PSB	PS	PR	PR	
	Laf.	Ames	*				Vickery	Laf.	
	Ind.	Ia.	Lafayette,	Indiana			Ohio	Ind.	
	n	n	d	a	d	a	a	a	
	%	Reac.	%	Score	%	Score	-----Reaction-----		
Century	60	S	91	4M	9	3E	2.2	R	R
Corsoy 79 (II)	0	S	96	5S	1	4E	2.2	R	R
Gnome	100	S	97	1	2	2E	3.0	S	S
Hardin (I)	0	S	84	4E	12	3S	3.0	R	R
Pella (III)	0	S	82	3M	11	4E	2.0	R	R
A75D15	0	S	92	4M	2	4E	3.1	S	S
A75D29	40	S	84	4E	16	2S	3.4	S	S
A75D44	20	S	78	4E	16	3E	3.4	S	S
A79-138035	80	S	82	4E	17	2E	2.1	R	R
A79-232005	60	S	91	1	5	5S	2.1	R	R
A79-232026	60	S	89	1	4	4S	3.5	S	S
A79-232027	0	S	84	3E	16	5S	2.3	S	S
A79-235002	0	S	85	5S	9	3S	2.2	R	R
A79-236002	40	S	93	5S	4	3E	2.3	R	R
A79-236003	80	S	70	4E	26	2E	2.5	R	S
A79-237014	40	S	80	5S	15	3E	2.3	R	R
A79-237034	20	S	80	3M	15	4E	2.5	R	S
A79-238034	60	S	86	1	10	5S	2.5	S	S
HC77-869	40	S	97	1	0	2E	3.1	S	S
HC77-870	40	S	93	1	5	2E	3.0	S	S
HC77-874	80	S	95	1	6	3S	3.2	R	R
HC77-876	80	S	89	1	11	3E	3.3	R	R
HC77-878	40	S	90	1	8	2E	3.1	H	H
HC77-951	20	S	96	1	2	2E	3.1	S	S
HC77-955	40	S	89	2M	7	1	3.8	S	H
HC77-1489	20	S	88	1	8	2M	4.4	S	S
HW79015	20	S	81	3E	10	3E	2.4	S	S
HW79022	40	S	74	1	15	3E	2.1	H	H
HW79054	0	S	78	4E	18	4S	2.3	R	R
U59218	40	S	61	3M	33	4E	2.6	H	H

* Petri dish germination on potato dextrose agar

Strain No. of Tests	Yield	Rank	Regional Summary				Seed Quality	Seed Size	Composition	
			Matur- ity	Lodg- ing	Height				Protein	Oil
			12	12	11	12			12	9
	bu/a	No.	Date	Score	In.	Score	g/100	%	%	
Century	46.7	16	+3	1.9	37	2.1	18.2	43.4	21.5	
Corsoy 79 (II)	46.9	15	9-21*	2.8	40	2.1	14.9	41.4	23.0	
Gnome	42.7	24	+6	1.5	24	1.7	15.8	42.2	22.3	
Hardin (I)	44.4	18	-3	2.6	37	2.3	14.8	40.8	22.9	
Pella (III)	48.8	7	+7	2.0	41		19.1	39.7	22.1	
A75D15	43.2	20	-4	2.1	34	2.5	16.9	40.6	22.7	
A75-D29	44.0	19	-3	1.8	31	2.0	15.4	41.9	21.2	
A75D44	45.6	17	0	3.0	41	2.2	16.7	41.0	21.9	
A79-138035	47.4	13	+1	2.2	38	2.0	17.2	40.5	22.5	
A79-232005	49.6	5	+5	2.8	36	2.2	15.9	41.7	22.2	
A79-232026	50.5	2	+6	2.5	36	2.2	15.9	40.0	22.7	
A79-232027	47.8	9	+6	2.5	39	2.4	17.4	39.7	23.3	
A79-235002	47.6	11	+6	2.1	40	2.1	17.9	41.7	21.6	
A79-236002	50.5	2	+6	2.3	40	2.3	17.9	40.4	22.5	
A79-236003	48.5	8	+3	2.7	36	1.7	16.2	40.6	22.8	
A79-237014	47.6	11	+3	2.0	37	2.1	19.9	43.1	21.2	
A79-237034	47.7	10	+4	2.0	38	1.8	22.2	43.2	22.5	
A79-238034	50.3	4	+7	2.5	37	2.0	14.8	40.9	22.5	
HC77-869	42.9	23	+5	1.6	28	1.7	14.0	37.7	22.9	
HC77-870	43.0	21	+4	1.4	27	1.7	14.3	38.0	23.2	
HC77-874	41.3	28	+6	1.5	26	1.6	16.7	43.4	21.7	
HC77-876	41.9	25	+6	1.5	26	1.7	14.9	43.7	21.2	
HC77-878	41.7	26	+2	1.2	22	1.7	16.5	40.9	22.0	
HC77-951	43.0	21	+3	1.3	26	1.8	14.7	37.7	23.2	
HC77-955	38.0	30	+1	1.3	20	2.0	20.3	42.1	21.7	
HC77-1489	40.5	29	0	1.7	24	2.0	15.4	41.6	21.5	
HW79015	51.9	1	+6	1.9	40	2.0	18.0	40.5	22.2	
HW79022	41.4	27	-3	2.0	39	2.0	20.9	43.0	20.1	
HW79054	49.2	6	+6	1.6	37	2.1	19.4	41.1	22.8	
U59218	47.0	14	-1	1.7	36	2.0	16.4	40.1	22.9	

* 128 days after planting

Strains in this test that matured 5 days or more later than Corsoy 79 should be evaluated in the Group III test if continued testing is warranted. Five strains within the maturity group, A79-138035, A79-236003, A79-237014, A79-237034, and U59218 averaged higher yields than the Group II checks. All of the A strains were resistant to phytophthora races 1 and 2 in the Indiana tests, but two were rated susceptible to phytophthora in Iowa tests. U59218 was heterogeneous in reaction to phytophthora, but had excellent lodging resistance, and was earlier in maturity than most Group II strains in the test. Two semideterminate strains A75D15 and A75D29 were early maturing, had good lodging resistance, but were not equal to the check varieties in yield. None of the determinate strains yielded well in these tests, but all of them had excellent lodging resistance and several of them were fairly tall compared to the determinate check Gnome.

PRELIMINARY TEST II, 1980

Strain	Mean 12 Tests	Iowa		Wisc.	Mich.	
		Marshall-	Ames	Arling-	Dundee	Ithaca
		town		ton		
		YIELD (bu/a)				
Century	46.7	51.3	57.7	26.5	47.4	43.4
Corsoy 79 (II)	46.9	50.2	55.8	22.1	45.2	59.8
Gnome	42.7	48.4	51.1	28.1	55.4	45.8
Hardin (I)	44.4	51.3	52.1	25.4	49.7	53.1
Pella (III)	48.8	50.9	57.1	26.6	51.2	50.1
A75D15	43.2	46.7	49.2	28.5	44.9	53.5
A73D29	44.0	47.6	50.5	25.6	48.2	58.4
A75D44	45.6	50.1	51.2	20.2	50.7	47.0
A79-138035	47.4	51.0	56.5	31.1	53.7	46.8
A79-232005	49.6	52.4	59.7	34.3	53.2	36.0
A79-232026	50.5	51.4	59.6	31.6	54.4	52.8
A79-232027	47.8	52.1	60.2	28.4	41.6	48.2
A79-235002	47.6	51.4	55.5	30.4	48.3	53.3
A79-236002	50.5	54.8	56.9	28.3	50.2	52.6
A79-236003	48.5	53.0	59.0	32.5	46.4	48.5
A79-237014	47.6	50.7	57.2	27.1	45.4	50.0
A79-237034	47.7	50.6	54.2	27.2	49.0	52.8
A79-238034	50.3	52.2	57.7	29.6	47.0	53.5
HC77-869	42.9	45.0	51.2	29.1	49.9	50.7
HC77-870	43.0	49.9	49.7	31.3	54.3	51.4
HC77-874	41.3	46.4	50.0	27.4	46.3	42.7
HC77-876	41.9	46.4	50.1	24.2	46.3	40.4
HC77-878	41.7	48.8	50.2	34.9	57.6	54.4
HC77-951	43.0	45.7	52.3	31.1	53.1	46.1
HC77-955	38.0	45.4	51.9	29.8	42.8	52.7
HC77-1489	40.5	42.6	56.5	29.2	40.4	44.8
HW79015	51.9	53.5	54.7	26.9	52.8	58.9
HW79022	41.4	44.5	42.8	29.5	43.1	49.3
HW79054	49.2	51.7	60.5	30.6	49.0	52.6
U59218	47.0	48.5	56.8	30.6	45.7	51.8
C.V. (%)		4.6	4.7	10.4	11.1	12.2
L.S.D. (5%)		4.6	5.2	6.1	11.0	12.5
Row sp. (in.)		27	27	30	30	28
Rows/plot		4	4	4	4	4
Reps		2	2	2	2	2

PRELIMINARY TEST II, 1980

N.J.	Minn.	Ill.		Ohio	Ind.	Neb.
Adelphia	Lamberton	Dekalb	Urbana	Hoytville	Lafayette	Mead
YIELD (bu/a)						
40.4	47.6	43.8	42.1	57.2	53.6	49.9
40.4	49.2	44.0	39.9	54.2	52.1	50.3
33.5	46.9	49.5	35.6	59.8	18.1	39.8
38.6	37.2	41.9	36.6	52.6	40.5	53.3
38.4	50.4	45.3	46.9	51.8	58.0	59.2
39.7	43.6	37.2	34.0	44.7	44.8	51.5
34.8	43.0	43.6	31.9	56.3	39.0	48.9
43.6	46.6	41.1	35.6	56.4	44.1	60.2
34.6	46.1	52.4	37.3	51.1	52.8	55.8
43.6	52.2	49.3	45.3	54.0	61.8	53.7
45.2	46.1	49.6	41.2	51.9	58.9	63.2
41.3	39.3	50.1	42.0	54.8	53.4	61.6
37.9	55.6	43.4	42.0	48.7	58.7	46.3
46.4	48.0	52.4	42.6	54.2	58.6	60.4
44.5	42.7	54.0	41.1	52.0	57.4	50.8
46.2	43.7	50.2	37.4	56.6	57.0	50.2
43.4	50.9	47.9	41.5	47.4	56.2	51.5
43.2	51.9	51.0	45.6	56.2	59.9	55.3
37.2	41.0	46.1	35.2	55.6	22.0	51.3
41.6	43.7	47.4	33.1	55.1	16.7	41.6
38.4	46.6	45.2	32.9	51.6	25.8	42.0
38.6	47.4	44.3	40.0	50.8	29.7	44.9
38.1	44.1	43.5	29.6	55.9	14.7	28.6
38.2	43.6	47.5	33.9	54.8	21.3	48.9
43.2	36.6	38.8	29.8	51.6	19.5	13.4
42.8	44.6	40.8	29.7	46.2	21.0	47.6
42.2	57.4	49.3	45.7	56.5	65.8	58.7
40.2	38.5	37.5	30.0	50.6	49.4	41.0
41.9	48.6	44.8	41.9	57.0	57.4	54.9
42.6	46.8	42.9	34.3	53.8	51.1	58.5
12.5	7.9	5.9	5.8	6.1	13.8	10.7
10.2	7.5	5.6	4.5	6.5	12.2	10.3
30	30	30	30	30	24	30
3	2	4	4	4	4	4
2	2	2	2	2	2	2

PRELIMINARY TEST II, 1980

Strain	Mean 11 Tests	Ia.	Wisc.	Mich.		N.J.
		Ames	Arlington	Dundee	Ithaca	Adelphia
MATURITY (date)						
Century	+3	+4	-2	+2	+4	+4
Corsoy 79 (II)*	9/21	9/14	10/7	9/24	10/7	9/13
Gnome	+6	+8	-4	+6	+4	+9
Hardin (I)	-3	-4	-10	-2	-8	-2
Pella (III)	+7	+8	0	+7	+6	+10
A75D15	-4	-4	-13	-5	-5	-4
A75D29	-3	-6	-5	-2	-6	-3
A75D44	0	0	-10	+3	-2	+5
A79-138035	+1	-1	0	+1	-1	+3
A79-232005	+5	+8	-4	+7	-1	+9
A79-232026	+6	+9	-2	+6	+3	+10
A79-232027	+6	+12	-1	+4	+5	+9
A79-235002	+6	+6	-2	+6	+4	+9
A79-236002	+6	+5	0	+6	+4	+9
A79-236003	+3	+4	-5	0	-1	+9
A79-237014	+3	+4	0	+4	+6	+4
A79-237034	+4	+4	-4	+6	+7	+8
A79-238034	+7	+8	0	+9	+7	+9
HC77-869	+5	+8	-2	+5	+3	+5
HC77-870	+4	+8	-6	+4	-1	+5
HC77-874	+6	+10	-4	+7	+3	+7
HC77-876	+6	+11	-2	+7	+4	+8
HC77-878	+2	+6	-8	+2	-4	+1
HC77-951	+3	+6	-6	+3	+1	+5
HC77-955	+1	+7	-8	-1	-2	+1
HC77-1489	0	+4	-10	-3	-2	-1
HW79015	+6	+8	+1	+4	+1	+9
HW79022	-3	-6	-11	-4	-5	0
HW79054	+6	+8	-1	+6	+5	+9
U59218	-1	+2	-10	0	-1	0
Date planted	5/16	5/13	5/20	5/14	5/22	5/29
Days to maturity*	128	124	140	133	138	107

PRELIMINARY TEST II, 1980

<u>Minn.</u>	<u>Ill.</u>		<u>Ohio</u>	<u>Ind.</u>	<u>Neb.</u>
Lamberton	Dekalb	Urbana	Hoytville	Lafayette	Mead
<u>MATURITY (date)</u>					
+1	+3	+6	+3	+3	+2
9/24	9/24	9/12	9/20	9/10	9/21
+5	+7	+5	+6	+9	+8
-2	-2	-2	-4	0	0
+6	+6	+10	+7	+8	+9
-2	-3	-4	-5	0	+1
-2	-2	-4	-4	+1	0
-2	0	-1	+2	+1	+1
0	+4	0	+3	+3	+3
+6	+7	+8	+7	+9	+4
+6	+7	+7	+5	+9	+7
+6	+6	+5	+3	+7	+8
+6	+5	+6	+6	+8	+7
+6	+6	+7	+5	+6	+7
+2	+4	+3	+4	+5	+5
+6	+3	0	+4	+4	+2
+2	+3	+3	+5	+5	+3
+6	+8	+9	+5	+9	+5
+4	+7	+7	+5	+6	+4
+4	+6	+8	+3	+8	+5
+6	+8	+7	+4	+9	+4
+6	+8	+7	+3	+8	+1
+2	+5	+5	+4	+5	+5
+4	+5	+7	+4	+6	+3
+3	0	+6	0	+3	+5
+3	-1	+3	0	+1	+1
+4	+5	+9	+5	+10	+7
0	-2	-3	-1	+2	-4
+4	+1	+10	+6	+9	+10
0	-1	-1	-1	+2	+2
5/14	5/21	5/7	5/6	5/7	5/27
133	126	128	137	126	117

PRELIMINARY TEST II, 1980

Strain	Mean 12 Tests	Iowa		Wisc.	Mich.	
		Marshall- town	Ames	Arling- ton	Dundee	Ithaca
LODGING (score)						
Century	1.9	1.8	1.9	1.5	2.8	3.8
Corsoy 79 (II)	2.8	3.2	3.0	2.5	3.0	3.3
Gnome	1.5	1.6	1.6	2.2	1.4	2.5
Hardin (I)	2.6	3.0	3.0	2.2	3.0	3.0
Pella (III)	2.0	2.0	1.6	1.8	2.6	4.0
A75D15	2.1	3.2	2.4	1.8	2.5	2.8
A75D29	1.8	2.5	1.6	1.8	2.3	2.0
A75D44	3.0	3.1	3.2	3.2	3.7	4.0
A79-138035	2.2	2.0	1.8	2.2	3.0	4.0
A79-232005	2.8	3.2	2.4	1.5	3.5	4.0
A79-232026	2.5	3.6	2.7	1.8	2.9	3.5
A79-232027	2.5	3.3	3.0	1.8	2.7	3.5
A79-235002	2.1	2.2	1.9	1.5	2.8	3.5
A79-236002	2.3	2.0	1.8	2.2	3.0	4.0
A79-236003	2.7	2.8	2.2	1.8	2.9	3.5
A79-237014	2.0	2.0	2.2	1.8	2.1	3.5
A79-237034	2.0	1.9	2.4	1.5	2.3	3.8
A79-238034	2.5	2.8	2.9	2.5	2.6	3.5
HC77-869	1.6	2.0	1.6	1.8	1.8	2.5
HC77-870	1.4	1.9	1.4	1.5	1.5	2.5
HC77-874	1.5	1.9	1.8	1.5	1.8	3.3
HC77-876	1.5	2.0	1.6	1.8	1.4	3.3
HC77-878	1.2	1.5	1.3	1.2	1.4	2.5
HC77-951	1.3	1.4	1.3	1.2	1.4	2.3
HC77-955	1.3	1.8	1.3	1.5	1.1	1.8
HC77-1489	1.7	1.6	2.1	2.8	1.6	2.5
HW79015	1.9	2.6	2.2	2.2	2.1	3.3
HW79022	2.0	2.0	1.8	2.5	2.4	3.3
HW79054	1.6	1.7	1.4	1.5	2.1	3.2
U59218	1.7	2.0	2.0	1.5	2.4	3.0

PRELIMINARY TEST II, 1980

N.J.	Minn. Lamber- ton	Ill. Dekalb	Urbana	Ohio Hoytville	Ind. Lafayette	Neb. Mead
LODGING (score)						
1.5	1.0	3.0	1.3	1.9	1.0	1.3
2.5	3.0	4.5	2.3	3.5	1.5	1.5
1.0	1.0	2.3	1.0	1.0	1.0	1.0
1.5	2.5	4.7	1.5	3.1	1.8	1.5
1.0	2.5	2.5	1.3	1.8	1.0	1.3
2.0	2.5	3.9	1.0	1.4	1.0	1.0
1.0	1.5	3.3	1.0	2.5	1.0	1.0
2.5	2.5	4.4	2.0	3.8	1.8	2.0
1.0	2.0	3.8	1.3	2.4	1.0	1.3
3.0	3.0	3.3	1.5	3.6	2.8	1.3
2.5	1.0	4.4	1.5	2.4	2.0	1.3
1.5	2.0	4.4	2.0	2.3	2.0	1.5
1.5	2.0	3.5	1.5	1.9	1.3	1.0
1.0	2.5	4.3	1.8	2.9	1.5	1.0
2.5	3.5	4.4	1.5	3.4	1.8	1.5
1.0	2.5	3.3	1.3	2.5	1.0	1.0
1.0	2.0	3.5	1.3	2.3	1.0	1.5
2.0	2.5	4.7	1.3	2.1	1.5	1.3
1.0	1.0	2.5	1.3	1.4	1.0	1.3
1.0	1.5	1.8	1.0	1.1	1.0	1.0
1.0	1.5	1.5	1.0	1.1	1.0	1.0
1.0	1.0	1.8	1.0	1.2	1.0	1.3
1.0	1.0	1.0	1.0	1.0	1.0	1.0
1.0	1.0	2.0	1.0	1.0	1.0	1.0
1.0	1.0	1.5	1.0	1.0	1.0	1.0
2.5	1.5	1.8	1.0	1.0	1.0	1.0
1.0	1.5	2.8	1.3	1.9	1.0	1.0
1.0	2.0	3.5	1.3	1.6	1.0	1.5
1.0	1.5	2.5	1.3	1.4	1.0	1.0
1.0	1.0	2.3	1.3	1.8	1.0	1.5

PRELIMINARY TEST II, 1980

Strain	Mean 12 Tests	Iowa		Wisc.	Mich.	
		Marshall- town	Ames	Arling- ton	Dundee	Ithaca
HEIGHT (inches)						
Century	37	36	44	31	41	47
Corsoy 79 (II)	40	40	46	39	43	43
Gnome	24	26	26	25	23	29
Hardin (I)	37	38	39	35	39	46
Pella (III)	41	40	44	36	44	47
A75D15	34	34	36	30	33	42
A75D29	31	34	34	28	32	40
A75D44	41	41	46	34	43	46
A79-138035	38	36	44	36	41	45
A79-232005	36	38	38	32	38	40
A79-232026	36	38	40	36	39	40
A79-232027	39	43	46	34	37	45
A79-235002	40	40	44	34	44	44
A79-236002	40	40	43	36	44	47
A79-236003	36	36	40	29	38	40
A79-237014	37	36	45	32	38	42
A79-237034	38	40	44	32	42	44
A79-238034	37	38	40	36	37	41
HC77-869	28	31	35	27	29	33
HC77-870	27	30	28	28	30	35
HC77-874	26	30	34	26	27	33
HC77-876	26	30	32	26	24	34
HC77-878	22	26	26	25	26	30
HC77-951	26	28	28	27	28	33
HC77-955	20	24	22	24	19	31
HC77-1489	24	26	28	26	22	32
HW79015	40	42	47	37	41	45
HW79022	39	38	47	40	38	44
HW79054	37	36	42	36	39	42
U59218	36	36	41	28	37	43

PRELIMINARY TEST II, 1980

<u>N.J.</u>	<u>Minn.</u>	<u>Ill.</u>		<u>Ohio</u>	<u>Ind.</u>	<u>Neb.</u>
Adelphia	Lamberton	Dekalb	Urbana	Hoytville	Lafayette	Mead
<u>HEIGHT (inches)</u>						
36	36	40	34	38	35	25
38	37	43	36	44	39	29
27	26	21	21	28	16	17
36	34	40	33	38	35	28
38	43	41	37	41	43	32
36	38	34	28	34	28	30
28	33	32	27	34	27	24
40	41	44	35	44	41	35
36	40	42	33	44	36	27
37	40	39	33	38	37	26
38	32	40	33	37	37	27
38	40	43	35	38	41	32
40	43	42	37	41	41	33
37	44	41	35	44	38	30
37	40	36	33	40	38	28
37	44	37	32	40	35	25
40	38	38	33	43	37	27
36	38	45	33	42	36	27
30	32	27	22	31	17	24
28	31	27	23	27	18	21
26	28	26	22	27	16	19
28	28	23	23	26	18	18
22	24	18	16	23	12	13
30	27	28	20	28	14	19
26	18	19	17	24	10	10
24	25	22	19	26	14	18
39	40	41	36	41	40	34
40	42	43	35	39	39	27
36	34	38	32	37	36	31
34	36	36	32	38	35	31

PRELIMINARY TEST II, 1980

		<u>Ia.</u>	<u>Wisc.</u>	<u>N.J.</u>	<u>Minn.</u>
Strain	Mean 9 Tests	Ames	Arlington	Adelphia	Lamberton
QUALITY (score)					
Century	2.1	2.0	2.5	1.0	2.0
Corsoy 79 (II)	2.1	2.2	3.5	1.0	2.0
Gnome	1.7	1.3	3.5	1.0	1.5
Hardin (I)	2.3	1.9	4.0	1.0	2.5
Pella (III)	2.1	1.3	3.5	1.0	3.0
A75D15	2.5	2.1	4.5	1.0	4.0
A75D29	2.0	1.9	4.0	1.0	2.5
A75D44	2.2	1.9	3.5	1.0	2.5
A79-138035	2.0	1.7	2.5	1.0	2.5
A79-232005	2.2	2.0	3.0	1.0	3.0
A79-232026	2.2	2.0	4.0	1.0	3.0
A79-232027	2.4	2.2	4.0	1.0	3.0
A79-235002	2.1	1.9	2.0	1.0	3.0
A79-236002	2.3	1.6	3.0	1.0	2.5
A79-236003	1.7	1.3	2.0	1.0	2.0
A79-237014	2.1	1.8	4.0	1.0	2.0
A79-237034	1.8	1.5	2.0	1.5	1.5
A79-238034	2.0	1.8	3.0	1.0	2.0
HC77-869	1.7	1.3	3.0	1.0	2.0
HC77-870	1.7	1.6	2.5	1.0	2.0
HC77-874	1.6	1.4	3.0	1.0	2.0
HC77-876	1.7	1.8	4.0	1.0	2.0
HC77-878	1.7	1.4	2.0	1.0	1.5
HC77-951	1.8	1.4	3.5	1.0	2.0
HC77-955	2.0	2.1	2.0	1.0	2.0
HC77-1489	2.0	1.4	3.0	1.0	1.5
HW79015	2.0	1.5	2.5	1.0	2.0
HW79022	2.0	1.6	2.5	1.0	2.5
HW79054	2.1	1.4	2.5	1.0	2.0
U59218	2.0	2.1	4.0	1.0	1.5

PRELIMINARY TEST II, 1980

<u>Ill.</u>		<u>Ohio</u>	<u>Ind.</u>	<u>Neb.</u>
<u>Dekalb</u>	<u>Urbana</u>	<u>Hoytville</u>	<u>Lafayette</u>	<u>Mead</u>
<u>QUALITY (score)</u>				
1.2	3.6	2.1	1.5	2.8
1.3	2.5	2.2	1.5	2.5
1.6	1.5	1.5	1.5	2.0
1.4	3.5	2.5	1.5	2.0
1.4	2.7	2.0	1.5	2.5
1.8	2.8	2.3	1.5	2.8
1.6	2.0	1.9	1.5	2.0
2.0	2.3	1.7	2.0	2.5
2.0	2.0	2.5	1.5	2.3
2.0	2.5	1.6	1.5	2.8
2.7	2.0	1.4	1.5	2.5
2.7	2.4	2.8	1.5	2.0
1.7	2.3	2.4	1.5	2.8
1.9	2.7	3.1	1.5	3.0
1.6	2.0	2.5	1.5	1.8
1.7	2.3	1.8	1.5	2.8
1.4	2.4	2.1	1.5	2.5
2.2	1.9	2.5	1.5	2.3
1.3	1.5	1.7	2.0	1.8
1.4	1.8	1.6	1.5	1.8
1.3	1.4	1.5	1.0	1.8
1.6	1.5	1.3	1.0	1.5
1.4	1.9	2.5	1.5	2.3
1.4	1.6	1.8	1.5	1.8
1.5	2.3	1.3	3.5	2.5
1.2	2.2	2.4	3.0	2.0
1.3	2.9	2.6	1.5	2.8
1.7	2.0	2.7	1.5	2.3
1.3	4.0	1.6	2.0	3.0
1.4	2.9	1.3	1.5	2.3

PRELIMINARY TEST II, 1980

		Ia.	Wisc.	N.J.	Minn.
Strain	Mean 9 Tests	Ames	Arlington	Adelphia	Lamberton
		SIZE (g/100)			
Century	18.2	20.0	15.7	11.0	19.9
Corsoy 79 (II)	14.9	17.0	13.3	10.0	17.1
Cnome	15.8	15.4	13.7	11.0	16.9
Hardin (I)	14.8	15.9	13.0	11.0	16.3
Pella (III)	19.1	21.0	16.9	16.0	20.0
A75D15	16.9	17.4	16.1	13.0	19.2
A75D29	15.4	16.4	16.1	11.0	16.4
A75D44	16.7	19.0	13.8	13.0	18.0
A79-138035	17.2	18.1	15.7	12.0	18.6
A79-232005	15.9	16.8	14.8	12.0	18.4
A79-232026	15.9	16.8	15.2	14.0	16.4
A79-232027	17.4	19.0	16.3	14.0	18.7
A79-235002	17.9	20.2	16.2	12.0	19.7
A79-236002	17.9	19.0	15.8	15.0	19.6
A79-236003	16.2	17.4	14.2	12.0	18.8
A79-237014	19.9	22.0	17.6	16.0	20.8
A79-237034	22.2	24.4	19.8	16.0	25.2
A79-238034	14.8	16.4	13.9	10.0	15.4
HC77-869	14.0	15.0	12.8	10.0	13.8
HC77-870	14.3	15.5	12.4	9.0	14.9
HC77-874	16.7	15.6	15.5	12.0	17.9
HC77-876	14.9	15.8	13.8	10.0	15.4
HC77-878	16.5	16.4	14.7	13.0	17.7
HC77-951	14.7	16.0	13.1	10.0	14.9
HC77-955	20.3	21.7	17.1	17.0	20.7
HC77-1489	15.4	16.0	13.2	11.0	16.8
HW79015	18.0	18.9	16.3	15.0	19.0
HW79022	20.9	21.4	19.5	17.0	23.9
HW79054	19.4	20.7	19.9	16.0	20.9
U59218	16.4	18.2	14.6	14.0	19.0

PRELIMINARY TEST II, 1980

Ill		Ohio	Ind.	Neb.
Dekalb	Urbana	Hoytville	Lafayette	Mead
QUALITY (score)				
17.4	19.0	18.6	18.1	24.2
13.6	14.3	16.3	14.8	17.3
16.5	14.8	16.7	16.7	20.6
13.7	14.1	15.5	14.1	19.4
19.1	20.6	19.2	18.8	20.5
15.6	14.8	17.7	17.0	21.5
15.0	14.5	15.2	14.7	19.2
15.9	16.0	17.5	16.4	20.3
18.4	16.2	17.2	17.4	21.5
15.7	15.2	15.9	16.6	17.7
16.7	14.8	16.1	16.2	16.5
17.8	15.9	17.8	17.6	19.1
17.9	18.8	17.6	19.6	19.4
18.1	18.0	16.8	17.8	20.7
16.4	14.8	15.9	17.9	18.3
19.8	18.4	19.7	20.2	24.6
22.4	20.6	23.2	23.0	25.1
15.3	14.6	15.1	15.4	16.8
13.7	14.0	14.3	15.1	16.9
14.6	14.7	14.4	15.5	17.5
18.2	16.2	18.0	17.0	19.6
15.1	15.0	15.8	15.3	17.7
16.7	16.0	17.4	17.0	20.0
14.4	15.0	15.9	14.9	18.2
19.9	21.2	20.6	18.2	26.0
13.7	16.0	15.4	15.7	21.2
18.5	17.9	17.5	18.6	19.7
20.8	19.4	21.3	21.3	23.5
18.5	19.9	18.0	19.7	21.0
15.4	15.2	15.5	17.2	18.3

PRELIMINARY TEST II, 1980

		<u>Iowa</u>	<u>Ill.</u>	<u>Ohio</u>	<u>Ind.</u>	<u>Neb.</u>
Strain	Mean 5 Tests	Ames	Urbana	Hoyt- ville	Lafay- ette	Mead
				<u>PROTEIN (%)</u>		
Century	43.4	42.4	42.4	44.4	43.1	44.9
Corsoy 79 (II)	41.4	40.7	40.5	41.7	42.2	41.9
Gnome	42.2	41.7	40.2	43.2	43.2	42.7
Hardin (I)	40.8	38.7	41.6	40.7	41.4	41.5
Pella (III)	39.7	37.3	39.4	40.8	41.0	40.2
A75D15	40.6	40.5	39.2	40.4	41.8	41.3
A75D29	41.9	40.8	41.6	42.2	42.2	42.5
A75D44	41.0	38.7	41.8	41.6	42.3	40.6
A79-138035	40.5	38.6	38.8	40.9	42.3	41.8
A79-232005	41.7	40.7	41.3	43.8	42.3	40.4
A79-232026	40.0	39.1	38.2	41.9	40.0	40.9
A79-232027	39.7	40.1	37.0	41.2	41.1	39.0
A79-235002	41.7	40.3	40.6	43.5	42.3	41.7
A79-236002	40.4	38.6	39.1	42.1	41.2	41.1
A79-236003	40.6	39.7	39.3	42.8	41.5	39.6
A79-237014	43.1	42.7	42.2	42.7	43.4	44.7
A79-237034	43.2	42.0	43.4	44.1	42.8	43.5
A79-238034	40.9	41.8	37.8	42.3	41.9	40.9
HC77-869	37.7	37.5	37.1	37.6	38.8	37.7
HC77-870	38.0	36.8	37.0	38.7	39.8	37.5
HC77-874	43.4	42.7	42.2	44.2	44.2	43.5
HC77-876	43.7	43.8	43.6	43.6	44.3	43.0
HC77-878	40.9	41.1	40.2	41.8	40.4	40.9
HC77-951	37.7	36.2	36.7	39.2	37.9	38.6
HC77-955	42.1	40.2	41.5	42.4	42.2	44.1
HC77-1489	41.6	40.0	42.3	41.3	43.2	41.3
HW79015	40.5	38.3	40.4	41.7	42.3	39.9
HW79022	43.0	41.6	41.6	42.7	45.3	43.6
HW79054	41.1	40.3	42.1	40.1	41.2	41.8
U59218	40.1	38.9	38.5	41.3	41.4	40.5

PRELIMINARY TEST II, 1980

Mean 5 Tests	<u>Ia.</u> Ames	<u>Ill.</u> Urbana	<u>Ohio</u> Hoytville	<u>Ind.</u> Lafayette	<u>Neb.</u> Mead
OIL (%)					
21.5	21.4	21.3	22.0	21.5	21.2
23.0	22.5	23.3	23.7	22.9	22.5
22.3	22.3	22.3	22.6	22.5	21.7
22.9	22.4	23.6	23.0	22.8	22.7
22.1	20.7	22.6	22.8	22.5	21.7
22.7	22.0	23.8	23.8	21.6	22.2
21.2	20.2	21.0	22.4	20.8	21.7
21.9	21.9	21.8	23.4	21.0	21.6
22.5	22.4	23.2	23.7	22.1	21.0
22.2	22.4	22.3	22.3	22.0	22.2
22.7	21.7	23.5	23.1	23.0	22.2
23.3	23.1	24.4	23.6	22.3	23.1
21.6	22.2	21.3	22.1	21.0	21.3
22.5	22.8	22.9	24.0	21.8	20.8
22.8	22.3	23.4	24.1	22.2	22.1
21.2	20.6	21.4	21.8	21.4	20.6
22.5	22.0	23.2	23.1	22.3	21.9
22.5	22.7	23.5	22.6	21.7	21.9
22.9	22.9	22.3	24.6	21.9	22.6
23.2	23.6	23.2	23.9	22.2	23.0
21.7	21.1	21.8	23.0	21.2	21.3
21.2	20.7	21.1	21.6	21.3	21.5
22.0	21.7	21.9	22.4	22.3	21.8
23.2	23.8	23.7	23.7	22.1	22.7
21.7	22.3	21.6	22.2	21.0	21.5
21.5	20.9	21.2	22.7	21.4	21.3
22.2	22.0	22.4	22.8	22.4	21.4
20.1	19.7	20.0	21.4	20.0	19.2
22.8	22.1	22.3	23.5	23.3	22.6
22.9	22.1	24.2	23.5	22.4	22.5

UNIFORM TEST III, 1980

Strain	Parentage	Previous Testing*	Generation Compositd
BSR 302	(Beeson x AP68-1016) x L15 x Calland)	2	F ₄
Century (II)	Calland x Bonus	1	F ₆
Cumberland (III)	Corsoy x Williams	4	F ₄
Pella	L66L-137 x Calland	4	F ₄
Union (IV)	Williams ⁵ x SL12 (Wayne <u>Rpm</u> <u>Rps</u>)	2	F ₃
Williams 79	Williams ⁶ x Lee 68	3	10F ₃
A78-227012	Pride B-216 x Ax901-40-2	PII	F ₄
A78-321011	Pride B-216 x Agripro 25	PII	F ₄
A78-322024	Williams x Sloan	PIII	F ₄
A78-324002	A72-512 x Pride B-216	PIII	F ₄
A78-325028	A72-512 x Williams	PIV	F ₄
A78-326032	BSR intermating of 9 parents	PIII	F ₄
HC76-4030	L72U-2567 x Essex	PIII	F ₅
HW74-3384 Sprite	Williams x Ransom	2	F ₅
HW74-3385	Williams x Ransom	2	F ₅
L24A	Williams ⁷ x Kingwa	--	4F ₃
L25A	Williams ⁶ x PI 96.983	--	4F ₃
L26	Williams ⁷ x Harrel	--	2F ₃
L75-8121	Williams x L70-2283	PIII	F ₄
L77-994	Williams ² x PI 88.788	--	F ₃
U36276	Bonus x Wayne	1	F ₄
U57073	Wayne x Cutler	PIII	F ₄

* Number of years in this test or name of 1979 test

UNIFORM TEST III, 1980

Descriptive and Other Data

Strain	Descriptive Code	Chlorosis		Hypocotyl	Shattering	
		Score		Score	Manhattan	Texas
		Ames	Lamberton	Ames	2 Weeks	Lubbock
BSR 302	PTBr DYB1	3.5	4.0	1	2	5
Century (II)	PTBr DYB1	3.0	3.5	5	1	3
Cumberland (III)	PGBr SYIb	3.7	5.0	5	2	3
Pella	PTTn DYB1	3.2	4.0	1	3	2
Union (IV)	WTTn SYB1	3.3	3.5	4	2	2
Williams 79	WTTn SYB1	3.5	2.5	4	2	2
A78-227012	P+WGBr DYIb	3.0	4.5	1	2	3
A78-321011	WGBr SYBf	4.0	5.0	1	2	3
A78-322024	WTTn SYBr	1.7	2.0	5	2	3
A78-324002	WGBr DYBf	4.2	4.5	5	3	5
A78-325028	WTBr SYBr	3.2	4.5	5	2	3
A78-326032	PGBr DYIb	3.3	4.0	5	2	3
HC76-4030	PTTn DYB1	2.2	1.5	1	1	2
HW74-3384 Sprite	WTTn DYB1	3.2	4.0	1	1	1
HW74-3385	WTTn DYB1	3.0	4.0	1	1	2
L24A	WTTn SYB1	3.0	3.5	3	2	2
L25A	WTTn SYB1	3.0	4.0	1	1	3
L26	WTTn SYB1	2.3	4.0	5	2	2
L75-8121	P+WGBr+Tn					
	SYBf+Ib	3.5	5.0	1	1	2
L77-994	WTTn SYB1	3.8	4.0	2	3	5
U36276	PTBr DYB1	2.5	4.5	5	3	4
U57073	PTBr SYB1	3.5	4.5	5	2	3

UNIFORM TEST III, 1980

Disease Data

Strain	BSR		GERM	SMV	PSB	PS	PR	PR	Race 1
	Laf.	Ames	*				Vickery	Laf.	Ames
	Ind.	Ia.	Lafayette,	Indiana			Ohio	Ind.	Ia.
	n	n	d	a	d	a	a	a	a
	%	Reac.	%	Score	%	Score	Reaction		
BSR 302	40	R	75	3E	3	4S	1.8	R	R
Century (II)	60	S	68	4M	20	3E	2.5	R	R
Cumberland (III)	0	S	85	3M	10	3E	2.1	S	S
Pella	0	S	70	4M	24	3E	2.2	R	R
Union (IV)	100	S	77	3E	16	3E	1.8	R	R
Williams 79	80	S	76	2M	13	4E	1.6	R	R
A78-227012	0	R	69	3E	18	5E	2.3	S	S
A78-321011	20	S	64	3E	21	5E	2.8	S	S
A78-322024	60	S	71	5E	12	2E	2.2	-	S
A78-324002	100	S	85	3M	8	5E	1.9	H	S
A78-325028	100	S	96	4E	0	4E	1.9	S	R
A78-326032	40	R	79	2S	10	3E	2.1	S	S
HC76-4030	0	S	93	1	1	2M	3.0	S	S
HW74-3384 Sprite	60	S	89	1	9	2E	3.1	S	S
HW74-3385	80	S	89	1	6	2M	2.9	S	S
L24A	80	S	77	3M	14	3E	2.0	R	R
L25A	40	S	91	1	3	3E	1.8	S	S
L26	80	S	79	2M	12	4E	1.5	R	R
L75-8121	20	S	71	2M	17	4S	2.1	R	R
L77-994	20	S	80	3M	11	3E	1.8	S	S
U36276	20	S	72	3E	20	5S	2.4	H	S
U57073	40	S	77	3M	12	5S	2.3	S	S

* Petri dish germination on potato dextrose agar

UNIFORM TEST III, 1980

Regional Summary

Strain	Yield	Rank	Matu- rity	Lodg- ing	Height	Seed Quality	Seed Size	Composition	
No. of Tests	23	23	20	22	23	21	21	5	5
	bu/a	No.	Date	Score	In.	Score	g/100	%	%
BSR 302	41.1	15	+1	2.4	41	2.3	16.5	42.7	19.8
Century (II)	41.4	14	-6	1.6	33	2.4	17.4	42.7	20.8
Cumberland (III)	45.7	1	9-23*	1.9	36	2.1	17.5	41.2	22.3
Pella	42.8	9	-2	1.5	36	2.2	18.0	40.3	21.9
Union (IV)	40.8	18	+5	2.2	43	1.9	17.6	43.3	20.5
Williams 79	43.1	8	+1	1.8	40	1.9	16.6	41.7	20.6
A78-227012	41.1	15	-7	1.9	30	2.3	14.5	42.0	20.5
A78-321011	42.7	10	-5	1.6	33	2.0	15.1	41.6	20.5
A78-322024	45.4	3	-1	2.2	41	2.5	17.4	43.0	21.7
A78-324002	45.6	2	+2	2.2	42	1.9	14.1	42.4	20.3
A78-325028	44.5	4	+3	2.4	42	1.9	16.4	44.0	20.2
A78-326032	41.0	17	+3	2.5	37	2.3	15.3	42.3	20.1
HC76-4030	37.4	22	+3	1.3	20	1.8	15.3	43.9	20.5
HW74-3384 Sprite	39.9	19	+1	1.3	21	1.6	17.4	41.1	22.6
HW74-3385 Hobbit	41.7	13	+1	1.2	21	1.6	15.6	40.3	22.5
L24A Williams 82	43.3	5	+2	1.7	39	1.9	16.3	41.9	21.2
L25A	43.3	5	+2	2.0	38	2.4 1-8	16.4	41.3	21.2
L26	43.2	7	+2	1.7	40	1.9	16.5	42.5	20.8
L75-8121	39.0	20	+3	1.9	40	1.9	15.2	40.4	20.8
L77-994	38.5	21	+1	1.9	39	2.0	15.0	42.9	20.3
U36276	42.1	11	-3	1.4	31	2.3	14.6	42.3	20.6
U57073	41.9	12	+4	1.8	39	2.1	16.5	43.5	19.8

* 126 days after planting

Cumberland has had the highest mean yield in the 1-year, 2-year, and 3-year analyses of this test. L24A, resistant to races 1-9 of phytophthora rot, L25A with resistance to SMV, and L26 with resistance to races 1, 3, 4, 5, 6, 7, 8, and 9 of phytophthora rot are all backcross derivatives of Williams and were very similar to Williams 79 in performance. The cyst nematode races 3 and 4 resistant L77-994 was somewhat lower in yield than the check varieties. Two strains, A78-227012 and A78-321011, performed well in Preliminary II in 1979, but because of their late maturity, were moved to UT III for 1980 evaluation. These strains matured early for the Group III test in 1980, so a separate comparison was made of these strains with Century and Pella in a table at the bottom of page 116. Both strains were intermediate in yield between Century and Pella, had excellent lodging resistance but were susceptible to phytophthora rot.

UNIFORM TEST III, 1980

Regional Summary

Strain	bu/a	Yield	Rank	Matu- rity	Lodg- ing	Height	Seed Quality	Size Seed g/100	Composition Protein %	Oil %
No.	Date	Score	In.	Score	g/100	%	%			
1979-1980, 2-year mean										
No. of Tests	47	47	41	46	47	43	42	10	10	
BSR 302	43.1	8	+2.3	2.8	42	2.4	17.0	41.8	19.8	
Century (II)	42.5	9	-6.1	1.7	35	2.3	18.0	42.4	20.2	
✓ Cumberland (III)	47.4	1	9-24.5*	2.1	37	2.1	18.1	40.8	21.8	
✓ Pella	45.6	2	-2.4	1.7	38	2.3	19.0	39.8	21.4	
Union (IV)	43.3	6	+5.1	2.4	44	1.9	18.0	42.4	20.2	
Williams 79	44.6	4	+1.4	2.0	41	2.0	17.1	41.4	20.6	
HW74-3384 Sprite	43.6	5	+0.3	1.4	22	1.7	17.8	40.5	22.0	
✓ HW74-3385 Hobbit	45.4	3	+0.8	1.3	23	1.8	16.0	40.4	21.6	
U36276	43.2	7	-2.8	1.6	32	2.2	15.0	42.8	19.9	

* 127 days after planting

1978-1980, 3-year mean										
No. of Tests	73	73	64	69	73	68	62	25	25	
BSR 302	42.3	7	+2.0	2.9	42	2.3	16.8	42.2	19.8	
Cumberland (III)	45.6	1	9-23.8*	2.1	36	2.2	18.0	41.4	21.7	
Pella	44.5	3	-2.1	1.8	37	2.2	19.0	40.3	21.3	
Union (IV)	43.1	6	+4.9	2.4	43	1.8	18.1	42.5	20.4	
Williams 79	43.9	4	+1.3	2.0	40	1.8	16.9	42.1	20.6	
HW74-3384 Sprite	43.5	5	0.0	1.4	22	1.8	17.6	40.5	22.4	
HW74-3385 Hobbit	45.3	2	+0.2	1.3	22	1.7	15.9	40.1	22.0	

* 125 days after planting

1979 PII and 1980 UTIII mean										
No. of Tests	34	34	30	33	34	31	31	10	10	
Century	44.6	4	9-20.5	1.8	35	2.2	18.1	41.8	20.4	
Pella	46.1	1	9-24.2	1.9	38	2.1	19.1	39.8	21.4	
A78-227012	44.9	3	9-20.2	2.1	32	2.2	15.1	41.8	20.2	
A78-321011	45.8	2	9-21.3	1.9	34	2.1	16.0	41.6	20.2	

UNIFORM TEST III, 1980

Strain	Mean 23 Tests	N.J.	Penn.	Md.	Ohio		S. Charles- ton	Ky.
		Adel- phia	Landis- ville	Clarks- ville	Hoyt- ville	Wooster		Lexing- ton
YIELD (bu/a)								
BSR 302	41.1	34.9	36.7	25.1	52.0	45.3	59.2	34.3
Century (II)	41.4	42.8	36.7	32.2	60.8	45.4	55.6	39.5
Cumberland (III)	45.7	40.1	39.1	35.3	57.2	50.1	62.5	39.8
Pella	42.8	41.4	33.1	32.1	57.0	49.7	61.7	37.0
Union (IV)	40.8	37.2	40.7	26.7	48.9	42.1	50.2	35.3
Williams 79	43.1	39.6	36.4	28.6	52.4	42.9	57.6	37.9
A78-227012	41.1	45.9	31.4	21.5	56.2	50.3	64.7	36.5
A78-321011	42.7	46.8	38.2	34.8	53.7	46.6	64.6	34.9
A78-322024	45.4	38.8	37.4	36.7	52.4	48.3	60.7	39.5
A78-324002	45.6	40.4	35.6	39.4	55.1	49.2	63.2	40.0
A78-325028	44.5	39.6	37.7	35.6	53.4	44.0	58.8	37.1
A78-326032	41.0	40.4	38.8	32.9	49.7	44.8	58.4	35.9
HC76-4030	37.4	47.4	34.9	33.2	59.8	52.6	60.1	33.2
HW74-3384 Sprite	39.9	45.2	47.0	30.6	58.4	53.2	58.2	40.1
HW74-3385	41.7	47.9	37.5	29.9	58.8	52.0	61.8	39.9
L24A	43.3	37.1	30.5	29.4	52.5	44.4	57.1	36.7
L25A	43.3	41.4	33.0	33.0	48.4	46.1	55.9	36.0
L26	43.2	35.8	35.5	30.8	48.4	47.3	57.9	35.9
L75-8121	39.0	36.4	33.0	28.8	45.6	40.2	49.7	34.5
L77-994	38.5	37.1	33.4	34.4	45.4	39.5	43.5	36.3
U36276	42.1	40.9	30.9	32.0	51.1	47.2	53.9	34.0
U57073	41.9	39.5	42.5	37.3	48.6	44.1	54.0	45.7
C.V. (%)		6.6	16.2	12.5	7.3	6.0	7.7	5.7
L.S.D. (5%)		5.4	NS	5.5	6.4	4.6	7.3	1.2
Row sp (in.)		30	30	30	30	30	30	30
Rows/plot		3	4	4	4	4	4	4
Reps		4	3	3	3	3	3	3

UNIFORM TEST III, 1980

Strain	Ind.			Illinois				
	Lafayette	Greenfield	Sullivan	Urbana	Girard	Brownstown	Belleville	Eldorado
YIELD (bu/a)								
BSR 302	56.6	48.1	49.5	50.9	38.1	28.0	37.4	46.5
Century (II)	53.3	38.0	52.3	48.3	38.2	32.8	35.9	45.1
Cumberland (III)	58.1	49.9	55.7	50.1	42.1	37.1	47.6	51.4
Pella	49.8	47.1	--	52.8	37.8	37.9	41.9	46.3
Union (IV)	58.1	46.7	45.6	50.6	38.7	35.6	47.7	49.2
Williams 79	58.5	48.6	51.7	50.3	42.6	38.3	45.1	46.9
A78-227012	59.2	39.0	39.2	49.2	28.5	34.5	40.5	46.6
A78-321011	52.0	47.0	44.0	50.8	36.8	35.0	38.6	48.6
A78-322024	60.4	53.4	46.9	52.1	43.9	38.3	47.9	52.2
A78-324002	58.5	50.1	48.8	52.5	47.3	40.9	44.8	51.9
A78-325028	62.2	49.0	58.2	54.0	42.5	44.1	42.0	53.1
A78-326032	56.2	46.4	40.3	51.9	34.4	31.0	35.2	44.5
HC76-4030	32.5	37.7	25.0	47.7	37.4	30.1	45.1	43.1
HW74-3384 Sprite	33.5	38.9	30.2	49.1	32.9	28.4	48.7	46.0
HW74-3385	36.3	39.6	44.5	52.8	38.6	22.0	42.9	48.7
L24A	64.7	51.6	52.0	52.4	41.1	40.5	45.8	47.6
L25A	63.7	45.0	48.2	56.6	39.1	37.7	46.9	47.5
L26	61.1	46.9	54.9	53.1	39.2	42.8	48.5	48.2
L75-8121	51.7	39.2	48.2	46.1	37.3	36.2	39.6	46.6
L77-994	46.4	40.3	40.6	48.9	37.9	35.4	42.9	45.4
U36276	53.6	48.2	48.4	52.2	37.5	35.4	42.2	50.0
U57073	56.7	45.8	38.5	48.8	40.1	36.9	48.8	48.8
C.V. (%)	7.6	13.5	17.5	4.5	7.5	9.2	6.6	5.1
L.S.D. (5%)	6.3	10.0	12.8	3.8	4.8	5.3	4.7	4.1
Row sp (in.)	24	30	30	30	30	30	30	30
Rows/plot	4	3	3	4	4	4	4	4
Reps	3	3	3	3	3	3	3	3

UNIFORM TEST III, 1980

Ill.	Iowa		Mo.	S.D. Elk	Kansas		Neb.
Pontiac	Ottumwa	Stuart	Novelty	Point	Manhattan	Powhattan	Mead
YIELD (bu/a)							
30.9	51.0	56.8	36.5	30.6	26.9	20.3	49.2
37.7	48.4	53.5	32.3	36.1	18.9	17.6	51.4
35.3	57.2	57.5	32.1	40.8	32.2	20.0	59.0
31.4	53.7	54.6	32.0	37.4	30.5	24.4	52.2
32.5	50.5	49.4	30.4	31.7	29.3	19.1	42.7
33.0	53.0	54.5	38.3	33.5	29.4	23.0	49.2
28.1	49.1	57.2	36.0	30.8	23.7	19.0	58.0
32.0	48.7	54.0	38.6	38.3	26.6	18.3	52.9
33.5	54.8	58.8	37.7	37.5	31.3	25.0	57.0
33.3	57.9	55.0	35.9	34.8	29.9	26.2	58.6
33.2	50.3	51.7	36.3	36.7	33.8	25.2	44.3
33.4	54.4	56.7	35.7	29.0	22.9	21.8	47.2
28.3	42.4	44.9	16.4	32.0	15.8	15.4	45.7
34.4	48.5	50.8	24.6	27.6	21.9	19.2	49.4
35.8	47.1	54.2	28.6	29.9	32.9	21.8	55.1
34.6	53.6	53.9	34.5	34.3	29.1	22.4	49.5
35.1	50.1	53.3	37.4	32.2	31.9	23.4	53.5
30.8	52.2	54.0	--	35.6	31.4	21.9	38.4
30.8	50.7	51.9	26.7	28.2	27.6	22.3	46.1
30.4	47.2	47.4	31.1	32.8	27.5	20.9	40.3
30.0	52.0	55.8	37.0	34.6	24.5	21.3	55.9
30.7	51.8	50.4	32.9	26.4	30.7	22.0	43.7
8.3	5.4	4.4	18.1	14.4	12.2	21.4	12.1
4.4	4.0	3.4	8.5	7.8	5.6	8.4	9.9
30	27	27	30	30	30	30	30
4	4	4	2	4	4	4	4
3	4	4	4	3	3	3	3

UNIFORM TEST III, 1980

Strain	Mean 20 Tests	N.J.	Penn.	Md.	Ohio		S. Charles- ton	Ky.
		Adel- phia	Landis- ville	Clarks- ville	Hoyt- ville	Wooster		Lexing- ton
MATURITY (date)								
BSR 320	+1	+3	+3	-1	+1	+4	0	+5
Century (II)	-6	-7	-3	-1	-4	-6	-10	-2
Cumberland (III)	*9/23	9/23	9/24	9/23	9/28	9/29	9/26	9/18
Pella	-2	-1	+1	-1	-3	-2	-7	0
Union (IV)	+5	+6	+8	-1	+4	+5	+1	+12
Williams 79	+1	+1	+2	-1	+1	+2	-1	+5
A78-227012	-7	-3	-5	-5	-4	-4	-10	-2
A78-321011	-5	0	-2	-2	-5	-3	-8	-2
A78-322024	-1	+1	-1	0	-3	-1	-5	+5
A78-324002	+2	+4	-1	+1	+1	+3	0	+5
A78-325028	+3	+5	+1	+1	+1	+3	-1	+9
A78-326032	+3	+9	+3	-1	+1	+5	+1	+9
HC76-4030	+3	+3	-1	-1	0	0	+2	+9
HW74-3384 Sprite	+1	0	-1	-1	-1	-1	0	0
HW74-3385	+1	+3	-1	0	-1	-3	0	-2
L24A	+2	+3	+2	-1	+1	+3	0	+5
L25A	+2	+4	+1	-1	0	+3	-1	+5
L26	+2	0	+2	0	0	+3	0	+9
L75-8121	+3	+4	+2	+6	+3	+5	+1	+9
L77-994	+1	+3	+3	-1	+1	+4	-1	+9
U36276	-3	-1	-1	-1	-3	-2	-3	0
U57073	+4	+6	+5	+6	0	+4	-2	+9
Date planted	5/18	5/28	6/5	6/11	5/6	5/16	4/30	6/8
*Days to maturity	126	118	111	104	145	136	149	102

UNIFORM TEST III, 1980

Strain	Ind.			Illinois				
	Lafayette	Greenfield	Sullivan	Urbana	Girard	Brownstown	Belleville	Eldorado
	MATURITY (date)							
BSR 302	+2	0	-3	+2	0	+1	-2	-1
Century (II)	-10	-5	-14	-3	-11	0	-11	-11
Cumberland (III)	*9/22	9/25	9/20	9/22	9/16	9/14	9/18	9/14
Pella	-3	-3	--	+1	-6	+2	-4	-4
Union (IV)	+5	+2	+1	+7	+4	+8	+7	+4
Williams 79	+2	+1	-2	+1	+1	+1	-1	0
A78-227012	-8	-5	-14	-7	-12	-1	-12	-11
A78-321011	-6	-4	-13	-3	-9	-1	-10	-7
A78-322024	-2	-3	-4	0	-2	+2	-2	-1
A78-324002	+3	+1	0	+3	+1	+5	+1	+1
A78-325028	+4	+2	-2	+4	+4	+6	+2	+3
A78-326032	+2	+2	0	+3	+3	+3	+1	+2
HC76-4030	+4	-2	+3	+7	+5	+1	+3	+2
HW74-3384 Sprite	+3	-2	-2	+4	+3	+2	+3	0
HW74-3385	+2	-2	-1	+6	+1	0	0	-1
L24A	+5	+3	-3	+5	+3	+4	+1	+2
L25A	+3	+1	-1	+4	+2	+2	0	+1
L26	+3	0	+4	+2	+2	+4	+1	+2
L75-8121	+3	+2	-1	+5	+3	+2	+1	0
L77-994	+2	-1	-2	+1	+1	+4	0	+1
U36276	-5	-4	-9	+1	-5	+1	-6	-3
U57073	+3	+2	-1	+7	+3	+6	+3	+2
Date planted	5/7	5/29	5/5	5/7	5/6	5/15	5/27	5/21
Days to maturity*	138	119	138	138	133	122	114	116

UNIFORM TEST III, 1980

Ill.	Iowa		Mo.	S.D.	Kansas		Neb.
Pontiac	Ottumwa	Stuart	Novelty	Elk Point	Manhattan	Powhattan	Mead
MATURITY (date)							
+1		-1		+1	0		+2
-1		-9		-5	-4		-7
9/21		9/29		10/1	9/21		9/30
+1		-5		-2	-3		-1
+4		+2		+6	+8		+4
+1		-1		+1	+3		+3
-4		-8		-9	-5		-5
-2		-8		-5	-2		-2
+1		-2		-3	+6		0
+1		+2		0	+9		+3
+2		+1		0	+7		+3
+2		+2		+1	+6		+1
+1		+1		+2	+11		+2
+1		+1		0	+12		0
+2		+1		0	+5		+3
+2		0		+2	+7		+3
+3		0		+3	+8		+3
+1		0		+2	+6		+3
+1		+1		0	+6		+3
+1		0		+1	+1		+2
0		-5		-2	-3		-1
+6		+1		+3	+5		+3
5/27	5/2	5/10	5/15	5/21	5/19	5/22	5/27
117	-	142	-	133	125	-	126

UNIFORM TEST III, 1980

Strain	Mean 22 Tests	N.J.	Penn.	Ohio			S. Charles- ton	Ky.
		Adel- phia	Landis- ville	Clarks- ville	Hoyt- ville	Wooster		Lexing- ton
LODGING (score)								
BSR 302	2.4	3.0	1.0	1.2	3.2	4.6	3.5	2.2
Century (II)	1.6	1.8	1.0	1.0	1.8	2.7	2.3	1.0
Cumberland (III)	1.9	3.0	1.0	1.0	2.0	2.8	2.3	1.0
Pella	1.5	1.0	1.0	1.0	1.9	2.5	1.7	1.0
Union (IV)	2.2	3.0	2.0	1.7	2.9	3.6	2.7	1.5
Williams 79	1.8	2.2	1.0	1.2	2.5	3.0	2.0	1.0
A78-227012	1.9	2.5	1.0	1.0	2.1	2.4	2.2	1.2
A78-321011	1.6	1.8	1.0	1.0	1.7	2.7	2.0	1.0
A78-322024	2.2	3.8	2.0	1.3	2.7	3.8	3.0	1.3
A78-324002	2.2	3.2	1.0	1.0	2.7	3.0	2.8	1.8
A78-325028	2.4	3.0	1.0	1.8	3.3	3.7	2.8	2.3
A78-326032	2.5	3.0	1.0	1.7	3.1	4.2	3.5	2.2
HC76-4030	1.3	1.0	1.0	1.2	1.0	1.7	1.7	1.0
HW74-3384 Sprite	1.3	2.2	1.0	1.3	1.1	2.0	1.7	1.0
HW74-3385	1.2	2.0	1.0	1.0	1.2	1.8	1.3	1.0
L24A	1.7	2.0	1.0	1.2	2.1	2.6	2.2	1.0
L25A	2.0	3.8	1.0	1.0	2.5	3.2	2.7	1.2
L26	1.7	2.5	1.0	1.0	2.4	2.6	2.2	1.0
L75-8121	1.9	3.2	1.0	1.2	2.6	3.9	2.2	2.2
L77-994	1.9	2.5	1.0	1.2	2.7	3.3	2.2	1.3
U36276	1.4	1.5	1.0	1.3	1.4	1.5	1.5	1.3
U57073	1.8	2.2	1.0	1.5	2.7	2.8	2.3	1.5

UNIFORM TEST III, 1980

Lafayette	Ind.		Illinois				
	Greenfield	Sullivan	Urbana	Girard	Brownstown	Belleville	Eldorado
LODGING (score)							
2.2	1.5	3.3	2.1	2.2	1.3	3.4	3.1
1.7	1.0	2.7	1.3	1.5	1.0	1.4	3.2
1.8	1.0	2.8	1.5	2.1	1.0	2.6	3.9
1.5	1.0	-	1.4	1.4	1.2	1.2	1.8
1.8	1.2	3.2	1.7	1.9	1.7	2.4	2.6
1.5	1.0	2.8	1.3	1.8	1.0	1.7	2.2
2.3	1.0	3.3	1.4	1.8	1.0	1.8	4.2
1.5	1.0	2.2	1.4	1.7	1.0	1.2	2.8
2.2	1.7	3.0	1.7	2.3	1.3	2.7	3.1
2.3	1.2	2.7	1.9	2.2	1.7	2.4	3.5
2.0	1.5	3.2	1.9	2.3	1.7	2.7	3.2
2.2	1.5	3.3	2.2	2.4	1.8	3.2	3.3
1.5	1.0	1.0	1.7	1.2	1.0	1.2	2.2
1.5	1.0	1.0	1.7	1.3	1.0	1.2	1.8
1.5	1.0	1.0	1.5	1.1	1.0	1.1	1.7
1.7	1.0	2.3	1.4	1.8	1.2	1.7	2.2
1.8	1.0	3.0	1.6	2.0	1.2	2.0	3.6
1.8	1.0	2.7	1.5	1.5	1.0	1.7	2.4
1.7	1.0	2.5	1.5	1.8	1.0	1.5	1.9
1.8	1.0	2.7	1.6	2.0	1.2	1.9	1.9
1.3	1.0	2.3	1.2	1.3	1.0	1.1	1.2
1.7	1.0	2.3	1.5	1.6	1.0	1.9	1.8

UNIFORM TEST III, 1980

Strain	Ill.	Iowa		Mo.	S.D.	Kansas		Neb.
	Pontiac	Ottum- wa	Stuart	Novelty	Elk Point	Manhat- tan	Powhat- tan	Mead
LODGING (score)								
BSR 302	1.5	3.0	2.4	2.1		3.7	1.0	1.7
Century (II)	1.0	2.2	1.7	1.6		2.0	1.0	1.2
Cumberland (III)	1.0	2.8	1.7	1.8		2.7	1.0	1.0
Pella	1.0	2.3	1.5	1.5		3.0	1.0	1.0
Union (IV)	1.3	2.7	2.0	2.1		3.7	1.0	1.5
Williams 79	1.0	2.7	1.7	1.6		3.0	1.0	1.3
A78-227012	1.0	2.7	1.7	1.8		3.7	1.0	1.0
A78-321011	1.0	2.1	1.6	1.7		3.3	1.0	1.2
A78-322024	1.0	2.9	2.0	2.2		3.0	1.0	1.2
A78-324002	1.3	2.8	2.3	2.5		4.0	1.0	1.3
A78-325028	1.5	3.1	2.6	2.1		3.3	1.0	2.2
A78-326032	1.8	2.9	2.8	2.2		4.3	1.0	1.3
HC76-4030	1.2	1.6	1.4	1.0		1.0	1.0	1.0
HW74-3384 Sprite	1.0	1.6	1.3	1.0		1.0	1.0	1.0
HW74-3385	1.0	1.6	1.4	1.0		1.0	1.0	1.0
L24A	1.0	2.6	1.7	1.8		2.0	1.0	1.2
L25A	1.2	2.6	1.8	2.0		3.0	1.0	1.3
L26	1.0	2.6	1.7	-		2.7	1.0	1.2
L75-8121	1.2	2.5	1.8	2.0		3.0	1.0	1.2
L77-994	1.2	2.7	1.7	2.2		3.0	1.0	1.0
U36276	1.0	2.3	1.5	1.6		2.0	1.0	1.2
U57073	1.0	2.6	1.6	2.1		2.7	1.0	1.5

UNIFORM TEST III, 1980

Strain	Mean 23 Tests	N.J.	Penn.	Md.	Ohio		Ky.	
		Adel- phia	Landis- ville	Clarks- ville	Hoyt- ville	Wooster	S. Charles- ton	Lexing- ton
HEIGHT (inches)								
BSR 302	41	44	32	30	50	49	44	37
Century (II)	33	39	29	28	43	38	39	33
Cumberland (III)	36	37	30	30	39	37	38	34
Pella	36	34	31	29	39	43	41	36
Union (IV)	43	46	36	32	53	47	45	38
Williams 79	40	41	32	29	46	44	42	37
A78-227012	30	35	23	21	36	35	36	28
A78-321011	33	36	27	24	38	37	38	29
A78-322024	41	45	32	35	51	50	45	39
A78-324002	42	41	28	32	54	42	45	37
A78-325028	42	44	31	34	53	46	45	37
A78-326032	37	41	29	32	42	40	38	34
HC76-4030	20	24	21	24	28	28	24	21
HW74-3384 Sprite	21	23	25	22	28	28	25	26
HW74-3385	21	25	22	21	29	30	23	25
L24A	39	40	27	29	46	43	42	37
L25A	38	41	27	30	43	41	40	36
L26	40	40	30	30	44	41	42	39
L75-8121	40	44	30	34	43	46	41	38
L77-994	39	41	29	31	44	41	38	38
U36276	31	32	22	23	40	35	37	29
U57073	39	38	29	33	44	42	41	35

UNIFORM TEST III, 1980

Strain	Ind.			Ill.				
	Lafay- ette	Green- field	Sull- ivan	Ur- bana	Girard	Browns- town	Belle- ville	Eldorado
	HEIGHT (inches)							
BSR 302	41	37	51	35	46	40	48	48
Century (II)	33	27	38	31	39	32	32	38
Cumberland (III)	34	32	42	32	41	35	42	43
Pella	36	31	--	36	39	33	42	44
Union (IV)	44	36	52	42	47	47	53	51
Williams 79	41	34	47	39	44	38	45	50
A78-227012	29	23	39	29	33	29	34	35
A78-321011	32	27	38	32	37	34	36	41
A78-322024	43	36	47	41	45	39	47	50
A78-324002	41	33	51	43	48	41	49	50
A78-325028	43	39	50	44	47	44	48	51
A78-326032	39	32	41	37	39	36	44	45
HC76-4030	12	19	20	18	20	16	23	16
HW74-3384 Sprite	12	20	16	23	19	16	21	17
HW74-3385	17	19	19	20	20	18	22	17
L24A	40	33	49	40	43	40	42	47
L25A	39	31	42	38	42	37	41	46
L26	41	34	46	39	43	40	45	48
L75-8121	40	35	50	40	45	40	48	50
L77-994	43	33	46	38	43	36	43	48
U36276	31	28	38	32	36	31	34	39
U57073	37	36	46	40	47	38	46	47

UNIFORM TEST III, 1980

<u>Ill.</u>	<u>Iowa</u>		<u>Mo.</u>	<u>S.D.</u>	<u>Kansas</u>		<u>Neb.</u>
<u>Pontiac</u>	<u>Ottumwa</u>	<u>Stuart</u>	<u>Novelty</u>	<u>Elk Point</u>	<u>Manhattan</u>	<u>Powhattan</u>	<u>Mead</u>
<u>HEIGHT (inches)</u>							
30	48	44	34	39	42	30	43
29	37	32	27	31	36	23	28
29	40	37	30	38	42	24	38
28	42	36	29	35	43	26	35
34	53	44	37	41	50	30	41
30	46	39	35	39	49	26	38
25	32	29	27	28	36	21	27
28	34	34	33	31	40	23	30
28	44	40	34	32	50	28	37
35	50	46	35	40	49	32	41
36	50	42	34	39	49	31	39
32	44	37	33	37	45	26	39
22	18	20	13	25	20	17	18
21	19	21	14	23	18	17	19
20	20	25	16	22	18	19	20
33	46	40	33	38	46	27	38
33	46	37	33	36	45	26	38
31	48	40	--	38	48	28	37
34	46	39	35	38	47	27	40
32	46	41	33	37	42	30	36
25	36	33	25	28	38	22	30
32	46	41	32	39	46	28	37

UNIFORM TEST III, 1980

Strain	Mean 21 Tests	N.J.	Penn.	Md.	Ohio		S. Charles- ton	Ky.
		Adel- phia	Landis- ville	Clarks- ville	Hoyt- ville	Wooster		Lexing- ton
QUALITY (score)								
BSR 302	2.3	1.0	2.0	3.3	1.7	1.8	1.5	2.0
Century (II)	2.4	2.0	2.2	2.3	1.7	2.9	1.5	2.0
Cumberland (III)	2.1	1.2	2.0	2.0	1.7	2.0	1.0	1.0
Pella	2.2	1.2	2.2	2.5	1.5	2.0	1.5	2.0
Union (IV)	1.9	1.0	2.0	2.0	1.8	1.6	1.0	1.0
Williams 79	1.9	1.0	2.0	2.0	1.8	2.4	1.5	1.0
A78-227012	2.3	1.0	2.2	2.7	1.4	2.1	1.5	2.0
A78-321011	2.0	1.0	2.2	2.2	1.4	2.3	1.0	1.0
A78-322024	2.5	1.2	2.0	2.2	1.6	2.3	1.5	2.0
A78-324002	1.9	1.0	2.0	1.8	1.5	1.8	1.5	1.0
A78-325028	1.9	1.0	2.0	1.5	1.8	1.6	1.0	2.0
A78-326032	2.3	1.0	2.0	2.3	1.7	1.7	1.5	1.0
HC76-4030	1.8	1.0	1.7	1.5	1.7	1.2	1.0	1.0
HW74-3384 Sprite	1.6	1.0	1.8	2.2	1.4	1.3	1.0	1.0
HW74-3385	1.6	1.2	1.8	2.2	1.5	1.4	1.0	1.0
L24A	1.9	1.0	2.0	2.0	2.0	2.5	1.5	1.0
L25A	2.4	1.0	2.0	1.5	1.5	2.2	1.0	1.0
L26	1.9	1.0	2.0	2.0	1.6	2.1	1.0	2.0
L75-8121	1.9	1.0	2.0	2.0	1.6	2.0	1.0	2.0
L77-994	2.0	1.5	2.0	2.0	1.6	1.7	1.0	2.0
U36276	2.3	2.0	2.0	2.5	1.8	2.9	1.5	2.0
U57073	2.1	1.2	2.0	2.5	1.5	2.4	1.0	2.0

UNIFORM TEST III, 1980

Lafayette	Ind.	Illinois					
	Greenfield	Sullivan	Urbana	Girard	Browns-town	Belleville	Eldorado
QUALITY (score)							
1.0	1.5	2.0	3.2	2.8	2.9	4.2	4.2
1.5	2.0	1.5	4.2	3.0	2.2	3.3	3.3
1.0	1.0	2.0	3.0	2.5	2.5	3.3	3.5
1.5	1.5	-	3.3	2.8	2.5	2.7	3.0
1.0	1.0	1.5	2.0	1.8	3.2	2.8	3.3
1.0	1.5	1.5	2.0	1.5	1.9	2.2	2.8
1.0	1.5	1.5	2.3	3.3	2.1	3.5	3.7
1.0	1.0	2.0	1.8	2.7	2.1	3.0	2.5
2.0	1.5	2.5	3.8	3.0	3.2	3.3	3.5
1.0	1.5	1.5	2.3	1.7	2.1	2.7	2.8
1.5	1.5	1.5	2.2	1.7	2.3	2.7	2.8
1.5	1.5	3.0	2.2	3.0	2.7	3.3	4.0
1.5	1.0	2.5	2.2	1.7	1.7	2.0	1.7
1.5	1.5	1.0	1.7	1.7	1.7	2.3	2.5
1.0	1.0	1.5	1.7	1.5	1.7	2.2	2.0
1.0	1.5	2.0	1.7	1.5	2.1	2.2	2.5
1.5	1.5	2.0	1.5	1.5	2.0	2.2	2.8
1.5	1.5	1.5	1.5	1.3	1.9	2.2	2.5
1.5	1.0	1.5	1.7	1.7	2.3	2.7	2.7
1.0	1.5	1.5	1.7	1.7	2.5	3.0	2.8
1.5	1.5	1.5	2.3	2.5	3.1	3.3	3.0
1.0	1.0	1.5	1.8	1.8	3.0	2.8	2.8

UNIFORM TEST III, 1980

Strain	Ill.	Iowa		Mo.	S.D.	Kansas		Neb.
	Pontiac	Ottum- wa	Stuart	Novelty	Elk Point	Manhat- tan	Powhat- tan	Mead
QUALITY (score)								
BSR 302	1.8	1.8			2.0	3.0	3.5	2.0
Century (II)	2.3	1.7			3.0	3.0	3.0	2.2
Cumberland (III)	2.1	1.5			3.0	3.0	3.0	2.5
Pella	2.1	1.7			2.0	3.0	2.5	2.2
Union (IV)	1.9	1.5			3.0	2.0	1.5	2.2
Williams 79	1.7	2.0			3.0	3.0	2.0	2.0
A78-227012	1.6	2.0			3.0	4.5	3.5	1.7
A78-321011	1.7	1.7			3.0	3.0	3.0	2.3
A78-322024	2.7	2.2			4.0	3.0	2.5	2.2
A78-324002	1.7	1.6			4.0	3.0	1.5	1.8
A78-325028	1.9	1.3			4.0	3.0	1.5	2.0
A78-326032	1.7	1.3			4.0	3.0	3.0	2.0
HC76-4030	1.6	1.4			3.0	4.5	1.5	1.8
HW74-3384 Sprite	1.7	1.5			2.0	2.0	2.0	1.7
HW74-3385	1.5	1.3			2.0	2.5	2.0	1.5
L24A	1.9	1.4			3.0	3.0	2.0	2.0
L25A	1.5	1.8			3.0	2.0	1.5	2.0
L26	1.6	1.3			4.0	3.0	2.5	2.2
L75-8121	1.9	1.4			2.0	3.0	2.0	2.0
L77-994	1.8	1.8			3.0	4.0	1.5	2.0
U36276	1.7	1.4			3.0	3.5	3.0	2.3
U57073	1.9	1.5			4.0	4.0	3.0	2.2

UNIFORM TEST III, 1980

		N.J.	Penn.	Md.	Ohio			Ky.
	Mean	Adel-	Landis-	Clarks-	Hoyt-		S. Charles-	Lexing-
Strain	21 Tests	phia	ville	ville	ville	Wooster	ton	ton
		SIZE (g/100)						
BSR 302	16.5	11.0	17.0	13.9	16.6	16.3	17.4	15.5
Century (II)	17.4	13.0	18.9	16.2	18.6	18.0	17.5	15.7
Cumberland (III)	17.5	15.0	16.8	15.4	18.2	18.9	19.2	14.1
Pella	18.0	15.0	18.6	15.8	18.5	19.1	18.7	14.4
Union (IV)	17.6	14.0	17.9	14.8	17.7	16.8	17.4	15.6
Williams 79	16.6	13.0	16.9	14.9	16.9	16.5	16.3	14.9
A78-227012	14.5	12.0	13.2	11.7	15.0	15.6	15.1	13.6
A78-321011	15.1	12.0	14.8	13.1	16.0	16.0	15.8	11.5
A78-322024	17.4	15.0	17.2	15.1	17.6	17.6	18.4	13.6
A78-324002	14.1	11.0	13.8	12.0	15.6	14.5	14.9	12.7
A78-325028	16.4	15.0	15.1	14.3	17.8	15.9	16.9	15.1
A78-326032	15.3	13.0	15.4	9.7	16.4	15.8	16.8	13.1
HC76-4030	15.3	12.0	12.7	12.7	14.6	14.7	16.1	11.9
HW74-3384 Sprite	17.4	14.0	16.7	14.4	18.7	17.8	19.2	13.4
HW74-3385	15.6	14.0	14.9	13.3	16.4	15.8	15.7	12.7
L24A	16.3	13.0	16.5	14.4	17.7	16.5	15.9	13.7
L25A	16.4	15.0	16.0	14.6	16.6	16.8	16.3	13.9
L26	16.5	16.0	15.3	15.1	16.2	16.9	15.8	14.1
L75-8121	15.2	14.0	15.2	14.0	15.5	15.0	15.4	14.1
L77-994	15.0	14.0	16.0	13.9	15.3	14.8	14.0	13.8
U36276	14.6	13.0	15.4	13.7	14.7	14.3	14.2	12.7
U57073	16.5	15.0	17.4	15.3	16.2	17.1	15.8	15.0

UNIFORM TEST III, 1980

Strain	Ind.			Illinois				
	Lafay- ette	Green- field	Sull- ivan	Ur- bana	Girard	Browns- town	Belle- ville	Eldorado
	SIZE (g/100)							
BSR 302	18.0	17.3	17.0	19.9	16.2	14.9	16.4	13.6
Century (II)	18.7	18.8	17.5	19.7	15.4	16.0	15.5	14.4
Cumberland (III)	18.6	19.0	17.9	21.0	16.8	15.1	16.7	14.8
Pella	19.9	20.6	--	20.3	17.0	16.5	16.1	15.4
Union (IV)	19.9	17.8	17.9	20.3	17.2	17.9	17.6	15.9
Williams 79	18.4	17.4	17.0	18.8	15.5	14.5	15.4	13.9
A78-227012	16.0	14.4	15.0	15.8	12.4	12.9	13.1	11.9
A78-321011	17.0	16.0	15.4	15.2	14.4	13.0	13.1	12.3
A78-322024	19.5	18.0	17.5	19.3	17.3	15.8	17.3	14.3
A78-324002	16.4	13.6	13.9	15.4	13.8	12.6	12.6	11.1
A78-325028	19.1	17.2	16.6	18.7	16.2	15.5	15.3	14.2
A78-326032	18.1	17.0	16.5	16.3	15.4	13.6	13.6	12.7
HC76-4030	18.2	15.2	18.9	15.6	15.4	15.6	14.5	13.3
HW74-3384 Sprite	20.7	16.2	18.5	19.5	16.0	15.6	17.3	15.2
HW74-3385	17.7	16.0	17.4	17.7	14.0	13.2	13.8	13.3
L24A	18.9	16.6	16.7	17.2	15.4	15.0	14.9	15.0
L25A	19.4	16.1	16.2	17.8	15.0	14.4	15.3	14.0
L26	18.1	16.4	15.9	17.9	15.9	16.1	15.5	14.8
L75-8121	16.8	15.5	15.2	16.3	15.0	13.4	14.4	12.8
L77-994	16.7	14.5	13.9	15.7	14.0	13.6	14.9	12.4
U36276	14.7	15.1	12.8	15.8	13.5	13.1	13.8	12.2
U57073	17.7	17.0	16.1	17.1	16.2	15.6	16.1	14.6

UNIFORM TEST III, 1980

<u>Ill.</u>	<u>Iowa</u>		<u>Mo.</u>	<u>S.D.</u>	<u>Kansas</u>		<u>Neb.</u>
Pontiac	Ottumwa	Stuart	Novelty	Elk Point	Manhattan	Powhattan	Mead
<u>SIZE (g/100)</u>							
20.3	17.7			17.4	16.5	16.5	16.6
18.8	18.0			22.5	14.4	16.3	22.0
21.2	18.6			19.7	15.3	16.6	18.8
21.5	19.6			20.0	16.5	16.7	20.5
21.0	19.8			19.9	16.4	16.4	17.3
20.8	19.1			17.8	16.5	16.4	17.8
16.1	14.8			17.2	13.3	15.4	19.1
18.0	14.6			20.3	14.4	16.2	17.6
22.0	18.4			19.9	16.8	16.7	17.7
15.7	15.7			15.3	14.8	14.8	15.0
19.1	18.4			15.6	16.6	16.3	16.1
17.6	17.0			15.9	15.6	16.2	16.0
15.8	15.8			17.1	16.4	16.2	18.3
19.4	17.6			22.1	16.7	16.7	20.3
17.8	15.8			16.4	16.6	16.3	18.4
19.6	18.1			17.7	16.9	16.4	15.9
20.2	17.6			18.4	16.4	16.4	17.1
20.3	18.4			17.3	16.2	16.7	17.1
16.7	16.8			15.1	16.2	16.2	14.6
18.2	15.4			16.5	15.6	16.2	15.4
17.3	15.0			16.5	15.8	16.4	15.9
19.4	17.7			17.7	16.5	16.4	16.7

UNIFORM TEST III, 1980

Strain	Mean 5 Tests	Ohio	Ind.	Ill.	Iowa	Kansas
		S. Charles- ton	Lafay- ette	Urbana	Ottumwa	Ashland
PROTEIN (%)						
BSR 302	42.7	41.4	42.1	41.8	42.1	46.3
Century (II)	42.7	42.8	43.2	41.8	40.6	45.2
Cumberland (III)	41.2	40.9	40.6	40.2	40.6	43.6
Pella	40.3	40.1	40.0	39.2	40.7	41.3
Union (IV)	43.3	44.4	42.1	44.3	42.2	43.7
Williams 79	41.7	41.9	41.6	39.5	41.1	44.5
A78-227012	42.0	41.9	42.8	40.7	41.0	43.7
A78-321011	41.6	42.8	42.5	39.4	40.7	42.7
A78-322024	43.0	43.2	42.3	42.4	41.1	46.0
A78-324002	42.4	42.7	42.2	41.3	43.4	42.5
A78-325028	44.0	43.7	42.9	44.1	44.5	44.9
A78-326032	42.3	43.3	41.8	40.8	42.3	43.3
HC76-4030	43.9	42.4	44.0	43.8	43.8	45.6
HW74-3384 Sprite	41.1	40.4	41.6	41.3	40.3	41.7
HW74-3385	40.3	41.6	40.8	39.5	39.3	40.4
L24A	41.9	41.2	41.2	41.3	42.8	42.8
L25A	41.3	40.8	41.8	40.2	40.8	42.8
L26	42.5	42.6	41.7	42.1	42.0	43.9
L75-8121	40.4	39.8	40.6	40.0	39.1	42.4
L77-994	42.9	42.5	43.4	40.9	43.0	44.7
U36276	42.3	42.3	41.6	42.2	39.5	45.8
U57073	43.5	43.4	43.0	43.4	43.4	44.2

UNIFORM TEST III, 1980

Strain	Mean 5 Tests	Ohio	Ind.	Ill.	Iowa	Kansas
		S. Charles- ton	Lafay- ette	Urbana	Ottumwa	Ashland
OIL (%)						
BSR 302	19.8	18.4	21.4	20.7	20.4	18.2
Century (II)	20.8	21.1	21.5	20.2	22.0	19.1
Cumberland (III)	22.3	21.7	23.2	22.9	23.5	20.4
Pella	21.9	22.4	22.0	22.2	23.2	19.9
Union (IV)	20.5	20.4	21.0	20.7	20.8	19.4
Williams 79	20.6	20.7	21.5	20.9	20.8	19.3
A78-227012	20.5	19.1	21.5	22.3	21.5	18.3
A78-321011	20.5	19.9	21.2	21.7	20.7	19.2
A78-322024	21.7	21.1	22.7	21.8	22.4	20.3
A78-324002	20.3	19.9	20.7	20.7	21.0	19.2
A78-325028	20.2	20.7	21.1	19.9	20.6	18.7
A78-326032	20.1	21.2	20.9	20.0	19.4	18.8
HC76-4030	20.5	19.3	22.0	21.0	21.5	18.5
HW74-3384 Sprite	22.6	22.1	22.7	23.1	23.4	21.5
HW74-3385	22.5	21.6	22.9	22.6	22.9	22.5
L24A	21.2	20.6	21.3	22.1	22.8	19.1
L25A	21.2	21.3	21.0	21.5	22.0	20.0
L26	20.8	21.4	21.6	20.4	20.7	19.7
L75-8121	20.8	18.5	22.4	21.4	22.1	19.5
L77-994	20.3	19.7	21.2	20.3	21.1	19.3
U36276	20.6	19.9	22.0	21.7	21.1	18.1
U57073	19.8	19.9	20.3	19.8	21.2	18.0

PRELIMINARY TEST III, 1980

Strain	Parentage	Generation Composited
Century (II)	Calland x Bonus	F ₆
Cumberland (III)	Corsoy x Williams	F ₄
Union (IV)	Williams ⁵ x SL12 (Wayne <u>Rpm</u> <u>Rps</u>)	F ₃
A79-237005	Pride B216 x Cumberland	F ₄
A79-331022	(L15 x AP68-1016) x Oakland	F ₄
A79-334010	Pride B216 x LL4102	F ₄
A79-336014	Pride x Oakland	F ₄
A79-337010	Pride B216 x LL4102	F ₄
A79-338015	Pella x Oakland	F ₄
A79-338021	A72-512 x A74-104030	F ₄
HC76-3711	L72U-2567 x Williams	F ₅
HC76-3863	L72U-2567 x Evans	F ₅
HC76-4374	L72U-2567 x Ransom	F ₅
HC76-4388	L72U-2567 x L72U-640	F ₅
HC77-1418	L72U-640 x Essex	F ₅
HC77-1419	L72U-640 x Essex	F ₅
HC77-5421	L72U-2567 x Wells	F ₅
HW79050	Cumberland x Pella (AX-1764M-23)	F ₅
HW79116	Cumberland x Pella (AX-1764E-30)	F ₅
HW79149	(A72-507 ⁶ x A1) x (A72-507 ⁵ x PI82.263-2)	F ₃
K1055	Tracy x Williams	F ₅
K1056	Tracy x Williams	F ₅
K1057	Tracy x Columbus	F ₅
K1058	Tracy x Bonus	F ₅
K1059	Tracy x Columbus	F ₅
K1060	Tracy x Pomona	F ₅
L77-443	Union x L75-8020	F ₄
L77-3014	L73-6626 x Williams	F ₅
L78-709	Williams x L70-2283	F ₇
L78-4883	L73-6626 x Williams	F ₆
LN1060	Tracy x Williams	F ₅
LN1062	Tracy x Williams	F ₅
LN1063	Tracy x Bonus	F ₅
LN78-246	Union x C1528	F ₅
LN78-537	Union x K1028	F ₅
LN78-575	Union x K1028	F ₅

PRELIMINARY TEST III, 1980

Descriptive and Other Data

Strain	Description		Chlorosis Score	Shattering Manhattan
			Ames	2 Weeks
Century (II)	PTBr	DYB1	3.0	2
Cumberland (III)	PGBr	SYIb	3.7	2
Union (IV)	WTTn	SYB1	3.3	1
A79-237005	WTTn	DYB1+Br	4.2	2
A79-331022	PTTn	DYBr	3.7	2
A79-334010	PGBr	DYIb	4.3	2
A79-336014	PTBr	DYB1	3.7	2
A79-337010	WG+TBr	DYBf+B1	4.0	3
A79-338015	PGTn	DYIb	2.8	2
A79-338021	PGBr	SYBf	1.7	2
HC76-3711	PTTn	DYB1	2.3	2
HC76-3863	WTTn	DYGr	3.7	1
HC76-4374	PTTn	SYB1	3.3	1
HC76-4388	PTTn	SYIb	2.8	1
HC77-1418	PGTn	DYIb	3.3	1
HC77-1419	PGTn	DYIb	3.5	1
HC77-5421	PTTn	SYB1	2.5	1
HC79050	PTBr	SYB1	4.3	1
HW79116	PGTn	SYIb	3.8	1
HW79149	WGTn	DYBf	3.8	2
K1055	WTTn	SYB1	3.8	2
K1056	WTTn	SYB1	4.5	1
K1057	P+WTTn	DYB1	3.2	2
K1058	PGTn	DYIb	3.0	2
K1059	WGBr	DYBf	2.8	2
K1060	WTTn	DYB1	3.3	1
L77-443	WG+TTn	SYB1	2.8	2
L77-3014	PTBr	SYB1	2.0	1
L78-709	WGTn	DYBf	3.3	1
L78-4883	WTBr	DYB1	3.7	1
LN1060	WTTn	DYB1	3.0	2
LN1062	WTTn	DYB1	3.7	2
LN1063	WGTn	DYBf	2.3	1
LN78-246	PGBr	DYIb	2.2	1
LN78-537	PGBr	DYB1	4.2	1
LN78-575	PTTn	SYB1	3.0	1

PRELIMINARY TEST III, 1980

Strain	Disease Data								
	BSR		GERM	SMV	PSB	PS	PR	PR	Race 1
			*				Vick-		
	Laf. Ind. n	Ames Ia. n	Lafayette, d	Ind. d			ery Ohio n	Laf. Ind. a	Ames Ia. a
	%	Reac.	%	Score	%	Score	-----Reaction-----		
Century (II)	60	S	73	4E	21	3E	2.4	R	R
Cumberland (III)	0	S	96	2M	1	3E	2.4	S	S
Union (IV)	100	S	96	2M	2	4E	2.0	R	R
A79-237005	20	S	93	5E	5	4E	2.2	S	S
A79-331022	0	R	92	4E	2	3E	2.2	S	S
A79-334010	0	S	88	5E	1	4S	2.0	S	S
A79-336014	20	S	85	4E	7	2M	2.3	S	S
A79-337010	40	S	81	5E	9	4E	2.0	S	H
A79-338015	80	S	91	3M	5	2E	2.6	S	S
A79-338021	20	S	99	3M	1	5E	2.7	S	S
HC76-3711	60	S	88	3M	3	3E	3.6	S	S
HC76-3863	20	S	83	2M	11	2E	3.7	S	S
HC76-4374	60	S	94	2M	2	2E	3.5	S	S
HC76-4388	40	S	95	2M	3	2E	3.4	S	S
HC77-1418	100	S	97	2M	0	2E	3.2	S	S
HC77-1419	80	S	99	1	0	3E	3.9	S	S
HC77-5421	60	S	93	2M	0	2E	3.6	S	S
HW79050	60	S	92	2M	7	4E	2.4	S	S
HW79116	60	S	86	2M	4	5S	2.2	R	R
HW79149	60	S	83	5E	1	5S	1.8	R	R
K1055	40	S	91	3M	0	3E	1.5	R	R
K1056	60	S	85	2E	2	3E	1.6	R	R
K1057	60	S	91	3E	5	5S	1.7	R	R
K1058	80	S	84	3E	0	4E	2.3	H	H
K1059	20	S	96	3E	2	3E	2.4	R	R
K1060	60	S	95	2M	1	4E	1.7	H	H
L77-443	60	S	87	2M	3	4E	2.4	R	R
L77-3014	20	S	73	4E	17	3E	2.4	S	S
L78-709	100	S	77	1	19	5E	2.6	S	S
L78-4883	100	S	90	3E	6	3E	2.2	S	S
LN1060	20	S	94	4E	3	2M	2.2	R	R
LN1062	0	S	86	3M	7	3E	2.4	R	R
LN1063	0	S	92	2E	4	3E	2.1	R	R
LN78-246	60	S	90	2E	9	5S	2.4	R	R
LN78-537	-	S	89	2M	14	3E	3.0	R	R
LN78-575	20	S	91	3M	2	4S	2.2	R	R

*Petri dish germination on potato dextrose agar.

PRELIMINARY TEST III, 1980

Regional Summary

Strain	Yield	Rank	Matu- rity	Lodg- ing	Height	Seed Quality	Seed Size	Protein	Oil
No. of Tests	10 bu/a	10 No.	9 Date	9 Score	10 In.	9 Score	9 g/100	5 %	5 %
Century (II)	46.5	14	-7	1.6	33	2.4	18.3	42.3	21.8
Cumberland (III)	48.9	3	9/24*	1.8	36	2.0	17.8	41.8	21.7
Union (IV)	46.2	18	+5	2.2	44	1.7	18.0	42.2	21.9
A79-237005	47.4	10	-1	1.7	37	2.0	15.3	42.2	21.5
A79-331022	47.2	11	+2	1.7	38	2.3	18.2	42.5	21.3
A79-334010	50.6	1	-2	2.3	39	2.2	15.7	42.2	20.6
A79-336014	50.1	2	+1	1.5	35	2.0	18.6	41.8	21.7
A79-337010	48.7	5	+3	2.3	41	2.1	16.6	43.0	20.1
A79-338015	48.9	3	+3	1.6	37	2.0	15.1	39.2	22.5
A79-338021	48.7	5	+4	2.2	40	2.5	16.1	42.5	20.9
HC76-3711	31.5	35	-1	1.3	19	1.9	17.6	42.9	21.8
HC76-3863	35.0	34	-6	1.5	21	2.2	17.0	41.2	22.9
HC76-4374	35.9	33	+1	1.2	19	2.0	14.5	42.1	21.5
HC76-4388	28.2	36	+1	1.3	19	2.0	16.4	43.9	21.1
HC77-1418	45.1	20	-3	1.3	26	1.8	15.0	43.9	21.3
HC77-1419	42.5	29	-5	1.2	22	1.6	14.4	42.2	22.0
HC77-5421	38.7	32	+3	2.3	20	2.0	13.6	42.3	21.4
HW79050	47.2	11	-2	1.5	34	2.3	19.3	39.5	22.0
HW79116	46.3	17	+2	1.6	37	2.1	18.1	39.6	22.4
HW79149	48.5	7	+3	2.6	38	2.7	17.5	41.2	21.0
K1055	44.0	26	+4	2.0	39	1.9	16.7	42.5	20.7
K1056	44.2	24	+4	1.9	39	1.7	16.7	41.5	20.4
K1057	45.8	19	+3	1.9	39	1.6	14.6	41.3	20.4
K1058	47.8	8	+6	2.3	39	1.8	18.2	43.8	20.4
K1059	44.7	21	+1	1.5	30	1.9	16.8	44.4	20.4
K1060	44.6	22	+4	1.8	39	1.7	16.6	42.6	19.7
L77-443	43.7	27	-2	2.0	38	1.9	16.3	40.5	21.8
L77-3014	42.0	30	-5	1.6	34	1.9	17.3	42.5	21.8
L78-709	44.2	24	-3	1.6	36	1.9	16.5	40.8	22.3
L78-4883	40.6	31	-1	1.7	35	1.9	17.9	41.5	21.7
LN1060	44.3	23	0	1.6	26	1.7	16.1	42.0	20.9
LN1062	43.2	28	-2	1.8	27	1.5	14.7	42.4	20.8
LN1063	46.4	16	+3	1.7	31	1.7	15.3	45.5	20.3
LN78-246	46.5	14	-1	1.9	34	2.2	16.2	42.3	20.9
LN78-537	46.7	13	0	1.8	39	1.7	17.9	42.7	21.7
LN78-575	47.7	9	0	1.6	37	2.2	18.3	40.9	21.4

*132 days after planting

Two strains, A79-334010 and A79-336014, were higher in average yield than Cumberland. Both strains were susceptible to phytophthora rot, A79-336014 had excellent lodging resistance. In general, the determinant strains were among the lowest yielding entries in these tests. The cyst nematode race 3 resistant strain L77-443 was not competitive in yield with the check varieties. Four of the Kansas strains, K1055 to K1058, were all resistant to races 1, 3, 4, 5, and 6 of phytophthora rot. The strain HW79149 was resistant to races 1 through 9 of phytophthora rot.

PRELIMINARY TEST III, 1980

Strain	Mean 10 Tests	Ohio	Ind.	Ill.	
		S. Charleston	Lafayette	Urbana	Girard
		YIELD (bu/a)			
Century (II)	46.5	59.9	52.8	52.3	39.0
Cumberland (III)	48.9	64.6	53.4	56.7	45.0
Union (IV)	46.2	52.3	59.8	52.0	44.2
A79-237005	47.4	60.3	54.7	50.3	34.2
A79-331022	47.2	63.1	55.4	52.3	43.8
A79-334010	50.6	62.6	60.5	56.3	41.8
A79-336014	50.1	60.4	53.3	57.0	47.4
A79-337010	48.7	61.5	53.1	55.5	40.7
A79-338015	48.9	60.1	54.2	52.5	44.3
A79-338021	48.7	58.0	62.7	51.0	44.3
HC76-3711	31.5	57.1	14.9	45.7	29.4
HC76-3863	35.0	58.6	20.5	45.0	32.0
HC76-4374	35.9	52.7	10.1	45.0	31.6
HC76-4388	28.2	52.9	14.4	39.4	28.6
HC77-1418	45.1	63.8	26.3	51.9	43.4
HC77-1419	42.3	63.7	21.7	41.6	38.0
HC77-5421	38.7	56.6	47.2	44.6	34.5
HW79050	47.2	63.7	49.7	52.5	41.5
HW79116	46.3	58.2	51.9	53.1	41.5
HW79149	48.5	65.3	48.6	49.5	42.6
K1055	44.0	54.3	58.9	48.4	42.3
K1056	44.2	54.6	49.8	48.6	44.8
K1057	45.8	50.7	49.5	47.9	45.4
K1058	47.8	53.8	57.8	50.7	46.0
K1059	44.7	53.4	43.2	48.8	46.0
K1060	44.6	55.7	52.6	47.4	40.9
L77-443	43.7	58.3	50.6	45.3	38.4
L77-3014	42.0	52.9	51.6	44.3	39.5
L78-709	44.2	55.3	49.4	52.3	39.3
L78-4883	40.6	48.1	49.1	42.9	38.0
LN1060	44.3	56.6	48.4	52.2	45.3
LN1062	43.2	52.9	44.1	49.2	40.8
LN1063	46.4	54.0	46.2	49.4	41.1
LN78-246	46.5	61.0	57.6	53.3	38.5
LN78-537	46.7	56.9	58.6	48.4	37.3
LN78-575	47.7	59.1	59.8	49.8	39.3
C.V. (%)		8.0	13.8	5.0	4.4
L.S.D. (5%)		9.4	13.1	5.1	3.6
Row sp (in.)		30"	24"	30"	30"
Rows/plot		4	4	4	4
Reps		2	2	2	2

PRELIMINARY TEST III, 1980

Iowa		S.D.	Neb.	Kans.	Ky.
Ottumwa	Stuart	Elk Point	Mead	Manhattan	Lexington
YIELD (bu/a)					
49.1	53.8	43.4	46.0	29.2	39.9
54.6	56.9	44.3	45.2	32.2	33.8
52.0	50.0	37.0	42.6	33.0	39.2
51.4	58.0	46.5	49.4	29.4	39.7
57.9	54.6	38.6	37.1	29.4	39.6
55.2	54.7	45.6	58.6	34.8	35.8
54.1	57.9	42.2	53.3	38.0	37.3
54.9	56.4	41.1	55.1	31.2	37.5
51.4	60.8	41.8	53.3	34.2	36.3
53.2	57.7	40.8	49.2	35.1	35.4
38.6	41.4	17.3	21.4	10.1	38.6
42.1	47.8	29.4	26.5	9.4	39.1
35.8	41.5	40.5	45.9	17.7	38.2
38.7	38.2	5.9	19.5	8.6	36.0
49.4	52.0	40.4	47.2	33.9	42.9
46.7	51.8	39.4	46.3	33.5	40.5
40.0	45.7	25.1	39.3	19.2	35.1
52.3	52.0	47.6	43.8	27.1	42.1
55.5	54.6	40.4	43.7	29.5	34.9
57.3	53.6	41.5	46.5	40.5	39.8
51.2	44.9	31.2	43.0	30.9	34.5
50.5	47.2	32.3	44.9	29.3	39.5
48.7	51.2	41.2	49.8	36.0	37.2
54.7	56.3	35.9	46.0	36.0	40.3
50.2	47.6	42.6	45.2	35.6	34.2
49.3	48.2	38.0	38.5	30.7	44.9
50.8	48.8	40.7	39.9	28.0	35.7
44.4	45.7	34.6	41.5	33.1	32.6
47.7	53.7	34.5	43.5	30.1	36.0
46.7	46.7	33.8	46.5	20.8	33.3
49.8	48.4	37.8	31.2	38.8	34.1
45.9	50.0	37.8	40.3	39.0	32.4
55.2	49.9	42.1	49.3	40.8	36.3
50.7	48.9	41.4	43.4	31.2	39.2
51.7	50.3	41.1	53.2	34.5	35.3
55.0	53.1	41.5	40.6	38.2	40.1
5.6	4.5	11.6	16.2	10.0	8.0
5.6	4.6	8.7	14.4	6.1	2.0
27"	27"	30"	30"	30"	30"
4	4	4	4	4	4
2	2	2	3	2	2

PRELIMINARY TEST III, 1980

Strain	Mean 9 Tests	Ohio	Ind.	Ill.	
		S. Charleston	Lafayette	Urbana	Girard
MATURITY (date)					
Century (II)	-7	-8	-9	-6	-8
Cumberland (III)*	9/24	9/24	9/21	9/25	9/15
Union (IV)	+5	+2	+7	+5	+7
A79-237005	-1	-4	+1	-1	-3
A79-331022	+2	+2	+3	+1	+2
A79-334010	-2	0	0	-2	-1
A79-336014	+1	-3	+1	-3	+1
A79-337010	+3	+2	+2	+3	+1
A79-338015	+3	0	+2	+3	+3
A79-338021	+4	+2	+7	+3	+7
HC76-3711	-1	-3	+3	-3	-2
HC76-3863	-6	-4	-7	-9	-7
HC76-4374	+1	+4	+1	+2	0
HC76-4388	+1	+2	+4	0	+3
HC77-1418	-3	-3	-4	-4	-4
HC77-1419	-5	-4	-6	-4	-6
HC77-5421	+3	+4	+3	+3	+6
HW79050	-2	-4	-3	-2	-1
HW79116	+2	0	+3	0	+2
HW79149	+3	+3	+3	+4	+4
K1055	+4	+2	+6	+4	+7
K1056	+4	0	+5	+2	+6
K1057	+3	+2	+6	+1	+3
K1058	+6	+4	+8	+5	+10
K1059	+1	+4	+2	+1	+2
K1060	+4	+2	+6	+5	+5
L77-443	-2	-5	-1	-4	-4
L77-3014	-5	-6	-2	-7	-7
L78-709	-3	-6	-4	-3	-6
L78-4883	-1	-2	+2	-4	-5
LN1060	0	+1	+2	-1	+1
LN1062	-2	-3	0	-2	-1
LN1063	+3	+3	+4	+2	+2
LN78-246	-1	+2	+3	-1	-2
LN78-537	0	-1	+3	-1	0
LN78-575	0	-2	+2	+1	0
Date planted	5/15	4/30	5/7	5/7	5/6
*Days to maturity	132	147	137	141	132

PRELIMINARY TEST III, 1980

Iowa		S.D.	Neb.	Kans.	Ky.
Stuart	Ottumwa	Elk Point	Mead	Manhattan	Lexington
MATURITY (date)					
-10		-7	-7	-5	-7
9/30		10/2	9/29	9/21	9/22
+1		+5	+5	+8	+7
-5		+1	+4	+1	0
0		+3	+4	+6	0
-5		-1	+1	+1	-7
0		+3	+2	+7	0
0		+1	+3	+7	+4
+1		+5	+4	+6	0
+2		+3	+5	+7	+4
-5		0	0	+5	0
-9		-6	-6	+1	-5
-4		-3	0	+6	0
-4		+3	0	+5	0
-6		+1	-1	-3	-5
-8		+1	-5	-3	-7
-1		+2	+2	+7	0
-5		-1	+4	-1	-7
0		+2	+5	+2	0
+1		+3	+4	+1	0
0		+5	+4	+8	+4
0		+2	+5	+9	+4
-1		+3	+3	+7	0
+2		+4	+4	+7	+7
-2		-2	-2	+4	+4
0		+3	+4	+7	+4
-6		+2	0	-2	0
-8		+1	-2	-5	-7
-6		-2	-1	-3	0
-5		+5	+2	-4	0
-3		+3	0	0	-7
-6		-2	-2	-1	-5
0		0	+1	+4	+11
-2		+1	+1	-2	-7
-5		0	+2	-2	+4
-4		+1	+2	0	0
5/10		5/21	5/27	5/19	6/8
143		134	125	125	106

PRELIMINARY TEST III, 1980

	Mean	Ohio	Ind.	Ill.	
Strain	9 Tests	S. Charleston	Lafayette	Urbana	Girard
		LODGING (score)			
Century (II)	1.6	2.0	1.5	1.1	1.4
Cumberland (III)	1.8	2.0	2.0	1.4	2.1
Union (IV)	2.2	2.8	2.0	1.6	2.1
A79-237005	1.7	2.0	2.0	1.2	1.6
A79-331022	1.7	1.8	1.5	1.2	1.9
A79-334010	2.3	2.8	2.3	2.1	2.2
A79-336014	1.5	1.5	1.5	1.2	1.5
A79-337010	2.3	2.8	2.3	2.2	2.0
A79-338015	1.6	1.5	1.5	1.3	1.6
A79-338021	2.2	2.5	2.3	2.0	2.2
HC76-3711	1.3	1.5	1.5	1.3	1.0
HC76-3863	1.5	1.5	1.5	1.8	1.3
HC76-4374	1.2	1.2	1.5	1.3	1.0
HC76-4388	1.3	1.8	1.5	1.3	1.1
HC77-1418	1.3	1.5	1.5	1.2	1.2
HC77-1419	1.2	1.2	1.5	1.2	1.1
HC77-5421	1.3	1.8	1.5	1.3	1.1
HW79050	1.5	1.5	1.5	1.2	1.4
HW79116	1.6	1.5	1.5	1.2	1.4
HW79149	2.6	2.2	3.0	1.8	2.7
K1055	2.0	2.0	2.0	1.4	2.1
K1056	1.9	1.8	1.8	1.3	2.2
K1057	1.9	2.0	2.0	1.4	2.0
K1058	2.3	2.5	2.0	1.7	2.2
K1059	1.5	1.8	1.5	1.3	1.6
K1060	1.8	2.2	1.5	1.3	1.9
L77-443	2.0	2.2	2.3	1.9	2.1
L77-3014	1.6	1.8	2.0	1.5	1.5
L78-709	1.6	1.5	2.0	1.3	1.7
L78-4883	1.7	2.2	2.3	1.4	1.9
LN1060	1.6	1.5	1.5	1.4	1.8
LN1062	1.8	2.0	1.5	2.0	2.0
LN1063	1.7	2.0	1.8	1.4	1.4
LN78-246	1.9	2.8	2.0	1.6	1.8
LN78-537	1.8	2.5	1.8	1.2	1.5
LN78-575	1.6	1.8	1.5	1.3	1.3

PRELIMINARY TEST III, 1980

Iowa		S.D.	Neb.	Kans.	Ky.
Stuart	Ottumwa	Elk Point	Mead	Manhattan	Lexington
LODGING (score)					
1.8	2.4		1.3	1.8	1.0
1.8	2.5		1.0	2.3	1.0
1.8	2.8		1.5	3.5	1.8
1.6	2.6		1.0	2.0	1.3
1.8	2.4		1.3	2.5	1.0
2.0	3.2		1.3	2.8	2.0
1.7	2.2		1.0	2.0	1.3
2.4	2.8		1.5	3.3	1.8
1.6	2.3		1.0	2.5	1.3
1.8	2.7		1.3	4.0	1.0
1.4	1.6		1.0	1.0	1.0
1.6	2.0		1.0	1.0	2.0
1.4	1.6		1.0	1.0	1.0
1.4	1.7		1.3	1.0	1.0
1.5	1.8		1.0	1.0	1.3
1.4	1.7		1.0	1.0	1.0
1.4	1.7		1.0	1.0	1.3
1.7	2.3		1.3	1.8	1.0
1.4	2.4		1.3	2.3	1.0
2.3	4.4		1.3	4.3	1.5
1.8	2.6		1.5	3.0	1.5
1.6	2.6		1.3	2.5	1.8
1.9	2.6		1.5	2.5	1.5
2.4	3.0		1.5	3.3	1.8
1.6	2.2		1.0	1.3	1.5
1.8	2.6		1.3	2.0	1.5
1.8	2.8		1.0	2.0	1.8
1.8	2.1		1.3	1.3	1.5
1.8	2.3		1.0	1.3	1.8
1.4	2.3		1.5	1.3	1.3
2.0	2.2		1.0	1.0	2.3
1.9	2.3		1.3	1.0	2.0
1.8	2.3		1.0	2.0	1.8
1.8	2.3		1.0	1.8	1.8
1.6	2.4		1.3	2.0	1.5
1.6	2.6		1.0	2.3	1.0

PRELIMINARY TEST III, 1980

Strain	Mean 10 Tests	Ohio	Ind.	Ill.	
		S. Charleston	Lafayette	Urbana	Girard
		HEIGHT (inches)			
Century (II)	33	38	31	33	35
Cumberland (III)	36	36	34	37	39
Union (IV)	44	44	44	46	46
A79-237005	37	38	34	37	37
A79-331022	38	40	34	38	41
A79-334010	39	39	36	41	40
A79-336014	35	37	31	35	37
A79-337010	41	43	39	43	43
A79-338015	37	38	34	37	39
A79-338021	40	41	39	41	42
HC76-3711	19	28	12	19	17
HC76-3863	21	24	12	22	20
HC76-4374	19	22	9	18	18
HC76-4388	19	24	10	19	18
HC77-1418	26	34	18	26	25
HC77-1419	22	30	11	20	21
HC77-5421	20	25	13	21	19
HW79050	34	38	31	32	35
HW79116	37	42	32	36	39
HW79149	38	38	31	36	40
K1055	39	38	39	40	41
K1056	39	39	36	37	40
K1057	39	38	37	41	41
K1058	39	40	36	39	43
K1059	30	33	33	30	29
K1060	39	38	37	38	42
L77-443	38	40	35	39	39
L77-3014	34	38	35	36	34
L77-709	36	39	32	36	39
L78-4883	35	38	31	35	36
LN1060	26	32	21	25	25
LN1062	27	32	24	26	26
LN1063	31	37	24	31	27
LN78-246	34	38	35	38	35
LN78-537	39	38	39	39	41
LN78-575	37	36	35	36	36

PRELIMINARY TEST III, 1980

Iowa		S.D.	Neb.	Kans.	Ky.
Stuart	Ottumwa	Elk Point	Mead	Manhattan	Lexington
HEIGHT (inches)					
30	36	32	23	38	34
34	38	37	28	45	29
44	47	43	37	52	40
34	39	38	33	45	38
36	42	41	34	42	35
36	40	40	34	47	35
34	36	35	31	42	30
40	42	39	39	50	32
35	38	37	31	47	32
40	40	41	32	46	33
19	18	20	15	15	23
24	22	20	17	19	29
17	18	24	18	19	25
20	21	20	16	18	22
25	24	31	23	25	28
26	18	27	22	22	27
22	22	20	18	20	23
30	34	33	27	41	34
34	40	36	31	44	31
40	41	40	28	48	36
40	42	40	35	46	33
36	42	41	34	44	38
39	42	40	35	43	34
38	40	40	31	47	37
31	26	36	26	26	31
40	38	42	35	43	36
37	38	40	31	44	37
34	30	38	26	35	33
33	35	35	31	41	34
36	37	40	31	32	34
26	28	30	20	26	29
28	27	31	25	25	30
32	31	36	30	29	32
33	30	38	29	31	34
39	41	37	36	42	39
36	38	39	31	44	38

PRELIMINARY TEST III, 1980

Strain	Mean 9 Tests	Ohio	Ind.	Ill.	
		S. Charleston	Lafayette	Urbana	Girard
QUALITY (score)					
Century (II)	2.4	1.5	1.5	3.0	3.5
Cumberland (III)	2.0	1.0	1.5	3.0	2.0
Union (IV)	1.7	1.0	1.5	2.3	1.5
A79-237005	2.0		1.5	1.8	2.0
A79-331022	2.3	2.0	1.5	2.3	2.5
A79-334010	2.2	1.5	1.5	2.5	2.5
A79-336014	2.0	1.0	2.0	2.3	2.0
A79-337010	2.1	1.5	1.5	2.5	2.3
A79-338015	2.0	1.0	1.5	2.0	2.0
A79-338021	2.5	2.0	2.0	2.8	3.3
HC76-3711	1.9	1.0	2.0	1.8	2.5
HC76-3863	2.2	1.0	1.5	2.0	2.5
HC76-4374	2.0	1.0	2.0	2.0	2.0
HC76-4388	2.0	1.0	1.5	3.0	2.3
HC77-1418	1.8	1.0	1.5	1.5	2.3
HC77-1419	1.6	1.0	1.5	1.8	2.0
HC77-5421	2.0	1.0	2.0	1.8	1.5
HW79050	2.3	1.0	1.5	2.8	2.5
HW79116	2.1	1.0	1.0	2.5	2.3
HW79149	2.7	2.0	1.5	2.5	3.3
K1055	1.9	1.0	1.0	1.8	1.5
K1056	1.7	1.0	1.0	1.8	2.0
K1057	1.6	1.5	1.0	2.3	1.5
K1058	1.8	1.5	1.0	2.5	1.8
K1059	1.9	1.5	1.0	2.3	1.8
K1060	1.7	1.0	1.5	2.0	1.5
L77-443	1.9	1.0	1.5	2.3	2.3
L77-3014	1.9	1.0	1.0	2.0	2.8
L78-709	1.9	1.0	1.5	2.0	2.3
L78-4883	1.9	1.5	1.5	1.5	2.5
LN1060	1.7	1.0	1.5	1.8	1.8
LN1062	1.5	1.0	1.5	1.8	1.8
LN1063	1.7	1.5	1.5	2.0	1.5
LN78-246	2.2	1.5	1.5	2.5	2.8
LN78-537	1.7	1.0	1.5	2.0	2.0
LN78-575	2.2	1.5	2.0	3.0	3.0

PRELIMINARY TEST III, 1980

Iowa		S.D.	Neb.	Kans.	Ky.
Stuart	Ottumwa	Elk Point	Mead	Manhattan	Lexington
QUALITY (score)					
	2.0	3.0	2.0	3.5	2.0
	1.3	2.0	2.5	3.0	2.0
	1.4	1.0	2.0	3.0	2.0
	1.3	3.0	2.3	3.0	2.0
	1.5	2.0	2.8	4.0	2.0
	1.6	2.0	2.0	3.5	3.0
	1.5	3.0	2.3	3.0	1.0
	1.5	2.0	2.3	3.0	2.0
	1.8	3.0	2.3	3.0	1.0
	2.0	3.0	2.8	3.0	2.0
	1.5	2.0	1.8	3.5	1.0
	2.2	3.0	2.0	3.5	2.0
	1.3	2.0	1.8	4.0	2.0
	1.4	3.0	1.8	3.0	1.0
	1.4	2.0	2.0	2.5	2.0
	1.3	2.0	2.0	2.0	1.0
	1.4	4.0	2.0	3.0	1.0
	1.4	3.0	3.3	3.5	2.0
	1.3	3.0	3.0	4.0	1.0
	2.0	4.0	3.3	4.0	2.0
	1.9	4.0	2.0	2.5	1.0
	1.3	1.0	1.8	3.0	2.0
	1.3	1.0	2.0	3.0	1.0
	1.6	1.0	2.0	3.0	2.0
	1.4	2.0	1.8	3.0	2.0
	1.4	1.0	2.0	3.0	2.0
	1.9	2.0	2.5	3.0	1.0
	1.4	3.0	2.0	2.0	2.0
	1.4	2.0	2.0	3.0	2.0
	1.6	3.0	1.8	2.0	2.0
	1.3	2.0	1.8	3.0	1.0
	1.3	1.0	2.0	2.0	1.0
	1.3	2.0	1.8	2.0	2.0
	1.6	2.0	2.8	3.0	2.0
	1.7	2.0	1.8	2.5	1.0
	1.6	2.0	2.3	3.0	1.0

PRELIMINARY TEST III, 1980

Strain	Mean 9 Tests	Ohio	Ind.	Ill.	
		S. Charleston	Lafayette	Urbana	Girard
SIZE (g/100)					
Century (II)	18.3	18.2	18.9	19.2	16.3
Cumberland (III)	17.8	18.2	18.6	19.2	16.3
Union (IV)	18.0	17.7	19.5	19.4	17.1
A79-237005	15.3	15.5	16.9	15.9	13.4
A79-331022	18.2	18.7	20.0	19.2	18.8
A79-334010	15.7	14.8	17.0	16.3	14.7
A79-336014	18.6	18.3	20.2	19.1	18.3
A79-337010	16.6	15.6	18.3	17.5	15.3
A79-338015	15.1	14.7	15.8	14.9	15.4
A79-338021	16.1	16.5	18.1	15.7	16.0
HC76-3711	17.6	17.7	20.7	16.1	14.2
HC76-3863	17.0	18.8	16.4	16.5	15.7
HC76-4374	14.5	14.7	16.9	13.7	12.1
HC76-4388	16.4	16.5	19.0	15.0	14.9
HC77-1418	15.0	15.2	16.7	15.6	14.7
HC77-1419	14.4	15.4	16.3	15.1	14.0
HC77-5421	13.6	14.4	13.7	12.5	12.8
HW79050	19.3	19.8	19.8	20.5	18.4
HW79116	18.1	16.9	19.3	19.0	17.4
HW79149	17.5	17.7	18.3	18.5	17.1
K1055	16.7	16.6	17.7	16.9	16.6
K1056	16.7	17.1	17.1	17.2	16.6
K1057	14.6	14.6	15.4	14.4	13.8
K1058	18.2	16.8	19.2	18.6	19.4
K1059	16.8	17.2	17.8	17.4	15.9
K1060	16.6	16.8	17.2	16.5	16.6
L77-443	16.3	16.3	17.9	16.3	14.5
L77-3014	17.3	19.2	18.9	16.3	16.8
L78-709	16.5	17.2	16.7	17.5	14.2
L78-4883	17.9	19.0	20.0	17.9	17.6
LN1060	16.1	17.2	16.7	16.8	15.4
LN1062	14.7	14.9	15.1	15.4	13.6
LN1063	15.3	15.6	16.8	15.5	13.8
LN78-246	16.2	15.8	17.6	16.7	13.8
LN78-537	17.9	19.3	19.0	17.8	16.1
LN78-575	18.3	20.0	20.1	18.8	16.5

PRELIMINARY TEST III, 1980

Iowa		S.D.	Neb.	Kans.	Ky.
Stuart	Ottumwa	Elk Point	Mead	Manhattan	Lexington
SIZE (g/100)					
	18.2	22.4	19.7	16.6	14.9
	17.9	19.8	18.4	16.4	15.8
	19.4	19.0	17.7	16.5	15.4
	15.0	17.3	16.9	15.5	11.5
	18.4	18.2	18.3	16.8	15.0
	16.4	17.0	17.1	16.6	11.5
	18.9	21.3	19.7	16.4	15.1
	17.4	17.4	17.2	16.4	14.1
	14.6	16.8	15.2	15.0	13.1
	15.9	17.2	15.1	16.4	13.8
	14.1	22.7	22.0	16.9	13.8
	15.1	20.5	20.5	16.4	13.4
	12.4	17.2	16.4	16.4	11.0
	14.5	17.0	19.7	16.7	14.2
	15.2	14.8	15.8	15.1	11.8
	13.2	15.0	15.3	14.0	11.2
	11.4	17.4	15.3	13.7	11.2
	18.4	23.2	20.4	16.5	16.6
	18.5	20.1	18.7	16.7	16.0
	20.0	19.7	16.6	16.7	12.9
	18.5	17.4	15.7	16.5	14.1
	18.1	16.3	17.1	16.5	14.0
	15.8	14.3	14.3	16.4	12.3
	20.2	19.1	17.6	16.6	16.3
	16.4	18.6	17.2	16.5	14.1
	18.2	16.6	15.8	16.4	15.1
	16.8	19.0	16.9	15.1	14.1
	17.8	18.2	18.2	16.4	14.1
	15.4	19.0	18.2	16.0	14.4
	19.2	17.9	18.1	16.6	15.2
	15.8	16.2	16.8	16.6	13.7
	15.4	15.2	15.4	15.3	12.0
	15.4	15.9	15.1	16.5	12.9
	16.1	19.0	18.3	15.6	13.0
	18.2	20.5	19.6	16.5	14.3
	18.8	19.2	19.3	16.6	15.0

PRELIMINARY TEST III, 1980

Strain	Mean 5 Tests	Ill.		Ohio	Iowa	Ind.
		Urbana	Girard	S. Charles- ton	Ottumwa	Lafay- ette
			PROTEIN (%)			
Century (II)	42.3	43.3	41.6	43.7	41.3	41.7
Cumberland (III)	41.8	41.5	40.9	42.9	39.6	43.9
Union (IV)	42.2	43.4	41.0	42.3	41.8	42.3
A79-237005	42.2	41.0	43.1	44.6	39.1	43.2
A79-331022	42.5	41.8	43.0	43.4	42.6	41.6
A79-334010	42.2	42.2	41.3	42.6	42.8	42.2
A79-336014	41.8	40.9	41.8	42.8	42.4	41.3
A79-337010	43.0	44.7	41.7	42.4	43.3	43.0
A79-338015	39.2	39.3	39.1	38.3	39.2	40.3
A79-338021	42.5	42.9	41.3	43.2	42.1	43.1
HC76-3711	42.9	41.9	42.7	44.0	41.3	44.7
HC76-3863	41.2	41.4	40.9	40.7	41.0	42.2
HC76-4374	42.1	42.0	40.6	43.0	41.8	43.3
HC76-4388	43.9	44.3	43.7	44.4	42.9	44.1
HC77-1418	43.9	44.8	42.2	45.2	43.0	44.5
HC77-1419	42.2	41.3	41.1	44.3	39.9	44.4
HC77-5421	42.3	43.0	42.5	43.1	41.1	42.0
HW79050	39.5	37.2	40.0	39.9	39.1	41.5
HW79116	39.6	38.8	38.9	40.1	39.7	40.6
HW79149	41.2	39.6	39.7	42.7	42.0	41.9
K1055	42.5	43.1	41.6	42.8	42.5	42.3
K1056	41.5	41.8	42.6	41.7	40.7	40.6
K1057	41.3	41.0	41.6	41.9	40.0	41.9
K1058	43.8	44.2	42.1	44.5	45.0	43.3
K1059	44.4	44.1	44.3	45.3	43.8	44.4
K1060	42.6	42.9	42.1	43.5	41.3	43.0
L77-443	40.5	41.6	37.8	41.4	39.3	42.6
L77-3014	42.5	42.6	40.8	44.1	42.2	42.6
L78-709	40.8	40.5	39.6	42.0	39.9	41.8
L78-4883	41.5	39.6	40.5	43.2	41.4	42.6
LN1060	42.0	41.7	43.2	42.3	39.7	43.2
LN1062	42.4	42.7	42.5	43.7	40.6	42.7
LN1063	45.5	46.5	44.6	47.0	43.7	45.7
LN78-246	42.3	43.2	40.3	43.9	41.5	42.5
LN78-537	42.7	42.0	42.5	43.4	42.4	43.2
LN78-575	40.9	42.1	39.5	41.7	39.2	42.0

Mean 5 Tests	Ill.		Ohio	Iowa	Ind.
	Urbana	Girard	S. Charleston	Ottumwa	Lafayette
			OIL (%)		
21.8	20.6	22.3	22.0	21.2	22.9
21.7	21.5	22.7	21.1	21.2	22.2
21.9	21.7	22.3	21.6	21.6	22.1
21.5	20.7	21.8	21.6	21.6	21.8
21.3	20.0	21.8	21.7	21.5	21.5
20.6	19.0	20.5	20.9	20.9	21.8
21.7	21.5	22.1	21.7	20.6	22.8
20.1	19.9	21.5	18.7	19.1	21.4
22.5	22.6	22.4	22.3	22.8	22.3
20.9	20.6	21.3	21.3	20.2	21.3
21.8	22.0	22.3	22.1	21.8	21.0
22.9	21.2	23.1	23.4	23.3	23.5
21.5	20.9	22.1	20.9	21.7	21.9
21.1	20.7	22.0	19.1	21.1	22.8
21.3	20.8	21.9	20.7	21.8	21.4
22.0	21.7	23.3	21.7	21.5	21.6
21.4	20.9	21.6	20.9	22.3	21.3
22.0	21.1	21.2	22.0	23.1	22.6
22.4	21.5	23.5	21.7	22.5	22.8
21.0	20.4	21.6	19.4	20.8	22.7
20.7	19.6	21.5	20.1	21.0	21.1
20.4	20.2	20.9	20.0	20.6	20.4
20.4	19.8	21.5	19.5	20.3	20.7
20.4	18.5	20.5	20.7	20.9	21.2
20.4	20.0	20.7	19.4	20.6	21.3
19.7	19.2	20.2	18.8	19.2	21.1
21.8	20.7	23.5	20.8	21.6	22.2
21.8	21.3	22.7	21.2	22.0	21.8
22.3	21.5	23.6	21.6	23.2	21.7
21.7	20.9	21.0	21.4	22.5	22.6
20.9	20.8	21.4	20.5	20.3	21.5
20.8	20.9	22.4	18.8	19.9	22.1
20.3	19.3	21.1	19.3	21.0	20.8
20.9	20.2	21.7	20.2	21.6	21.0
21.7	21.7	22.8	21.2	21.4	21.5
21.4	20.2	21.0	21.5	22.0	22.5

UNIFORM TEST IV, 1980

Strain	Parentage	Previous Testing*	Generation Compositd
Franklin	L12 x Custer	1	F ₃
Union (IV)	Williams ⁵ x SL12 (Wayne <u>Rpm</u> <u>Rps</u>)	4	F ₃
William79 (III)	Williams ⁶ x Lee 68	1	10F ₃
HC76-3840	L72D-2567 x Hodgson	PIV	F ₅
K1033 Douglas	Williams x Calland	2	F ₅
K1041	Williams x Calland	1	F ₆
K1044	Tracy x Williams	PIV	F ₅
K1045	Tracy x Williams	PIV	F ₅
K1046	Tracy x Williams	PIV	F ₅
Ky75-146-74	L66-1359 x Columbus	PIV	F ₅
L73-318	Williams ² x L69-5343(L6- <u>Ir</u> - <u>Im</u>)	PIV	F ₃
L74D-609 Pixie	Williams x Ransom	3	F ₅
L74L-125	Calland x Williams	2	F ₆
L74L-358	L68-4096 - L66L-177	PIV	F ₆

*Number of years in this test or name of 1979 test.

Descriptive and Other Data

Strain	Descriptive Code	Chlorosis Score		Hypocotyl	Shattering	
				Score	Texas	Manhattan
		Ames	Lamberton	Ames	Lubbock	2 Weeks
Franklin	PGBr DYIb	2.2	3.4	2	2.5	2
Union (IV)	WTTn SYB1	3.3	3.5	4	1.5	1
Williams79 (III)	WTTn SYB1	3.5	2.5	4	1.8	1
HC76-3840	PTTn SYBr	4.0	4.5	5	1.0	1
K1033 Douglas	WTBr DYB1	2.8	2.3	5	1.0	2
K1041	WTTn DYB1	3.0	2.5	5	2.0	3
K1044	WTTn SYB1	4.7	5.0	1	2.0	2
K1045	WTTn SYB1	3.0	5.0	1	2.2	2
K1046	WTTn SYB1	4.3	4.0	1	2.5	3
Ky75-146-74	P+WTBr DYB1	3.7	4.5	1	2.0	4
L73-318	WTBr SYBr+Y	3.3	4.0	5	2.0	1
L74D-609 Pixie	PTTn SYB1	3.2	4.0	1	1.0	4
L74L-125	PTTn DYB1	1.8	1.2	5	1.5	2
L74L-358	WTTn DYY	3.3	4.0	4	1.8	1

UNIFORM TEST IV, 1980

Disease Data

Strain	BSR		GERM*	SMV	PSB	PS	PR	PR	Race 1
	Laf.	Ames					Vickory	Laf.	Ames
	Ind.	Ia.	Lafayette, Ind.				Ohio	Ind.	Ia.
	n	n	d	a	d	a	n	a	a
	%	Reac.	%	Score	%	Score	-----Reaction-----		
Franklin	80	S	95	1	3	4S	2.5	R	R
Union (IV)	80	S	90	2M	2	4E	1.8	R	R
Williams79 (III)	100	S	95	3E	2	4E	1.6	R	R
HC76-3840	80	S	90	3M	2	3E	3.6	S	S
K1033 Douglas	100	S	84	1	8	4E	1.7	R	R
K1041	0	S	83	4E	12	3E	1.7	R	R
K1044	40	S	99	2M	0	2E	2.0	R	R
K1045	80	S	87	4M	3	2E	1.7	R	R
K1046	60	S	97	3E	0	4E	1.5	R	R
Ky75-146-74	80	S	89	2M	3	3E	1.5	S	S
L73-318	0	S	84	1	6	4E	1.6	R	R
L74D-609 Pixie	20	S	98	3E	1	2E	3.0	S	S
L74L-125	100	S	92	3E	5	4E	2.5	S	S
L74L-358	80	S	92	4E	5	5S	1.8	S	S

*Petri dish germination on potato dextrose agar.

The strain K1041 averaged 2 bushels per acre higher in the 1980 test and 1.5 bushels per acre higher in the 2-year average yield than Union. K1041 averaged 1 bushel per acre higher in yield than Douglas and averaged nearly five days earlier in maturity than Douglas. In other characteristics K1041 was very similar to Union. L74L-125 averaged about 1 bushel per acre higher in yield than Union, had excellent lodging resistance, and was very resistant to iron chlorosis in both the Iowa and Minnesota tests.

UNIFORM TEST IV, 1980

Regional Summary

Strain	Yield	Rank	Matu- rity	Lodg- ing	Height	Seed Quality	Seed Size	Protein	Oil
No. of Tests	18	18	17	20	21	19	17	5	5
	bu/a	No.	Date	Score	In.	Score	g/100	%	%
Franklin	32.0	13	+3	1.8	42	2.2	14.1	39.6	19.9
Union (IV)	36.7	6	9/26*	1.9	39	2.0	16.8	43.1	19.1
Williams79 (III)	36.0	9	-4	1.7	36	2.0	15.7	42.9	19.2
HC76-3840	25.0	14	+1	1.1	16	2.6	16.5	42.2	19.1
K1033 Douglas	37.9	2	+6	1.7	35	2.7	17.1	42.3	19.1
K1041	38.8	1	+1	1.9	37	2.2	15.8	41.2	19.2
K1044	37.3	4	+10	1.6	37	1.9	14.7	43.7	17.7
K1045	35.1	10	+4	2.3	40	2.0	15.9	43.4	18.2
K1046	37.1	5	+5	2.1	40	1.8	16.1	44.2	18.2
Ky75-146-74	36.4	7	+2	1.8	37	2.2	14.9	42.8	19.3
L73-318	34.3	11	+2	1.8	41	2.3	16.0	42.9	19.2
L74D-609 Pixie	36.3	8	-1	1.3	19	1.8	15.8	41.8	19.2
L74L-125 <i>Lawrence</i>	37.9	2	0	1.4	36	2.4	16.6	42.4	18.9
L74L-358	34.0	12	+1	1.9	38	2.4	14.7	41.6	19.0

*126 days after planting

1979-1980, 2-year mean

No. of Tests	41	41	36	43	44	42	36	10	10
Franklin	35.3	7	+2.7	2.0	41	2.2	14.6	38.8	20.5
Union	40.3	4	9/27*	2.1	39	2.1	17.5	42.4	19.6
Williams79	39.9	6	-3.8	1.7	36	2.0	16.6	41.8	20.1
K1033 Douglas	40.5	3	+5.8	1.8	35	2.6	17.8	41.0	20.0
K1041	41.8	1	+1.2	2.1	38	2.3	16.6	40.2	20.0
L74D-609 Pixie	40.1	5	+0.5	1.4	20	1.9	16.3	41.6	19.8
L74L-125 <i>Lawrence</i>	41.1	2	+0.4	1.5	35	2.3	17.4	41.8	19.6

*128 days after planting

1978-1980, 3-year mean

No. of Tests	64	64	56	63	67	65	56	19	19
Union (IV)	40.8	3	9/25.9*	2.2	40	2.1	17.6	42.4	20.2
K1033 Douglas	41.4	1	+6.6	1.9	36	2.6	17.9	41.6	20.4
L74D-609 Pixie	40.6	4	+0.7	1.4	20	1.8	16.4	42.2	20.5
L74L-125 <i>Lawrence</i>	41.4	1	+0.3	1.5	36	2.3	17.3	42.0	20.2

*125 days after planting

UNIFORM TEST IV, 1980

Strain	Mean 18 Tests	Del. George- town	N.J. Adel- phia	Md. Queens- town	Penn. Landis- ville	Ohio S. Charles- ton	Neb. Lincoln
YIELD (bu/a)							
Franklin	32.0	31.0	21.2	37.6	31.4	43.5	30.6
Union (IV)	36.7	32.2	33.0	36.5	31.0	54.0	40.0
Williams79 (III)	36.0	33.4	25.2	33.1	29.7	56.5	42.1
HC76-3840	25.0	27.6	22.8	26.3	31.7	61.6	26.4
K1033	37.9	33.1	26.8	41.1	30.7	60.3	41.2
K1041	38.8	35.2	29.5	35.8	37.3	63.6	44.0
K1044	37.3	32.6	23.4	42.0	29.2	48.1	42.4
K1045	35.1	33.0	29.0	41.6	29.0	52.4	38.0
K1046	37.1	36.3	24.4	37.9	32.2	59.9	39.0
Ky75-146-74	36.4	31.6	25.4	38.4	27.4	60.2	41.4
L73-318	34.3	33.9	25.2	33.8	25.9	52.8	36.2
L74D-609	36.3	30.1	32.0	41.6	34.8	63.2	42.2
L74L-125	37.9	35.2	29.6	41.0	33.8	54.0	36.1
L74L-358	34.0	33.4	25.4	34.3	31.6	52.4	38.2
C.V. (%)		11.7	17.2	9.6	13.4	7.3	9.3
L.S.D. (5%)		3.0	9.2	4.9	N.S.	6.9	6.0
Row sp (in.)		30"	30"	30"	30"	30"	30"
Rows/plot		4	3	4	4	4	4
Reps		3	4	3	3	3	3

	18 Tests	YIELD RANK					
Franklin	13	12	14	8	7	14	13
Union (IV)	6	10	1	9	8	8	7
Williams79 (III)	9	5	9	13	10	7	4
HC76-3840	14	14	13	14	5	3	14
K1033	2	7	6	4	9	4	6
K1041	1	2	4	10	1	1	1
K1044	4	9	12	1	11	13	2
K1045	10	8	5	2	12	11	10
K1046	5	1	11	7	4	6	8
Ky75-146-74	7	11	7	6	13	5	5
L73-318	11	4	9	12	14	10	11
L74D-609	8	13	2	2	2	2	3
L74L-125	2	3	3	5	3	8	12
L74L-358	12	6	7	11	6	11	9

*Not included in mean

UNIFORM TEST IV, 1980

Strain	Ky.	Ind.		Ill.			
	Lexing- ton	Lafay- ette	Sulli- van*	Belle- ville	Eldo- rado	Browns- town	Carbon- dale
	YIELD (bu/a)						
Franklin	37.6	55.4	43.9	36.6	43.6	29.2	22.3
Union (IV)	37.1	52.7	47.1	47.7	49.2	35.6	26.4
Williams79 (III)	38.5	52.0	55.9	45.1	46.9	38.0	32.0
HC76-3840	41.7	9.0	17.3	34.1	44.4	30.6	27.1
K1033	42.0	61.7	49.9	49.3	49.7	39.6	25.5
K1041	41.2	58.8	52.3	45.7	51.4	33.7	30.7
K1044	40.2	53.2	49.7	50.4	45.6	40.7	28.7
K1045	39.1	48.8	46.7	44.9	43.7	34.8	29.9
K1046	40.0	54.4	43.8	45.4	46.3	37.9	30.6
Ky75-146-74	37.5	64.7	57.1	40.5	44.4	36.4	26.0
L73-318	35.5	47.5	60.7	42.2	46.1	34.8	27.4
L74D-609	37.2	36.8	43.4	59.0	52.2	40.1	33.3
L74L-125	41.1	56.2	49.0	46.1	48.3	36.9	34.8
L74L-358	39.9	42.5	61.3	43.8	47.3	34.5	25.8
C.V. (%)	10.8	14.1	24.1	5.5	5.8	8.4	15.2
L.S.D. (5%)	2.4	11.3	19.0	4.1	4.6	5.3	7.2
Row sp (in.)	30"	24"	30"	30"	30"	30"	30"
Rows/plot	4	4	3	4	4	4	4
Reps	3	3	3	3	3	3	3

	YIELD RANK						
Franklin	13	5	12	13	14	14	14
Union (IV)	12	8	9	4	4	8	10
Williams79 (III)	9	9	4	8	7	4	3
HC76-3840	2	14	14	14	11	13	9
K1033	1	2	6	3	3	3	13
K1041	3	3	5	6	2	12	4
K1044	5	7	7	2	10	1	7
K1045	8	10	10	9	13	9	6
K1046	6	6	11	7	8	5	5
Ky75-146-74	10	1	3	12	11	7	11
L73-318	14	11	2	11	9	9	8
L74D-609	11	13	13	1	1	2	2
L74L-125	4	4	8	5	5	6	1
L74L-358	7	12	1	10	6	11	12

UNIFORM TEST IV, 1980

Mo.				Kan.			Texas
Novelty	Clinton*	Portageville		Manhattan	Powhattan	Ottawa	Lubbock
		Loam*	Clay				
YIELD(bu/a)							
34.3	8.7	22.4	20.1	19.9	21.0	19.4	41.8
38.0	13.7	19.5	23.1	31.4	21.4	23.1	48.0
36.8	12.1	29.5	23.0	34.7	16.4	22.5	41.2
14.0	6.1	1.4	1.1	12.6	11.1	7.0	20.6
36.5	17.4	15.2	21.3	32.7	18.8	24.7	47.3
39.1	11.2	19.8	24.9	35.4	18.5	21.1	53.2
36.6	15.5	23.0	35.6	31.9	24.9	22.3	42.9
25.4	14.6	29.9	28.1	31.1	19.3	23.2	41.3
37.8	13.1	25.8	26.4	30.5	18.9	23.9	46.1
38.0	12.9	26.7	25.8	28.3	19.6	21.9	47.3
35.2	12.9	28.3	25.4	27.0	21.7	19.6	47.9
26.7	11.4	19.4	16.2	38.7	17.8	16.1	34.7
40.8	13.6	27.7	27.3	27.4	23.5	20.3	50.3
33.4	12.1	28.2	18.8	25.2	19.4	18.1	47.4
13.4	20.5	24.8	18.3	15.8	13.5	11.9	9.2
6.5	3.7	9.4	6.9	7.5	4.3	3.9	6.7
30"	30"	30"	30"	30"	30"	30"	40"
2	2	4	4	4	4	4	4
4	4	3	3	3	3	3	3
YIELD RANK							
10	13	9	11	13	5	11	10
3	4	11	8	6	4	4	3
6	9	2	9	3	13	5	12
14	14	14	14	14	14	14	14
8	1	13	10	4	10	1	7
2	12	10	7	2	11	8	1
7	2	8	1	5	1	6	9
13	3	1	2	7	8	3	11
5	6	7	4	8	9	2	8
3	7	6	5	9	6	7	6
9	7	3	6	11	3	10	4
12	11	12	13	1	12	13	13
1	5	5	3	10	2	9	2
11	9	4	12	12	7	12	5

UNIFORM TEST IV, 1980

Strain	Mean 17 Tests	Del. George- town	N.J. Adel- phia	Md. Queens- town	Penn. Landis- ville	Ohio S. Charles- ton	Neb. Lincoln
MATURITY (date)							
Franklin	+3	+1	+1	+7	+3	+2	0
Union (IV)*	9/26	10/6	9/29	9/24	10/3	9/27	9/30
Williams79 (III)	-4	0	-6	-1	-4	-3	-2
HC76-3840	+1	+6	-7	+5	0	+1	-1
K1033	+6	+5	+4	+16	+6	+5	+5
K1041	+1	+1	-4	+1	0	-1	+2
K1044	+10	+6	+6	+18	+7	+10	+6
K1045	+4	0	+5	+10	+3	+4	+4
K1046	+5	0	+6	+10	+3	+6	+3
Ky75-146-74	+2	+3	-3	+5	0	+1	+1
L73-318	+2	0	-1	+7	+1	0	+2
L74D-609	-1	0	-2	+1	0	+3	-4
L74L-125	0	+5	-5	0	0	-1	0
L74L-358	+1	+1	-5	+3	0	-1	0
Date planted	5/23	6/7	5/29	5/30	6/5	4/30	5/21
*Days to maturity	126	121	123	117	120	150	132
Mean 20 Tests							
LODGING (score)							
Franklin	1.8	1.2	2.5	2.8	1.0	2.2	1.0
Union (IV)	1.9	1.0	2.5	3.2	1.0	2.0	1.7
Williams79 (III)	1.7	1.0	1.5	2.7	1.0	2.0	1.3
HC76-3840	1.1	1.0	1.0	1.0	1.0	1.0	1.0
K1033	1.7	1.0	1.8	3.0	1.0	2.0	1.0
K1041	1.9	1.0	2.5	2.8	1.0	2.7	1.7
K1044	1.6	1.0	1.2	2.0	1.0	2.2	1.0
K1045	2.3	1.3	2.8	3.7	1.0	2.8	2.2
K1046	2.1	1.2	2.2	3.2	1.0	2.3	2.0
Ky75-146-74	1.8	1.0	2.0	2.8	1.0	2.2	1.5
L73-318	1.8	1.0	2.2	3.2	1.0	1.5	1.3
L74D-609	1.3	1.0	2.2	1.3	1.0	1.5	1.0
L74L-125	1.4	1.0	1.0	2.2	1.0	1.3	1.0
L74L-358	1.9	1.0	1.8	3.0	1.0	2.0	1.3

UNIFORM TEST IV, 1980

Ky.	Ind.		Ill.			
Lexington	Lafayette	Sullivan	Belleville	Eldorado	Brownstown	Carbondale
MATURITY (date)						
+4	+2	+3	+5	+3	+8	0
9/29	9/28	9/21	9/25	9/18	9/23	9/24
-7	-4	-3	-8	0	-1	-8
0	+2	+7	+1	0	+4	-12
+4	+5	+7	+7	+6	+12	0
+4	-1	-1	-1	+1	+1	+1
+4	+5	+10	+15	+13	+14	+13
+4	+2	+6	+4	+6	+3	+8
+4	+3	+6	+6	+10	+7	+8
+4	+1	+2	+3	+2	+3	0
+4	+2	+1	+4	+4	+10	+1
0	+1	+3	-3	+1	+1	-3
-3	-2	0	-3	-1	+2	0
+4	0	+1	+2	+2	+8	-1
6/8	5/7	5/5	5/27	5/27	5/15	5/27
113	144	139	121	114	131	120
LODGING (score)						
2.2	2.7	2.5	2.2	2.0	1.3	1.4
1.8	2.5	2.8	2.4	2.6	1.7	1.7
1.3	1.8	2.5	1.7	2.6	1.2	1.4
1.2	1.5	1.7	1.3	2.0	1.0	1.0
1.7	2.3	2.7	1.8	1.9	1.7	1.2
1.8	2.2	3.0	2.2	3.2	1.5	1.0
1.3	2.2	3.2	1.9	1.9	1.7	1.4
2.2	2.3	3.7	3.3	3.6	2.2	1.8
2.0	2.3	3.4	2.9	2.7	2.0	1.7
1.3	2.3	2.5	2.2	2.0	1.5	1.5
1.8	2.8	2.8	1.9	3.2	2.0	1.5
1.5	1.2	1.2	1.6	2.9	1.7	1.0
1.2	1.5	2.2	1.6	1.9	1.0	1.4
1.5	2.3	3.3	2.4	3.0	2.0	1.5

UNIFORM TEST IV, 1980

Strain	Mo.		Portageville		Kan.		Texas	
	Novel- ty	Clin- ton	Loam	Clay	Man- hattan	Pow- hattan	Otta- wa	Lub- bock
MATURITY (date)								
Franklin			+1	-1	+3			+3
Union (IV)			9/19	9/26	9/29			9/18
Williams79 (III)			-7	-3	-2			-1
HC76-3840			-7	-2	+8			+17
K1033			+1	+7	+8			+9
K1041			-4	+7	0			+9
K1044			+12	+15	+9			+5
K1045			+1	+7	+6			0
K1046			+2	+7	+7			+1
K775-146-74			0	-1	+6			+4
L73-318			+1	-2	+4			+4
L74D-609			-5	-4	-3			+2
L74L-125			-2	-2	+10			+3
L74L-358			-2	-2	+7			+3
Date planted	5/15	5/28	5/9	5/28	5/19	5/22	5/29	5/24
Days to maturity	-	-	133	121	133	-	-	117
LODGING (score)								
Franklin	2.0		2.0	2.0	2.0	1.0	1.0	1.8
Union (IV)	2.0		2.0	1.5	2.0	1.0	1.0	2.0
Williams79 (III)	1.9		1.5	1.5	2.0	1.0	1.0	2.2
HC76-3840	1.0		1.0	1.0	1.0	1.0	1.0	1.0
K1033	1.7		1.5	1.5	2.0	1.0	1.0	1.2
K1041	2.0		1.5	1.5	2.0	1.0	1.0	2.0
K1044	1.9		1.5	1.5	2.0	1.0	1.0	1.5
K1045	2.4		2.0	2.5	2.7	1.0	1.0	2.0
K1046	2.3		2.0	2.5	3.0	1.0	1.0	1.8
K675-146-74	2.2		2.0	2.0	2.3	1.0	1.0	2.0
L73-318	2.0		1.5	1.5	2.0	1.0	1.0	1.5
L74D-609	1.1		1.0	1.0	1.0	1.0	1.0	1.0
L74L-125	1.5		1.5	1.5	2.0	1.0	1.0	1.2
L74L-358	2.0		2.0	2.0	2.7	1.0	1.0	2.0

UNIFORM TEST IV, 1980

Mean 21 Tests	Del. George- town	N.J. Adel- phia	Md. Queens- town	Penn. Landis- ville	Ohio S. Charles- ton	Neb. Lincoln
HEIGHT (inches)						
42	26	47	42	30	41	40
39	23	44	42	30	43	37
36	23	40	39	26	41	32
16	14	25	17	21	24	20
35	21	35	36	24	40	34
37	23	44	38	30	43	33
37	25	38	36	25	41	35
40	28	44	43	32	43	37
40	30	44	41	32	41	38
37	23	38	38	22	40	35
41	28	44	42	27	43	38
19	17	25	21	23	25	23
36	23	38	37	27	37	30
38	24	37	38	26	40	35

Mean 19 Tests	QUALITY (score)					
2.2	3.0	1.2	2.2	2.3	1.5	2.3
2.0	2.0	1.0	2.3	2.0	1.0	2.7
2.0	2.0	1.0	2.2	2.2	1.5	2.5
2.6	3.3	1.0	3.2	2.0	1.0	2.7
2.7	2.8	1.2	3.7	2.3	2.0	2.7
2.2	2.7	1.0	2.7	2.3	1.5	2.7
1.9	1.8	1.0	1.7	2.0	1.0	2.0
2.0	1.8	1.0	1.8	2.0	1.0	2.0
1.8	2.2	1.0	1.8	2.0	1.0	2.0
2.2	2.7	1.0	2.3	2.0	1.0	2.7
2.3	2.2	1.0	2.7	2.2	1.0	2.8
1.8	1.8	1.2	2.0	2.0	1.0	1.7
2.4	3.2	1.2	2.3	2.5	1.5	3.0
2.4	2.8	1.0	2.2	2.3	1.0	3.0

UNIFORM TEST IV, 1980

Strain	Ky.	Ind.		Ill.			
	Lexing- ton	Lafay- ette	Sulli- van	Belle- ville	Eldo- rado	Browns- town	Carbon- dale
HEIGHT (inches)							
Franklin	45	45	54	56	53	47	49
Union (IV)	42	42	54	53	51	47	43
Williams79 (III)	37	40	48	45	50	43	41
HC76-3840	26	10	19	20	18	17	16
K1033	40	39	43	46	50	43	36
K1041	42	40	46	49	51	44	40
K1044	39	42	46	44	49	41	38
K1045	43	43	49	54	51	46	47
K1046	43	44	49	52	50	44	46
Ky75-146-74	43	40	51	48	47	47	39
L73-318	44	44	57	53	51	52	47
L74D-609	25	14	20	25	17	22	22
L74L-125	39	39	47	43	49	43	41
L74L-358	41	38	49	52	53	47	45

QUALITY (score)							
Franklin	1.0	1.0	2.0	3.5	3.5	3.0	1.0
Union (IV)	1.0	1.0	2.5	2.8	3.3	3.2	1.0
Williams79 (III)	2.0	1.0	2.0	2.2	2.8	2.8	1.5
HC76-3840	2.0	1.0	3.0	2.7	2.8	3.3	1.5
K1033	3.0	3.0	2.0	3.5	3.7	3.7	1.5
K1041	2.0	1.0	1.5	2.8	3.0	3.1	1.0
K1044	1.0	1.0	1.5	2.3	2.0	3.0	1.5
K1045	2.0	1.0	1.0	2.7	2.2	3.1	1.5
K1046	1.0	1.0	1.0	2.2	2.2	3.0	1.5
Ky75-146-74	1.0	1.0	2.0	3.5	3.5	3.3	2.5
L73-318	2.0	2.0	1.5	2.7	3.3	3.5	1.5
L74D-609	2.0	1.0	1.5	1.2	2.3	2.3	1.0
L74L-125	2.0	1.0	2.0	3.2	3.2	3.3	1.5
L74L-358	2.0	1.0	2.0	3.3	3.7	3.9	2.5

UNIFORM TEST IV, 1980

Mo.				Kan.			Texas
Novelty	Clinton	Portageville		Manhattan	Powhattan	Ottawa	Lubbock
		Loam	Clay				
HEIGHT (inches)							
42	35	41	38	51	33	35	27
39	28	35	34	55	30	33	24
32	24	29	33	51	24	31	24
13	11	9	12	16	16	11	9
33	29	15	34	47	26	32	25
37	30	25	30	48	28	30	26
34	31	37	34	48	30	34	32
38	33	31	38	49	32	38	31
39	33	30	45	44	33	38	32
34	28	34	33	51	26	31	26
40	29	33	38	55	31	34	32
18	21	14	12	21	17	15	10
33	25	34	33	48	26	26	28
38	28	41	31	52	29	33	30

QUALITY (score)							
		2.5	3.0	3.0	1.5	2.5	2.7
		2.5	2.5	2.0	1.5	1.5	3.0
		2.5	2.5	2.0	1.5	2.0	2.7
		2.5	3.0	2.5	2.5	4.0	5.0
		3.0	2.5	2.5	3.0	2.0	4.0
		2.5	3.0	2.0	2.5	1.5	3.7
		2.5	2.0	2.0	3.0	3.0	2.5
		2.5	2.5	2.0	2.0	2.0	3.0
		2.0	2.5	2.0	1.5	1.5	2.7
		2.5	2.5	2.0	2.0	2.0	2.7
		2.5	2.5	2.5	2.0	2.0	3.0
		2.5	2.5	2.5	1.0	1.5	2.5
		2.5	2.5	2.5	2.5	2.5	3.5
		3.0	3.0	3.0	1.5	2.0	2.5

UNIFORM TEST IV, 1980

Strain	Mean 17 Tests	Del. George- town	N.J. Adel- phia	Md. Queens- town	Penn. Landis- ville	Ohio S. Charles- ton	Neb. Lincoln
SIZE (g/100)							
Franklin	14.1		11.0	15.3	14.3	13.9	14.1
Union (IV)	16.8		14.0	16.3	17.7	18.6	18.4
Williams79 (III)	15.7		13.0	14.0	17.2	16.6	18.5
HC76-3840	16.5		10.0	17.0	15.4	19.4	19.6
K1033	17.1		12.0	18.4	19.0	18.1	19.0
K1041	15.8		11.0	14.5	17.4	17.2	19.4
K1044	14.7		11.0	17.1	17.2	13.3	16.1
K1045	15.9		11.0	16.1	18.0	16.3	17.6
K1046	16.1		13.0	16.3	17.4	17.0	17.1
Ky75-146-74	14.9		12.0	14.4	14.6	16.2	16.6
L73-318	16.0		14.0	15.8	17.3	17.3	17.4
L74D-609	15.8		12.0	17.7	15.8	17.0	17.7
L74L-125	16.6		12.0	16.8	19.1	17.2	19.6
L74L-358	14.7		9.0	14.6	14.8	15.8	16.0

Strain	Mean 5 Tests	Md. Queens- town	Ind. Lafay- ette	Ill. Eldo- rado	Mo. Portageville Loam	Kan. Man- hattan
PROTEIN (%)						
Franklin	39.6	40.8	39.3	39.0	38.6	40.3
Union (IV)	43.1	44.1	42.2	43.3	42.0	44.0
Williams79 (III)	42.9	44.1	42.0	42.7	42.3	43.6
H76-3840	42.2	44.3	42.5	39.8	41.7	42.8
K1033	42.3	42.1	42.2	42.1	42.8	—
K1041	41.2	43.6	40.2	40.2	41.7	40.2
K1044	43.7	44.2	42.3	44.7	42.5	44.7
K1045	43.4	43.8	42.6	43.0	43.9	43.5
K1046	44.2	44.1	43.3	44.4	44.9	44.2
Ky75-146-74	42.8	43.3	42.3	43.3	41.7	43.4
L73-318	42.9	43.6	42.0	42.4	42.8	43.8
L74D-609	41.8	41.9	42.2	41.3	41.3	42.5
L74L-125	42.4	42.5	42.5	42.2	41.5	42.5
L74L-358	41.6	41.6	41.8	41.7	40.3	42.5

UNIFORM TEST IV, 1980

Ky.	Ind.		Ill.			
Lexington	Lafayette	Sullivan	Belleville	Eldorado	Brownstown	Carbondale
SIZE (g/100)						
14.2	16.2	14.8	13.7	12.0	14.6	12.2
15.0	19.3	18.0	17.6	15.9	17.9	13.7
14.3	18.2	15.9	15.4	13.9	15.4	13.7
13.0	20.2	19.3	16.0	14.2	16.1	11.1
16.0	19.0	19.5	15.9	15.6	18.5	11.3
15.5	17.2	17.2	15.0	14.3	16.1	11.5
13.5	15.5	15.4	13.8	12.0	15.9	12.0
15.2	17.2	15.5	15.8	13.7	17.5	13.8
16.2	18.4	17.0	15.8	13.7	17.4	13.0
12.9	16.4	16.1	14.3	13.3	15.5	12.0
14.2	18.6	16.8	15.7	14.5	17.7	11.9
14.8	17.3	18.8	15.3	14.2	15.1	11.4
15.1	19.4	16.7	16.0	15.3	16.5	14.1
13.8	16.6	16.8	14.5	13.3	15.4	12.3

Mean	Md.	Ind.	Ill.	Mo.	Kan.
5 Tests	Queenstown	Lafayette	Eldorado	Portageville Loam	Ashland
OIL (%)					
19.9	16.2	21.8	22.2	17.2	22.2
19.1	14.9	21.6	21.9	16.0	21.2
19.2	14.5	22.3	22.4	15.9	21.0
19.1	14.3	21.7	22.5	15.7	21.5
19.1	16.7	21.7	21.6	16.2	—
19.2	15.9	21.3	21.3	15.4	22.0
17.7	14.0	20.8	19.2	15.3	19.3
18.2	14.9	20.8	20.7	15.1	19.3
18.2	15.0	21.0	20.6	14.9	19.7
19.3	15.3	22.3	21.7	16.0	21.0
19.2	15.3	21.8	21.7	16.0	21.4
19.2	15.6	21.9	21.2	16.3	21.2
18.9	14.8	21.0	22.2	15.7	20.8
19.0	15.8	21.8	20.8	16.5	20.2

UNIFORM TEST IV, 1980

Strain	Mo.		Kan.			Texas
	Portageville		Manhattan	Powhattan	Ottawa	Lubbock
	Loam	Clay				
			SIZE (g/100)			
Franklin		12.3	14.2	15.6	13.6	17.8
Union (IV)		14.6	16.6	16.6	14.8	20.8
Williams79 (III)		13.8	16.7	16.5	13.5	19.5
HC76-3840		13.8	16.8	16.7	16.6	24.6
K1033		16.7	16.6	16.5	16.4	22.2
K1041		14.4	16.6	16.3	13.6	21.0
K1044		14.9	14.7	14.3	15.0	18.7
K1045		15.1	16.3	16.4	15.3	18.8
K1046		14.0	16.7	16.4	15.1	19.5
Ky75-146-74		13.4	14.7	16.3	14.7	19.2
L73-318		14.8	16.2	16.5	13.5	19.6
L74D-609		13.3	16.5	16.4	15.1	20.2
L74L-125		15.3	16.5	16.7	14.5	21.9
L74L-358		13.2	16.5	15.9	12.9	19.3

PRELIMINARY TEST IV, 1980

Strain	Parentage	Generation Composited
Union (IV)	Williams ⁵ x SL11 (Wayne <u>Rpm Rps</u>)	F ₃
Williams79 (III)	Williams ⁶ x Lee68	10F ₃
A79-331020	(L15 x AP68-1016) x Cumberland	F ₄
A79-335034	A72-512 x Pride B216	F ₄
A79-336007	NKS1492 x A72-512	F ₄
A79-337020	A72-512 x A74-20434	F ₄
C1585	Wells x UFV-1	F ₅
C1586	L72-844c-1 x CX456-90	F ₅
C1587	L72-844c-1 x Wells	F ₅
C1588	L72-844c-1 x Wells	F ₅
C1589	L72-844c-1 x Wells	F ₅
HC76-4449	L72U2567 x Essex	F ₅
HC77-982	Williams x Ransom	F ₇
HC77-1165	Wells x V68-1034	F ₅
HC77-5481	Evans x L72U-2567	F ₅
HC77-5686	L72U2567 x L72D-549	F ₅
K1033 Douglas	Williams x Calland	F ₅
K1061	Tracy x Columbus	F ₅
K1062	Tracy x Williams	F ₅
K1063	Tracy x Williams	F ₅
K1066	Tracy x Pomona	F ₅
K1067	K1001 x Bonus	F ₆
Ky78-405	EMS treated Williams	F ₃
Ky78-1214	EMS treated Williams	F ₃
L77-515	Union x L75-8020	F ₄
L77-546	Union x L75-8020	F ₄
L77-8043	Williams x Mitchell	F ₅
L77-8079	Williams x Mitchell	F ₅
L77-8209	Williams x Mitchell	F ₅
LN1053	Tracy x Pomona	F ₅
LN1057	Williams x D60-9647	F ₅
LN1058	Tracy x Columbus	F ₅
LN1059	Williams x D60-9647	F ₅
LS78-229	L71L-436 x J74-5	F ₄
LS78-335	L71L-436 x J74-5	F ₄
LS78-344	L71L-436 x J74-5	F ₄

PRELIMINARY TEST IV, 1980

Descriptive and Other Data

Strain	Descriptive Code		Chlorosis Score	Shattering Manhattan
			Ames	2 Weeks
Union (IV)	WTTn	SYB1	3.3	1
Williams79 (III)	WTTn	SYB1	3.5	1
A79-331020	PTBr	SYB1	4.2	2
A79-335034	WGBr	DYY	2.0	2
A79-336007	WGBr	DYBf	3.8	3
A79-337020	PGBr	DYIb	3.2	3
C1585	WTBr	DYB1	3.8	2
C1586	PTTn	SYB1	3.0	1
C1587	WGBr	SYBf	3.7	3
C1588	P+WGBr	SYBf	3.5	4
C1589	P+WGBr	DYBf+Ib	2.2	3
HC76-4449	PTTn	DYB1	2.2	1
HC77-982	WTTn	DYB1	3.5	1
HC77-1165	PGTn	DYIb	3.3	2
HC77-5481	WTBr	SYB1	2.8	2
HC77-5686	PTBr	DYB1	3.0	1
K1033 Douglas	WTBr	DYB1	3.5	1
K1061	WTTn	SYB1	2.7	2
K1062	WTTn	DYB1	2.3	2
K1063	WTTn	DYB1	3.2	2
K1066	WTTn	DYB1	3.5	1
K1067	PGBr	DYIb	3.3	1
Ky78-405	WTTn	DYB1	4.0	2
Ky78-1214	WTTn	DYB1	4.0	2
L77-515	WGTn	SYBf	3.8	2
L77-546	WTTn	SYB1	3.0	1
L77-8043	P+WTTn	DYBr	3.8	1
L77-8079	WTTn	SYB1	3.7	1
L77-8209	WTTn	SYB1	4.0	3
LN1053	WTBr	DYB1	2.7	2
LN1057	WTTn	DYB1	3.0	3
LN1058	PGBr+Tn	DYIb	2.3	2
LN1059	WTBr	DYB1	2.7	4
LS78-229	P+WTBr	DYB1	2.5	3
LS78-335	PTBr	SYB1	4.3	2
LS78-344	PTBr	DYB1	3.3	2

PRELIMINARY TEST IV, 1980

Disease Data

Strain	BSR		GERM*	SMV	PSB	PS	PR	Race 1		PR
	Laf.	Ames						Laf.	Ames	
	Ind.	Ia.						Ind.	Ia.	
	n	n						a	a	
	%	Reac.	%	Score	%	Reac.	-----Reac.-----			
Union (IV)	80	S	88	2E	3	5S	R	R		1.8
Williams79 (III)	100	S	94	2E	3	4E	R	R		1.8
A79-331020	40	R	97	4M	1	3E	R	R		1.8
A79-335034	0	S	79	3E	11	5E	H	S		1.8
A79-336007	20	S	93	4E	0	5E	S	S		1.8
A79-337020	60	S	92	1	1	5E	R	R		1.9
C1585	80	S	91	4E	6	4E	R	R		2.0
C1586	0	S	97	4E	0	2E	R	R		2.5
C1587	80	S	92	2M	4	5S	R	S		1.8
C1588	100	S	85	2M	4	5S	R	R		2.2
C1589	60	S	92	2M	3	4E	R	R		2.2
HC76-4449	20	S	97	3M	0	2E	S	S		2.7
HC77-982	20	S	80	2M	5	3E	S	S		2.6
HC77-1165	60	S	97	1	0	3E	H	R		2.1
HC77-5481	40	S	94	1	3	2M	R	R		3.5
HC77-5686	0	S	98	3E	1	3E	S	S		3.9
K1033 Douglas	-	S	88	1	3	3E	R	R		2.4
K1061	60	S	94	3M	2	4E	R	R		2.2
K1062	80	S	97	2M	0	2E	R	R		1.9
K1063	100	S	-	-	-	-	R	R		2.2
K1066	80	S	93	4E	2	3E	R	R		2.1
K1067	100	S	81	1	2	5E	R	R		1.7
Ky78-405	-	S	93	2M	6	4E	S	S		2.9
Ky78-1214	80	S	87	3M	4	3E	S	S		2.2
L77-515	80	S	86	2M	6	5S	S	S		2.2
L77-546	60	S	89	4E	3	4S	H	S		1.6
L77-8043	60	S	94	4E	7	3E	S	S		1.6
L77-8079	20	S	78	4M	14	4E	S	S		2.0
L77-8209	60	S	98	3E	2	3E	S	S		1.7
LN1053	40	S	96	4M	0	2E	R	R		2.0
LN1057	40	S	87	5S	10	3E	R	R		1.5
LN1058	40	S	94	3M	2	5E	R	R		1.6
LN1059	100	S	96	5S	0	2M	R	R		2.3
LS78-229	40	S	95	3M	1	2E	H	S		1.5
LS78-335	80	S	96	5S	1	1	R	H		1.5
LS78-344	0	S	99	3M	0	2E	H	S		1.5

*Petri dish germination on potato dextrose agar.

PRELIMINARY TEST IV, 1980

Regional Summary

Strain	Yield	Rank	Matu- rity	Lodg- ing	Height	Seed Quality	Seed Size	Composition	
No. of Tests	10	10	9	10	10	10	8	5	5
	bu/a	No.	Date	Score	In.	Score	g/100	%	%
Union (IV)	38.6	17	9/25*	2.2	42	2.4	16.5	44.5	19.5
Williams79 (III)	40.0	9	-3	1.8	39	2.1	16.3	42.6	20.3
A79-331020	37.9	23	+4	2.3	45	2.8	16.6	43.1	18.4
A79-335034	39.3	12	-2	2.5	43	2.7	14.3	44.5	18.3
A79-336007	38.6	17	-2	2.8	41	2.3	12.9	43.2	19.1
A79-337020	38.2	20	0	2.3	40	2.5	15.0	42.3	19.2
Cl585	36.7	28	0	2.0	42	3.1	16.6	45.2	18.7
Cl586	37.0	26	-1	1.2	22	2.1	15.8	42.9	19.2
Cl587	37.5	24	+1	1.8	41	2.3	14.8	44.2	19.4
Cl588	37.2	25	+1	2.4	43	3.0	16.0	43.5	19.2
Cl589	35.3	30	-1	2.1	40	2.8	15.5	43.6	19.0
HC76-4449	28.2	20	+4	1.1	23	2.0	14.5	44.4	19.3
HC77-982	36.8	27	-3	1.5	26	2.4	16.2	43.9	20.1
HC77-1165	36.7	28	-3	1.2	27	2.3	14.2	42.4	20.4
HC77-5481	26.3	36	-4	1.3	17	2.6	14.8	43.8	19.3
HC77-5686	26.8	35	-5	1.3	19	2.6	15.7	44.6	19.0
K1033 Douglas	41.2	4	+7	1.9	37	3.1	17.0	42.8	19.3
K1061	41.7	2	+7	2.2	37	2.0	15.0	45.0	17.9
K1062	41.9	1	+2	2.0	40	1.9	15.6	41.6	19.6
K1063	38.5	19	+8	1.6	25	2.2	15.7	43.6	18.4
K1066	38.0	22	+6	1.2	25	1.9	15.1	41.4	19.1
K1067	38.7	16	+3	1.8	44	3.3	14.1	42.4	19.6
Ky78-405	34.8	31	+6	2.3	39	2.6	15.4	43.8	18.6
Ky78-1214	41.2	4	+7	1.9	42	2.3	15.0	42.3	19.6
L77-515	39.4	11	-1	2.0	39	2.0	14.0	42.1	20.1
L77-546	38.8	14	-2	2.1	39	2.5	15.7	42.5	20.1
L77-8043	41.4	3	+2	2.1	40	2.4	14.0	41.6	19.8
L77-8079	39.6	10	+1	1.9	37	2.7	14.3	42.0	20.5
L77-8209	41.0	7	+6	1.9	40	2.3	15.0	41.7	20.5
LN1053	40.1	8	+3	1.4	24	2.0	15.1	43.6	20.0
LN1057	39.1	13	+8	1.9	35	2.8	18.6	45.0	19.9
LN1058	38.8	14	0	2.5	35	2.6	16.3	43.6	19.7
LN1059	41.2	4	+7	2.1	34	2.4	17.4	43.6	19.6
LS78-229	34.5	32	+23	2.1	39	2.2	12.0	39.1	20.4
LS78-335	32.7	33	+22	2.2	39	2.2	11.4	39.1	20.4
LS78-344	32.4	34	+17	2.5	43	2.2	11.7	39.2	21.0

*128 days after planting

Several strains in this test were resistant to specific pathogens. The strains K1061 through K1066 were resistant to races 1, 3, 4, 5, and 6 of phytophthora root rot. The strains L77-515 and L77-516 were resistant to race 3 of the soybean cyst nematode. The strains LN1053 through LN1059 were resistant to races 1 and 3 through 9 of phytophthora root rot. The three LS strains from Carbondale, Illinois, were all resistant to race 3 of the soybean cyst nematode.

PRELIMINARY TEST IV, 1980

Strain	Mean 10 Tests	Mo.		Ill.		Md.	Ky.
		Portageville Loam	Clay	Belle- ville	Eldo- rado	Queens- town	Lexing- ton
YIELD (bu/a)							
Union (IV)	38.6	32.9	22.4	46.7	48.3	35.6	37.3
Williams79 (III)	40.0	29.1	23.2	46.5	51.6	34.7	38.2
A79-331020	37.9	35.3	26.3	40.7	42.5	42.5	39.2
A79-335034	39.3	32.0	25.8	46.2	47.3	36.4	38.2
A79-336007	38.6	29.0	15.5	45.8	44.9	37.3	37.5
A79-337020	38.2	40.5	22.3	39.5	46.9	39.0	38.5
C1585	36.7	36.6	25.7	40.5	43.8	36.9	35.3
C1586	37.0	11.4	17.5	51.3	51.8	40.9	37.0
C1587	37.5	28.4	27.0	48.7	47.4	40.0	37.6
C1588	37.2	31.7	25.2	40.5	44.9	38.8	42.7
C1589	35.3	23.2	23.5	44.6	45.4	36.3	43.7
HC76-4449	38.2	19.1	22.9	54.8	48.3	43.9	38.3
HC77-982	36.8	21.2	23.7	50.7	49.5	38.5	38.3
HC77-1165	36.7	17.3	18.1	49.4	50.9	39.2	40.8
HC77-5481	26.3	7.3	8.4	33.6	31.5	32.1	36.7
HC77-5686	26.8	5.7	4.6	40.6	32.1	31.8	36.0
K1033 Douglas	41.2	32.5	28.1	51.6	50.0	48.6	40.3
K1061	41.7	34.5	30.3	47.2	46.9	38.0	38.1
K1062	41.9	37.5	27.5	50.0	50.4	39.1	40.3
K1063	38.5	22.9	31.9	52.2	52.1	44.5	39.8
K1066	38.0	19.9	17.9	54.6	45.7	40.8	41.5
K1067	38.7	33.4	28.0	43.8	47.5	36.7	38.9
Ky78-405	34.8	33.8	18.0	37.9	41.8	34.5	38.3
Ky78-1214	41.2	38.7	29.2	48.1	45.1	37.1	38.8
L77-515	39.4	28.7	24.1	44.5	43.8	33.8	37.1
L77-546	38.8	31.3	22.5	46.3	46.7	35.8	38.7
L77-8043	41.4	37.6	26.9	51.5	49.7	40.6	35.4
L77-8079	39.6	31.1	25.5	49.9	47.5	37.7	36.9
L77-8209	41.0	31.0	29.1	48.4	48.5	40.6	36.3
LN1053	40.1	12.1	26.7	55.2	50.5	36.6	37.7
LN1057	39.1	24.5	33.3	47.0	46.9	40.6	36.8
LN1058	38.8	24.0	26.7	47.7	49.4	37.8	34.8
LN1059	41.2	28.8	29.7	51.3	49.3	42.4	40.1
LS78-229	34.5	29.3	29.8	36.3	41.3	50.1	27.3
LS78-335	32.7	29.4	29.4	31.8	42.7	49.7	26.5
LS78-344	32.4	31.1	23.7	26.5	39.5	44.3	30.0
C.V. (%)		16.4	13.7	6.7	4.0	12.1	9.2
L.S.D. (5%)		9.2	6.8	6.2	3.8	NS	2.3
Row sp. (in.)		30"	30"	30"	30"	30"	30"
Rows/plot		4	4	4	4	4	4
Reps		2	2	2	2	2	2

PRELIMINARY TEST IV, 1980

	Kan.	Ohio	Ind.	Del.
Strain	Manhattan	S. Charleston	Sullivan	Georgetown
	YIELD (bu/a)			
Union (IV)	23.8	55.1	55.7	28.1
Williams79 (III)	33.7	57.8	59.2	26.3
A79-331020	25.0	51.5	46.1	29.4
A79-335034	27.2	62.3	50.0	27.9
A79-336007	34.1	61.0	49.9	30.6
A79-337020	23.4	55.5	47.3	29.3
C1585	21.4	53.8	46.3	26.5
C1586	34.2	54.2	44.7	26.8
C1587	25.6	49.8	42.7	27.5
C1588	27.1	52.2	42.2	26.8
C1589	20.1	50.3	39.7	25.8
HC76-4449	23.9	54.3	44.7	31.6
HC77-982	19.2	59.6	40.9	26.8
HC77-1165	28.3	51.5	46.3	25.5
HC77-5481	7.0	52.7	29.7	23.5
HC77-5686	8.2	56.2	26.4	26.2
K1033 Douglas	31.5	57.2	45.2	26.7
K1061	25.9	54.2	75.5	26.0
K1062	25.5	59.3	59.1	30.4
K1063	34.4	57.4	19.9	30.0
K1066	33.4	59.6	35.8	30.7
K1067	29.8	47.3	49.3	32.4
Ky78-405	12.9	51.2	51.0	28.7
Ky78-1214	31.2	56.7	51.2	36.3
L77-515	29.7	62.8	56.4	33.3
L77-546	34.8	51.6	49.1	30.9
L77-8043	30.3	59.3	53.2	29.4
L77-8079	27.0	61.0	52.6	26.9
L77-8209	28.4	57.9	57.9	31.9
LN1053	29.5	64.5	59.6	28.6
LN1057	27.6	57.3	47.1	29.7
LN1058	36.0	45.1	58.1	28.5
LN1059	28.4	55.8	61.4	25.2
LS78-229	25.8	36.4	38.7	30.4
LS78-335	22.1	31.9	30.6	32.5
LS78-344	22.5	40.0	41.8	24.9
C.V. (%)	21.2	6.4	19.1	11.1
L.S.D. (5%)	11.1	7.1	17.9	5.2
Row sp. (in.)	30"	30"	30"	30"
Rows/plot	4	4	3	4
Reps	2	2	2	3

PRELIMINARY TEST IV, 1980

Strain	Mean 10 Tests	Mo.		Ill.		Md.	Ky.
		Portageville Loam	Clay	Belle- ville	Eldo- rado	Queens- town	Lexing- ton**
MATURITY (date)							
Union (IV)*	9/25	9/15	9/24	9/27	9/18	9/25	9/29
Williams79 (III)	-3	-5	-1	-5	-3	-3	-3
A79-331020	+4	+4	+1	+10	+4	+8	Frost Kill
A79-335034	-2	-6	0	-2	-1	-5	-7
A79-336007	-2	-1	-1	-3	-1	-4	-1
A79-337020	0	-1	-1	-2	0	-4	-3
C1585	0	+3	0	+1	-1	0	-3
C1586	-1	-1	-1	-4	+1	-2	0
C1587	+1	0	0	+3	+2	-1	0
C1588	+1	+4	-2	0	0	0	+4
C1589	-1	+3	-4	-1	0	-4	-3
HC76-4449	+4	+7	0	+4	+3	+8	+4
HC77-982	-3	-4	-2	-5	-3	-5	-3
HC77-1165	-3	-4	-3	-6	-5	-3	-3
HC77-5481	-4	-4	-3	-8	-4	-4	-11
HC77-5686	-5	-4	-4	-14	-5	-3	-11
K1033 Douglas	+7	+8	+2	+7	+7	+12	+4
K1061	+7	+9	+8	+9	+9	+9	+4
K1062	+2	-4	0	+1	+5	+12	+4
K1063	+8	+10	+15	+7	+10	+12	+4
K1066	+6	+6	+7	+2	+9	+9	+4
K1067	+3	+3	+1	+2	+3	0	+4
Ky78-405	+6	+9	+2	+12	+7	+8	+4
Ky78-1214	+7	+9	+7	+7	+8	+11	+4
L77-515	-1	-3	-2	-3	0	+2	-3
L77-546	-2	-3	-2	-3	-2	0	-3
L77-8043	+2	+3	-1	+2	+3	+5	+4
L77-8079	+1	+3	0	-2	+1	-2	+4
L77-8209	+6	+10	+1	+6	+8	+11	+4
LN1053	+3	+6	0	0	0	+10	+4
LN1057	+7	+8	+8	+6	+9	+13	+4
LN1058	0	0	-2	-3	+1	-1	-3
LN1059	+3	+7	-1	0	+1	+12	0
LS78-229	+20	+23	+17	+21	+23	+23	Frost Kill
LS78-335	+20	+22	+15	+21	+23	+25	Frost Kill
LS78-344	+17	+17	+9	+19	+16	+20	Frost Kill
Date planted	5/24	5/9	5/28	5/27	5/21	5/29	6/8
*Days to maturity	128	129	119	123	120	119	113

** Not included in the mean.

PRELIMINARY TEST IV, 1980

	Kan.	Ohio	Ind.	Del.
Strain	Manhattan	S. Charleston	Sullivan	Georgetown
	MATURITY (date)			
Union (IV)	10/4	9/26	9/21	9/30
Williams79 (III)	-6	-2	-2	0
A79-331020	0	+4	+3	+6
A79-335034	-3	0	-1	+1
A79-336007	0	+2	0	0
A79-337020	-3	+2	+1	+7
C1585	+1	0	0	+1
C1586	-6	0	+1	+2
C1587	0	+1	+2	+1
C1588	0	0	+1	0
C1589	-3	0	0	-1
HC76-4449	-2	+4	+5	+5
HC77-982	-15	0	-2	+8
HC77-1165	-9	0	0	+8
HC77-5481	-12	0	-2	+7
HC77-5686	-10	-2	-1	+3
K1033 Douglas	+4	+6	+9	+6
K1061	+2	+8	+6	+9
K1062	-2	+4	+2	0
K1063	+4	+3	+6	+12
K1066	+2	+4	+3	+9
K1067	+1	+2	+6	+7
Ky78-405	+2	+5	+7	+6
Ky78-1214	+4	+5	+7	+11
L77-515	-4	0	-1	+2
L77-546	-5	-2	-2	-1
L77-8043	+1	+2	+1	-1
L77-8079	0	+4	+4	-1
L77-8209	+2	+5	+5	+4
LN1053	-6	+6	+2	+7
LN1057	+4	+6	+6	+7
LN1058	0	+2	+2	+7
LN1059	+2	+2	+1	+5
LS78-229	+7	+25	+12	+29
LS78-335	+10	+25	+14	+29
LS78-344	+7	+25	+13	+29
Date planted	5/19	4/30	5/5	6/7
*Days to maturity	138	149	139	115

PRELIMINARY TEST IV, 1980

Mean 10 Tests	Mo.		Ill.		Md.	Ky.
	Portageville Loam	Clay	Belleville	Eldorado	Queenstown	Lexington
LODGING (score)						
2.2	2.0	1.5	2.7	2.7	3.3	1.5
1.8	1.5	1.0	2.3	2.4	2.0	1.5
2.3	2.5	1.5	2.9	2.4	3.3	2.0
2.5	2.0	1.0	3.4	4.4	3.5	1.5
2.8	2.5	1.0	2.8	4.8	3.5	2.0
2.3	2.0	1.5	2.4	3.5	3.3	1.5
2.0	2.0	1.5	2.2	2.9	2.5	1.3
1.2	1.0	1.0	1.3	1.3	1.5	1.0
1.8	2.0	1.5	2.1	1.9	3.0	1.3
2.4	2.0	1.5	2.4	3.7	3.5	1.5
2.1	2.0	1.5	2.5	3.5	2.8	1.8
1.1	1.0	1.0	1.3	1.5	1.0	1.0
1.5	1.5	1.5	1.8	1.8	1.8	1.3
1.2	1.0	1.0	1.5	1.8	1.5	1.0
1.3	1.0	1.0	1.5	2.2	1.0	1.0
1.3	1.0	1.0	1.4	2.3	1.3	1.0
1.9	1.5	2.0	1.9	1.7	2.0	1.3
2.2	1.5	2.0	2.9	3.0	3.5	2.0
2.0	1.5	2.0	2.4	2.0	2.8	1.8
1.6	1.5	1.0	2.0	2.3	2.0	1.8
1.2	1.0	1.0	1.5	1.4	1.5	1.0
1.8	1.5	2.0	1.9	1.8	2.5	1.3
2.3	2.0	1.5	2.6	3.9	3.5	1.8
1.9	1.5	1.5	2.1	2.9	2.0	1.0
2.0	2.0	1.5	2.5	3.1	2.5	1.5
2.1	2.0	1.5	2.2	2.4	3.3	1.3
2.1	2.0	1.5	2.9	2.3	3.0	1.5
1.9	2.0	2.0	2.3	1.9	2.5	1.0
1.9	1.5	1.5	2.0	2.3	3.3	1.3
1.4	1.0	1.0	2.3	1.9	1.5	1.5
1.9	1.0	2.5	2.2	2.3	2.0	1.3
2.5	1.5	2.0	2.7	3.9	3.3	1.5
2.1	1.5	2.0	2.5	3.2	3.0	1.0
2.1	1.0	2.0	2.3	1.9	3.5	3.3
2.2	1.0	2.0	2.4	2.1	3.3	2.5
2.5	2.0	2.0	2.3	2.4	3.8	2.5

PRELIMINARY TEST IV, 1980

Strain	Kan.	Ohio	Ind.	Del.
	Manhattan	S. Charleston	Sullivan	Georgetown
	<u>LODGING (score)</u>			
Union (IV)	2.5	2.2	2.5	1.0
Williams79 (III)	2.5	1.8	2.3	1.0
A79-331020	2.3	2.5	2.8	1.0
A79-335034	2.8	2.5	2.8	1.0
A79-336007	3.0	3.5	3.5	1.0
A79-337020	2.3	2.2	3.0	1.0
C1585	2.0	2.2	2.8	1.0
C1586	1.0	1.5	1.3	1.0
C1587	2.0	1.5	2.0	1.0
C1588	2.8	2.5	2.8	1.2
C1589	2.0	1.8	2.3	1.0
HC76-4449	1.0	1.2	1.3	1.0
HC77-982	1.0	2.2	1.5	1.0
HC77-1165	1.0	1.5	1.0	1.0
HC77-5481	1.0	1.8	1.3	1.0
HC77-5686	1.0	1.8	1.3	1.0
K1033 Douglas	2.5	2.0	3.0	1.0
K1061	2.0	2.0	2.3	1.0
K1062	2.0	2.2	2.3	1.0
K1063	1.0	1.8	1.5	1.0
K1066	1.0	1.2	1.3	1.0
K1067	2.3	1.5	1.8	1.0
Ky78-405	2.0	2.0	2.3	1.0
Ky78-1214	2.5	1.8	3.0	1.0
L77-515	2.3	1.2	2.3	1.0
L77-546	2.5	1.5	2.8	1.0
L77-8043	2.8	1.8	2.5	1.0
L77-8079	2.0	1.5	2.5	1.0
L77-8209	2.3	1.5	2.5	1.0
LN1053	1.0	1.5	1.5	1.0
LN1057	2.3	1.8	2.8	1.0
LN1058	3.8	2.2	3.0	1.0
LN1059	2.3	2.0	2.5	1.0
LS78-229	2.3	1.8	1.5	1.8
LS78-335	3.0	2.5	1.5	1.7
LS78-344	2.8	2.0	3.0	1.8

PRELIMINARY TEST IV, 1980

Mean 10 Tests	Mo.		Ill.		Md.	Ky.
	Portageville Loam	Clay	Belleville	Eldorado	Queenstown	Lexington
HEIGHT (inches)						
42	37	38	56	50	37	38
39	28	30	50	48	35	39
45	45	39	57	52	45	43
43	45	33	54	48	41	41
41	35	31	53	51	37	38
40	32	36	52	49	40	39
42	41	36	53	49	41	36
22	14	17	25	22	24	30
41	34	36	54	52	40	41
43	28	43	57	51	42	38
40	28	38	51	49	38	38
23	17	16	28	20	27	26
26	17	19	33	28	28	33
27	16	15	36	29	33	34
17	12	10	21	16	20	24
19	12	14	23	17	23	23
37	23	35	48	48	33	40
37	34	36	44	43	36	34
40	28	36	51	47	39	44
25	23	20	30	28	25	33
25	16	17	32	26	26	36
44	34	30	64	58	37	45
39	34	29	51	45	41	39
42	35	37	50	52	43	39
39	35	34	47	48	36	36
39	28	35	52	49	37	36
40	35	33	48	49	37	39
37	34	34	44	49	36	34
40	28	38	53	50	40	37
24	15	22	30	27	25	29
35	20	40	45	42	33	34
35	22	31	45	43	35	38
34	18	35	45	43	37	34
39	24	36	45	42	44	38
39	24	36	49	44	39	41
43	27	38	49	46	49	45

PRELIMINARY TEST IV, 1980

Strain	Kan.	Ohio	Ind.	Del.
	Manhattan	S. Charleston	Sullivan	Georgetown
	HEIGHT (inches)			
Union (IV)	48	44	49	25
Williams79 (III)	47	40	51	19
A79-331020	49	42	55	25
A79-335034	52	46	50	20
A79-336007	50	44	48	21
A79-337020	45	40	48	19
C1585	48	41	55	20
C1586	21	31	22	16
C1587	45	42	48	21
C1588	53	44	51	22
C1589	48	38	49	20
HC76-4449	20	28	23	20
HC77-982	29	34	24	18
HC77-1165	25	37	30	19
HC77-5481	11	24	19	12
HC77-5686	17	23	20	14
K1033 Douglas	42	40	44	19
K1061	40	38	42	22
K1062	41	42	51	23
K1063	27	28	21	19
K1066	25	33	20	19
K1067	53	41	52	23
Ky78-405	45	41	43	23
Ky78-1214	49	40	53	24
L77-515	53	40	44	20
L77-546	47	42	41	20
L77-8043	46	40	51	22
L77-8079	43	37	41	18
L77-8209	36	46	53	21
LN1053	22	32	18	17
LN1057	36	40	39	21
LN1058	45	34	39	18
LN1059	36	37	40	19
LS78-229	42	40	48	28
LS78-335	41	39	45	30
LS78-344	44	41	56	34

PRELIMINARY TEST IV, 1980

Mean 10 Tests	Mo. Portageville		Ill.		Md.	Ky.
	Loam	Clay	Belleville	Eldorado	Queenstown	Lexington
QUALITY (score)						
2.4	2.5	2.5	3.0	3.3	3.0	1.0
2.1	2.5	2.5	2.3	3.0	1.8	1.0
2.8	2.5	2.5	3.5	3.8	3.5	2.0
2.7	3.0	3.0	3.5	3.5	3.0	1.0
2.3	2.5	3.5	2.3	3.3	2.0	1.0
2.5	2.0	3.0	2.8	3.3	2.5	2.0
3.1	2.5	3.0	3.8	4.3	3.5	3.0
2.1	2.5	2.5	2.0	2.3	1.5	2.0
2.3	2.5	2.5	2.8	3.8	2.3	1.0
3.0	2.5	3.0	3.8	4.3	3.3	1.0
2.8	2.5	2.5	3.8	4.3	2.8	1.0
2.0	2.5	2.5	2.3	1.5	2.3	1.0
2.4	2.5	3.0	3.0	2.8	3.0	2.0
2.3	3.0	2.5	2.8	2.5	2.8	2.0
2.6	3.5	2.5	2.8	2.8	3.3	2.0
2.6	3.0	3.0	2.5	2.5	2.5	2.0
3.1	2.5	2.5	3.5	3.0	3.8	4.0
2.0	2.5	2.5	2.5	2.0	1.5	1.0
1.9	2.5	2.5	2.5	1.8	2.0	1.0
2.2	2.5	2.5	2.5	1.8	3.3	2.0
1.9	2.5	2.5	2.0	1.8	2.3	1.0
3.3	2.5	2.5	4.0	3.5	3.5	3.0
2.6	2.5	2.5	3.3	2.5	1.8	2.0
2.3	2.0	2.5	3.5	2.0	1.0	1.0
2.0	2.5	2.5	2.3	2.5	1.8	1.0
2.5	3.0	2.5	3.0	3.0	3.0	3.0
2.4	2.5	2.5	2.8	2.3	1.5	2.0
2.7	2.5	2.5	2.8	3.0	3.3	2.0
2.3	2.5	2.5	2.8	2.8	2.3	1.0
2.0	2.5	2.5	2.5	2.8	2.5	1.0
2.8	3.0	2.5	2.8	3.5	2.5	2.0
2.6	3.0	2.5	3.3	3.3	2.8	1.0
2.4	2.5	2.5	2.8	3.0	2.0	2.0
2.2	3.0	2.5	3.0	2.5	2.5	1.0
2.2	2.5	2.5	3.0	2.5	1.0	1.0
2.2	2.5	2.5	3.0	2.3	1.5	2.0

PRELIMINARY TEST IV, 1980

Strain	Kan.	Ohio	Ind.	Del.
	Manhattan	S. Charleston	Sullivan	Georgetown
	QUALITY (score)			
Union (IV)	3.0	1.0	2.0	2.3
Williams79 (III)	2.0	1.5	2.0	2.7
A79-331020	4.0	1.5	2.5	2.2
A79-335034	3.5	1.0	2.0	3.2
A79-336007	3.0	1.0	2.0	2.0
A79-337020	3.0	1.5	2.0	2.8
C1585	4.0	1.5	3.0	2.5
C1586	3.0	1.0	1.5	2.3
C1587	2.5	1.0	2.0	2.7
C1588	3.5	1.5	4.0	2.8
C1589	4.0	1.5	3.5	2.3
HC76-4449	3.0	1.0	1.5	2.2
HC77-982	2.0	1.0	1.5	2.8
HC77-1165	2.0	1.0	2.0	2.7
HC77-5481	3.0	1.5	1.5	2.8
HC77-5686	4.5	1.0	3.0	2.3
K1033 Douglas	4.0	2.5	3.0	2.5
K1061	3.0	1.0	1.5	2.3
K1062	2.5	1.0	1.5	2.0
K1063	2.0	1.0	1.5	2.5
K1066	2.0	1.0	1.5	2.3
K1067	4.5	1.5	4.5	3.0
Ky78-405	3.5	2.0	3.5	2.8
Ky78-1214	3.0	1.5	3.0	3.0
L77-515	2.0	1.0	2.5	2.3
L77-546	2.0	1.0	2.0	2.5
L77-8043	4.0	1.5	2.5	2.5
L77-8079	3.0	1.5	4.0	2.5
L77-8209	3.0	1.0	2.5	2.5
LN1053	2.0	1.0	1.5	2.0
LN1057	3.5	1.5	4.0	3.0
LN1058	3.0	1.0	3.0	3.0
LN1059	2.5	1.0	2.5	2.8
LS78-229	2.0	1.0	1.5	2.5
LS78-335	3.0	1.5	2.0	2.5
LS78-344	3.0	1.0	1.5	2.5

PRELIMINARY TEST IV, 1980

Mean 8 Tests	Mo.	Ill.		Md.	Ky.
	<u>Portageville</u> Clay	Belleville	Eldorado	Queenstown	Lexington
		SIZE (g/100)			
16.5	15.5	18.2	15.7	17.3	16.0
16.3	14.5	16.3	21.6	14.9	14.7
16.6	15.6	16.6	14.1	18.0	16.5
14.3	12.6	15.1	12.6	14.1	14.0
12.9	14.6	13.2	11.4	12.3	11.1
15.0	14.1	14.9	14.0	14.4	12.8
16.6	16.5	15.8	15.2	16.9	15.2
15.8	14.6	15.1	14.8	17.5	13.6
14.8	14.2	15.0	14.4	14.8	14.3
16.0	14.8	16.1	14.0	15.9	15.8
15.5	14.6	16.0	14.4	15.3	15.8
14.5	12.7	14.4	13.3	15.7	12.8
16.2	14.7	16.3	15.4	16.9	14.4
14.2	12.0	15.0	13.5	14.8	13.2
14.8	11.5	16.4	13.4	14.9	12.3
15.7	11.9	15.1	16.2	15.5	13.1
17.0	15.5	16.8	17.8	18.6	16.0
15.0	14.7	15.4	14.0	15.7	14.5
15.6	13.0	16.4	14.0	16.5	15.0
15.7	15.8	16.0	12.9	18.6	13.9
15.1	14.8	15.0	12.5	15.3	14.1
14.1	13.6	13.5	13.6	14.6	13.8
15.4	14.7	16.4	13.5	17.1	15.9
15.0	13.7	14.5	13.8	16.0	14.4
14.0	11.8	14.2	12.9	14.6	13.3
15.7	13.1	17.1	14.6	15.3	14.8
14.0	13.8	14.9	13.1	14.4	12.0
14.3	12.8	14.4	13.0	14.9	13.5
15.0	13.2	15.8	13.2	16.2	13.4
15.1	13.0	16.0	12.8	17.4	12.7
18.6	17.5	18.0	17.8	20.1	18.6
16.3	14.8	16.0	15.5	16.2	14.3
17.4	17.0	17.3	16.3	18.3	16.6
12.0	11.7	11.7	10.9	15.0	12.2
11.4	12.1	11.2	10.7	14.5	11.6
11.7	11.1	12.5	10.2	15.0	12.0

PRELIMINARY TEST IV, 1980

Strain	Kan.	Ohio	Ind.
	Manhattan	S. Charleston	Sullivan
		SIZE (g/100)	
Union (IV)	14.7	17.7	17.2
Williams79 (III)	15.8	16.6	16.0
A79-331020	16.3	17.4	18.3
A79-335034	14.9	15.8	15.3
A79-336007	13.7	13.5	13.1
A79-337020	16.2	17.1	16.1
C1585	16.6	18.3	18.3
C1586	14.1	19.1	17.5
C1587	15.3	15.0	15.5
C1588	16.3	16.7	18.0
C1589	14.6	16.7	16.6
HC76-4449	15.3	15.3	16.1
HC77-982	16.3	17.7	17.5
HC77-1165	14.2	15.7	14.9
HC77-5481	16.1	17.6	16.4
HC77-5686	16.5	18.2	19.2
K1033 Douglas	16.4	17.6	17.4
K1061	14.1	15.7	16.2
K1062	15.8	17.8	15.9
K1063	14.3	16.8	17.1
K1066	16.6	15.8	16.4
K1067	14.3	14.6	14.7
Ky78-405	15.2	14.9	15.5
Ky78-1214	16.2	15.4	16.1
L77-515	14.8	15.6	14.8
L77-546	16.4	17.4	16.9
L77-8043	14.5	15.2	14.4
L77-8079	15.0	15.1	15.6
L77-8209	16.1	15.7	16.7
LN1053	15.5	16.8	16.6
LN1057	16.4	20.8	19.8
LN1058	16.4	18.3	18.6
LN1059	16.2	19.2	18.6
LS78-229	11.4	10.9	11.9
LS78-335	10.5	9.6	10.8
LS78-344	11.2	9.4	12.1

PRELIMINARY TEST IV, 1980

Mean	Mo.	Ill.	Ky.	Kan.	Ind.
5 Tests	Portageville	Eldorado	Lexington	Manhattan	Sullivan
	Loam				
PROTEIN (%)					
44.5	44.3	43.6	44.3	46.2	44.2
42.6	42.2	41.8	43.6	41.7	43.6
43.1	40.2	42.1	42.4	45.6	45.2
44.5	-	42.0	44.2	44.7	47.1
43.2	40.7	45.1	43.4	41.5	45.5
42.3	39.5	41.9	42.1	44.0	44.1
45.2	45.1	44.4	44.2	46.4	45.9
42.9	42.3	41.0	43.9	43.0	44.2
44.2	43.7	43.9	44.3	44.1	45.2
43.5	40.5	44.0	42.4	44.7	45.7
43.6	42.1	43.2	43.1	44.4	45.4
44.4	42.7	43.4	44.5	45.6	45.6
43.9	42.8	42.8	44.1	45.8	43.8
42.4	42.0	40.2	42.4	43.7	43.6
43.8	42.2	41.3	44.3	45.3	45.7
44.6	43.6	44.1	46.0	45.9	43.6
42.8	41.8	41.6	42.5	43.8	44.4
45.0	44.3	44.1	45.2	45.7	45.7
41.6	40.7	40.7	41.9	41.6	43.0
43.6	45.0	42.2	43.2	44.4	43.2
41.4	43.1	39.4	42.0	41.5	40.8
42.4	39.6	41.3	43.1	43.9	44.3
43.8	43.1	41.9	43.8	44.9	45.5
42.3	38.2	42.2	42.4	44.8	43.8
42.1	40.5	41.3	42.7	43.2	42.9
42.5	39.2	42.5	44.2	44.2	42.4
41.6	40.7	40.6	43.7	40.8	42.1
42.0	41.3	40.5	42.3	42.3	43.6
41.7	40.3	41.4	41.3	42.0	43.4
43.6	41.2	44.1	44.9	47.1	43.4
45.0	43.3	46.5	44.2	44.8	46.3
43.6	41.7	42.9	42.6	44.7	46.2
43.6	44.0	44.0	43.3	42.2	44.4
39.1	37.4	39.0	37.9	40.0	41.3
39.1	35.5	38.9	38.7	40.6	41.6
39.2	34.9	39.2	37.6	42.6	41.5

PRELIMINARY TEST IV, 1980

Strain	Mean 5 Tests	Mo.	Ill.	Ky.	Kan.	Ind.
		Portageville Loam	Eldo- rado	Lexing- ton	Man- hattan	Sulli- van
OIL (%)						
Union (IV)	19.5	15.8	20.9	18.5	19.7	22.5
Williams79 (III)	20.3	16.3	22.3	18.9	20.8	23.0
A79-331020	18.4	15.9	22.6	18.2	19.3	16.1
A79-335034	18.3	-	20.1	18.7	18.0	15.6
A79-336007	19.1	16.4	21.5	17.5	19.0	21.0
A79-337020	19.2	15.7	21.8	18.6	18.1	21.8
C1585	18.7	15.5	19.9	17.7	18.7	21.9
C1586	19.2	15.7	20.9	17.9	19.3	22.0
C1587	19.4	16.4	21.6	18.1	19.8	21.2
C1588	19.2	15.8	19.6	19.1	19.4	21.9
C1589	19.0	16.1	20.5	18.8	17.9	21.6
HC76-4449	19.3	15.7	21.4	18.2	18.9	22.5
HC77-982	20.1	16.3	23.0	17.8	20.2	23.3
HC77-1165	20.4	16.1	23.5	18.6	20.5	23.2
HC77-5481	19.3	16.0	22.2	17.4	19.1	22.0
HC77-5686	19.0	15.5	20.0	17.4	18.8	23.1
K1033 Douglas	19.3	16.5	20.9	18.0	18.7	22.2
K1061	17.9	14.2	19.6	17.0	17.8	20.9
K1062	19.6	16.3	21.4	17.9	20.4	22.2
K1063	18.4	14.2	20.3	17.8	18.2	21.6
K1066	19.1	15.6	21.4	17.9	19.4	21.4
K1067	19.6	16.7	20.3	18.8	19.8	22.2
Ky78-405	18.6	15.7	21.1	17.1	17.5	21.5
Ky78-1214	19.6	16.7	21.3	18.9	18.5	22.4
L77-515	20.1	17.2	22.0	18.8	19.9	22.7
L77-546	20.1	16.8	22.2	18.3	20.2	23.2
L77-8043	19.8	16.5	22.5	18.2	20.2	21.6
L77-8079	20.5	16.3	21.0	22.6	20.0	22.7
L77-8209	20.5	17.6	21.7	22.1	19.3	21.9
LN1053	20.0	15.9	21.6	20.8	16.8	22.4
LN1057	19.9	16.1	21.6	21.7	18.6	21.4
LN1058	19.7	15.8	21.1	21.9	18.9	20.7
LN1059	19.6	15.6	21.0	21.4	19.2	20.8
LS78-229	20.4	17.2	21.1	23.5	19.1	21.0
LS78-335	20.4	18.0	22.2	21.5	18.4	21.8
LS78-344	21.0	17.7	22.0	23.5	19.7	22.2